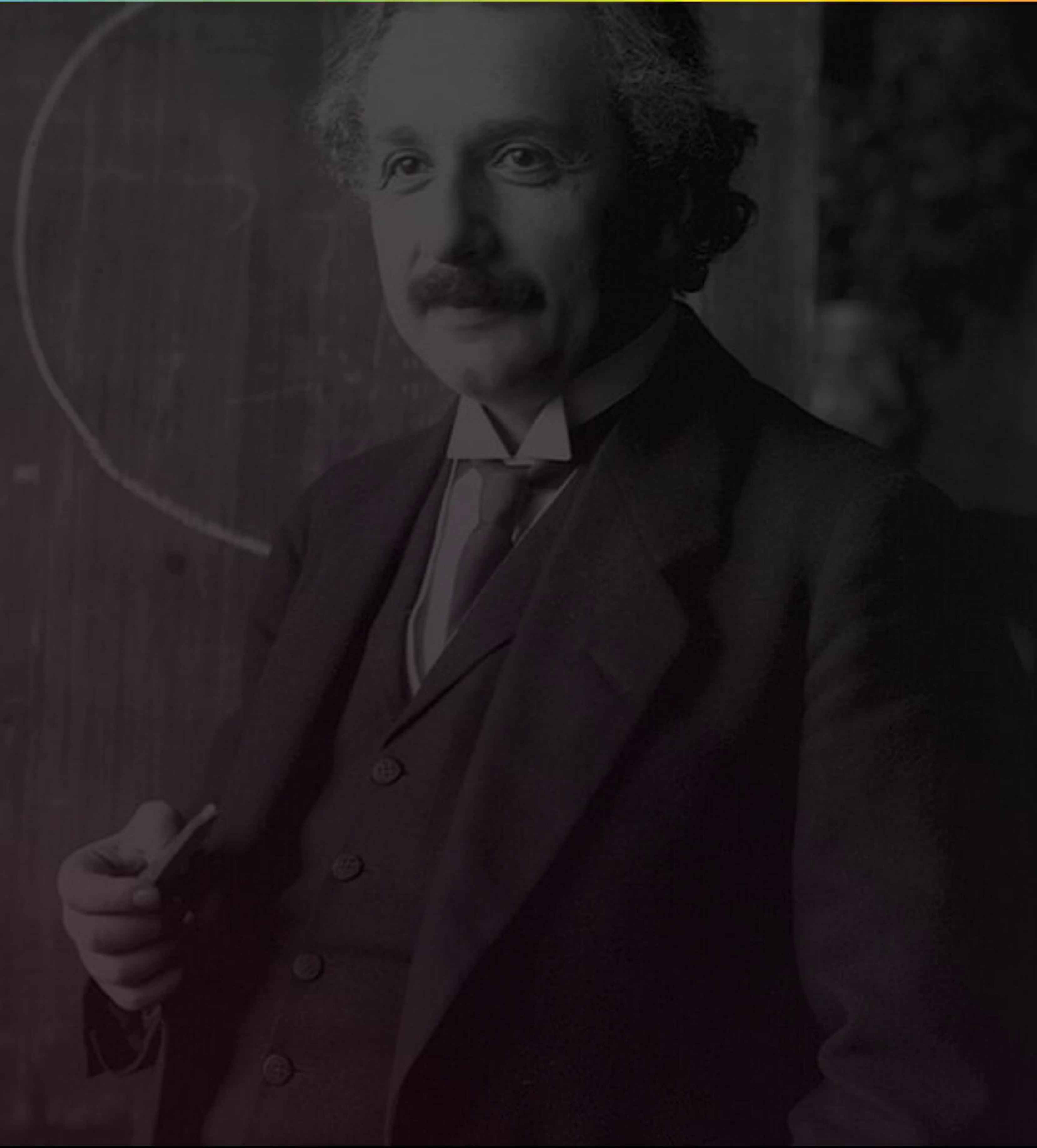
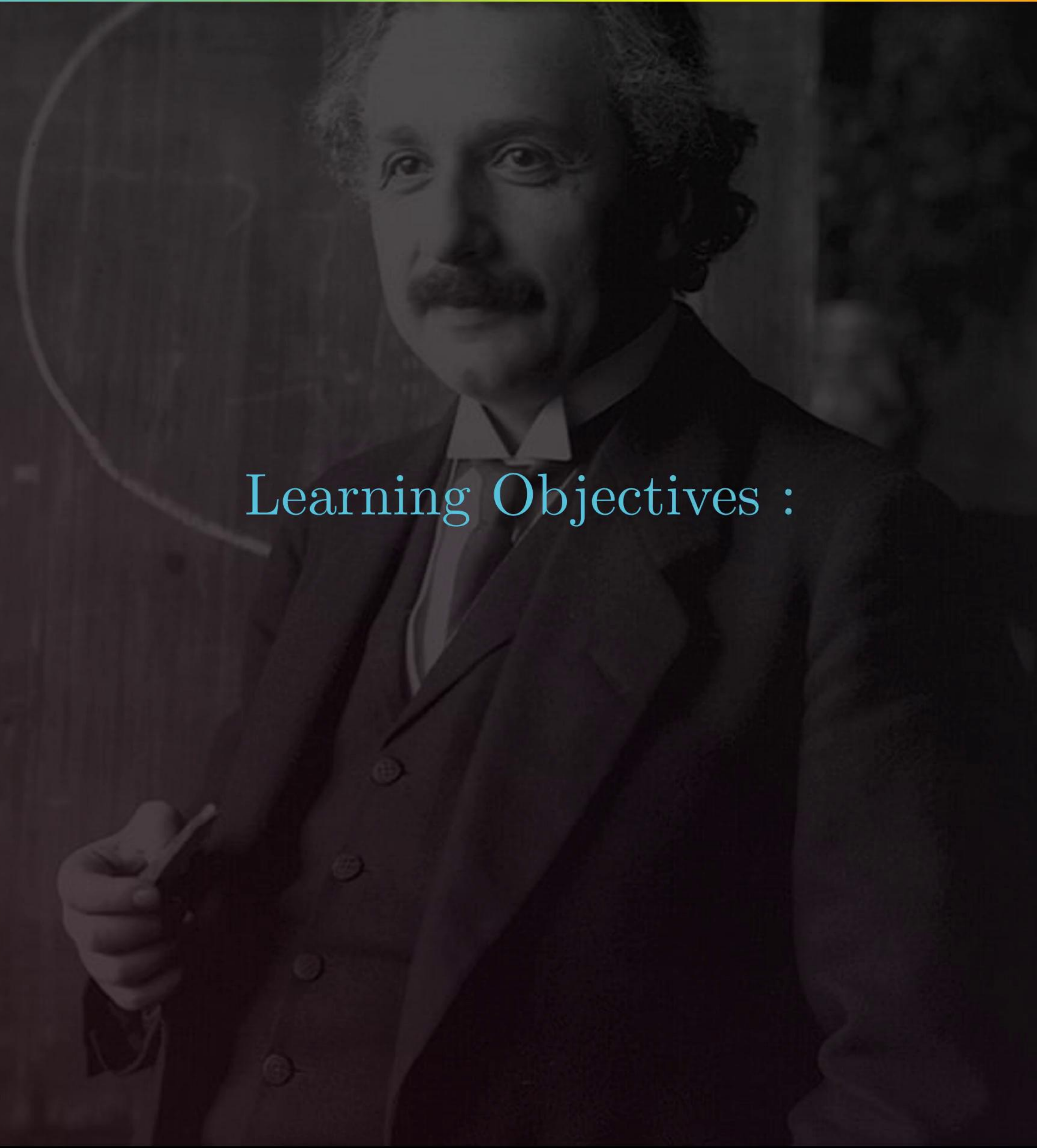


CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER



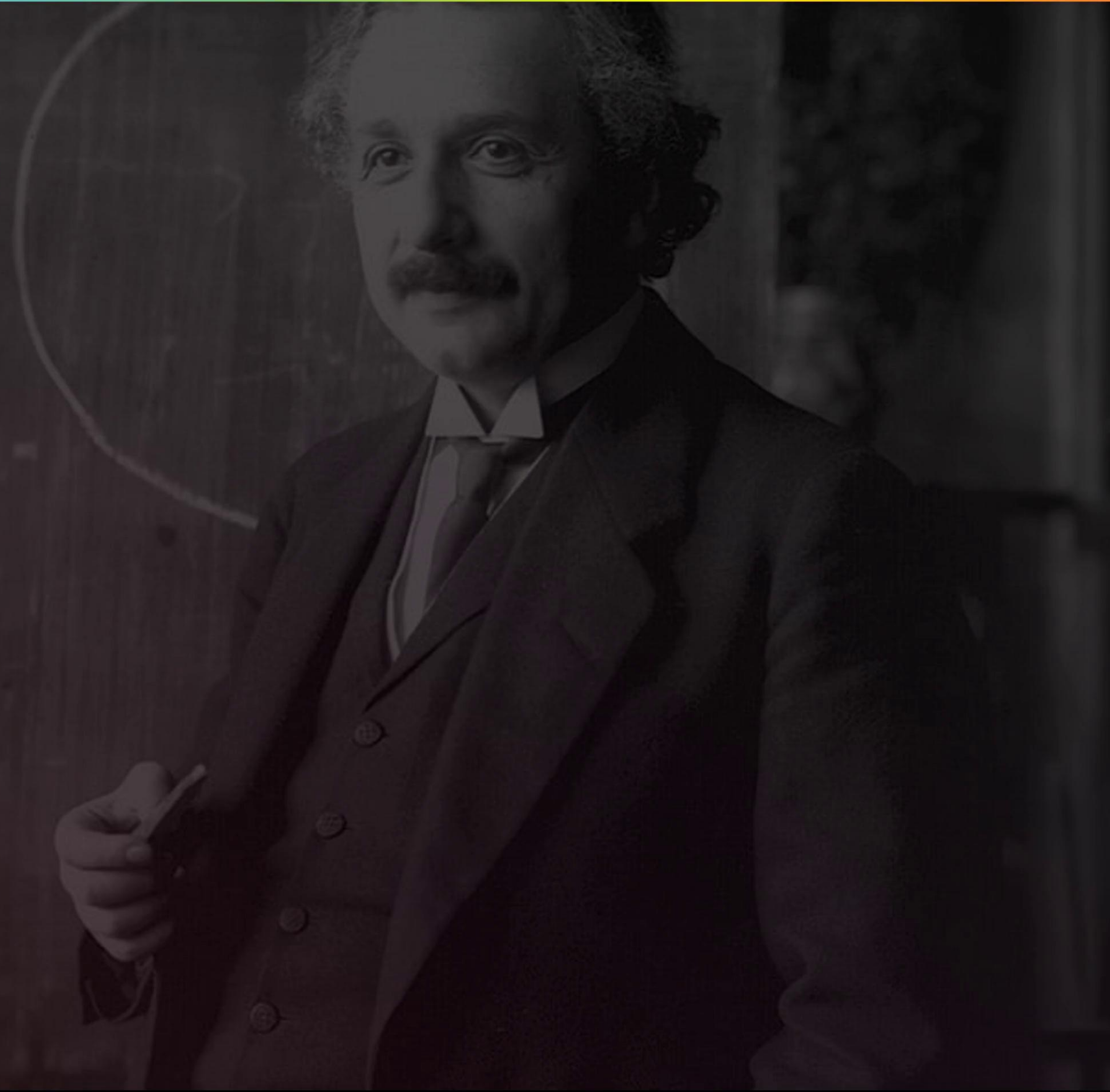
CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :



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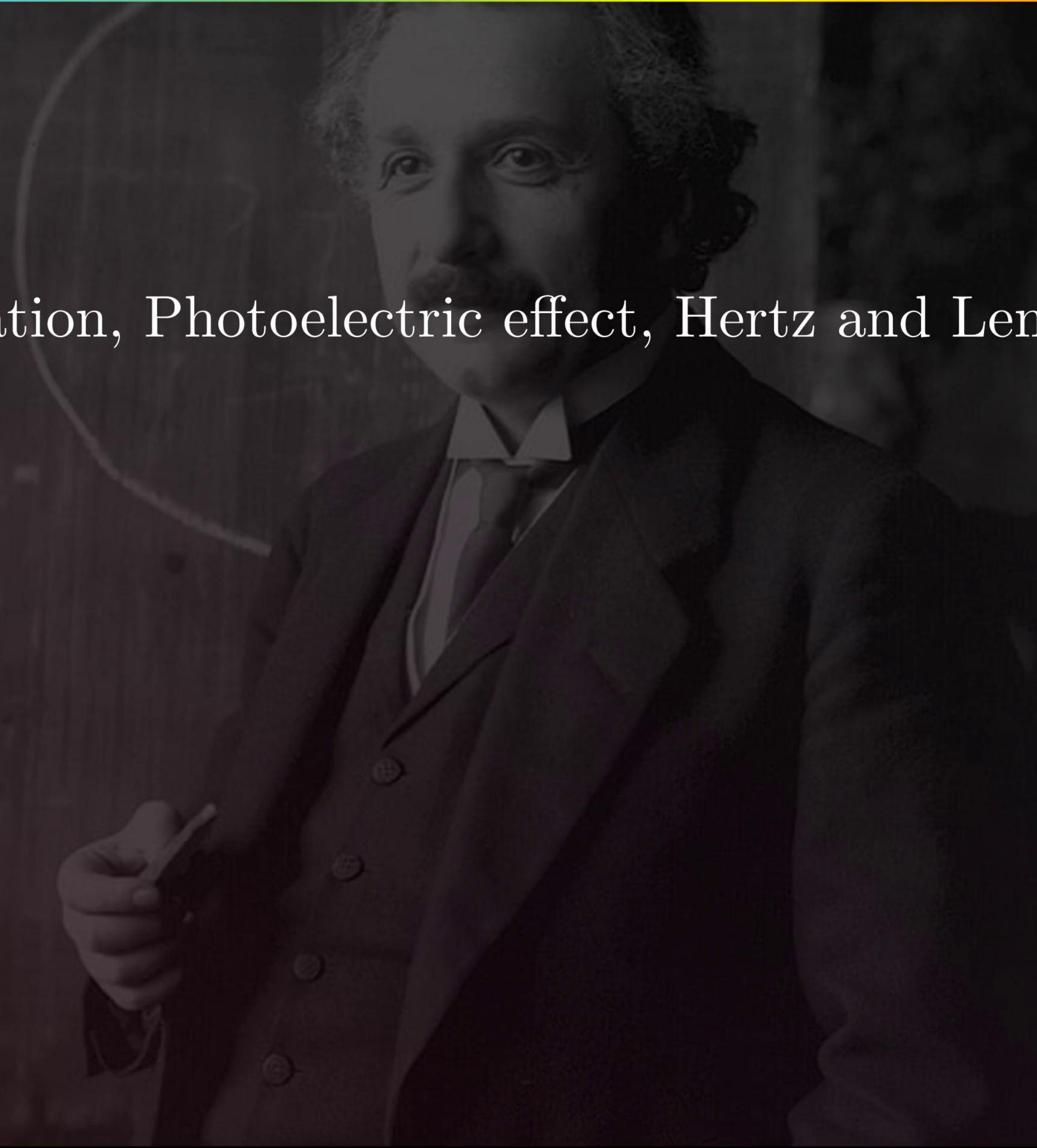
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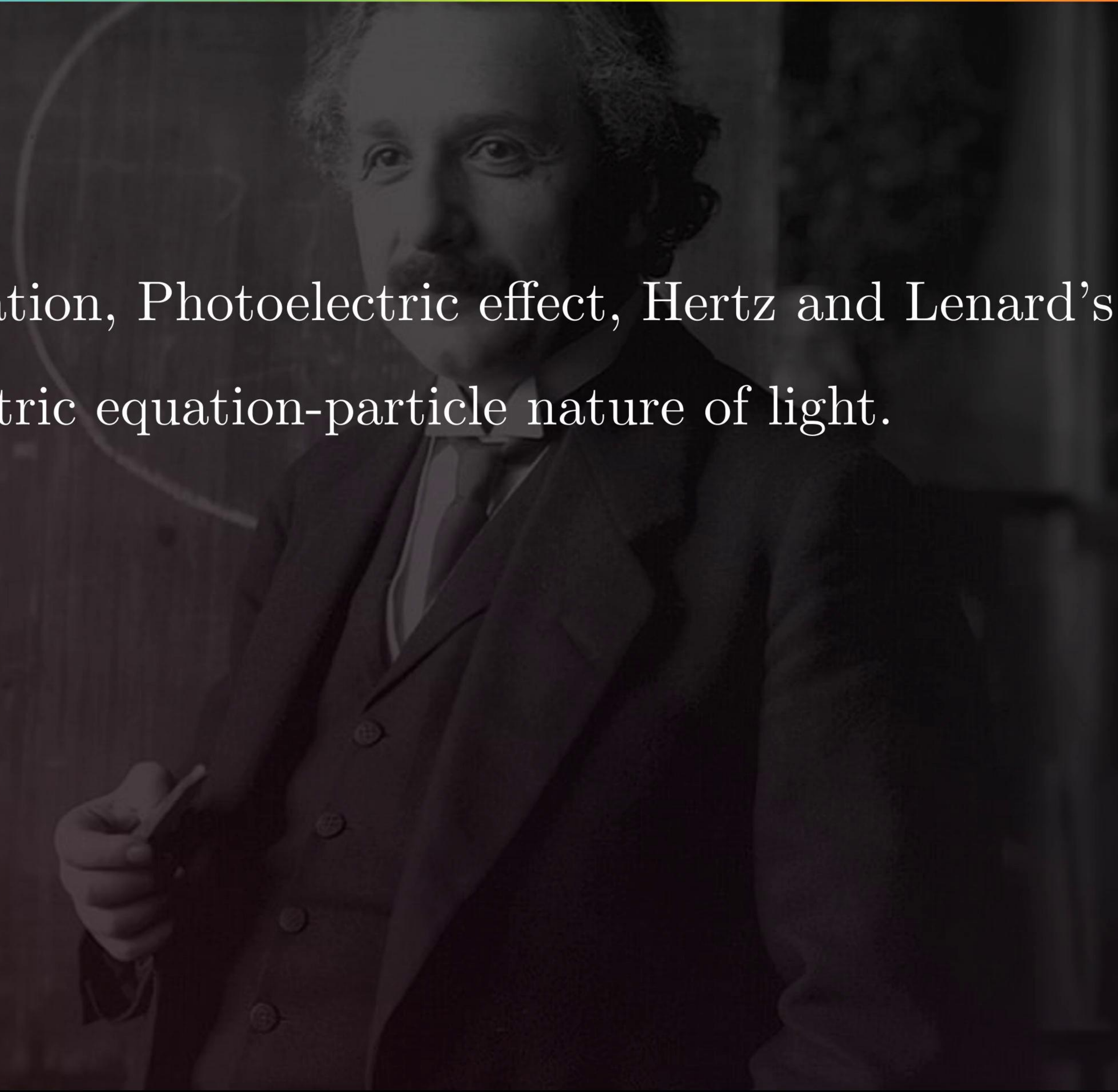
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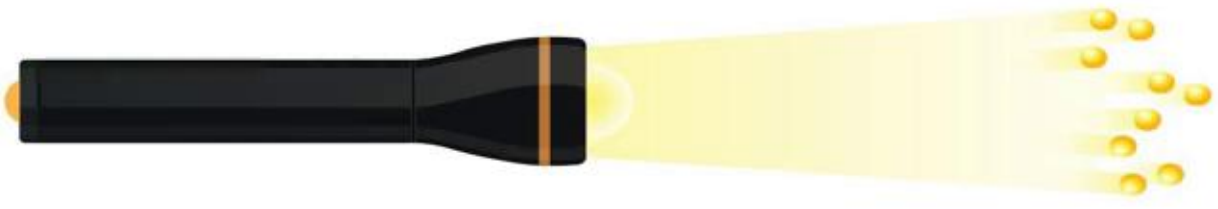
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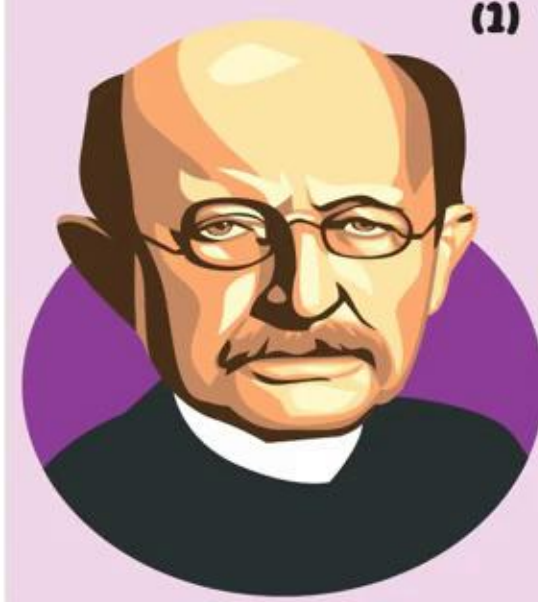
- Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations;
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- Matter waves-wave nature of particles, de-Broglie relation.

DUAL NATURE OF RADIATION AND MATTER



Particles

PLANCK'S QUANTUM THEORY OF LIGHT



- (1) The energy of one photon is proportional to its frequency
- (2) $E \propto \nu$, $E = h\nu$
 $h = \text{Planck's constant} = 6.62 \times 10^{-34} \text{ JS}$
- (3) Energy of any light or radiation is one integral multiple of $h\nu$.
 $E = nh\nu$
- (4) Energy of one photon.
 $E = h\nu = \frac{1240}{\lambda(\text{nm})} \text{ eV}$

PROPERTIES OF PHOTONS

- (1) Photon is just a packet of energy.
- (2) Energy of photon does not change with medium.
- (3) Photon can not be deflected by electric field and magnetic field.
- (4) Momentum of photon $|p| = m \times c = \frac{h}{\lambda} = \frac{E}{c}$
- (5) Intensity of light beam = $\frac{\text{Energy}}{\text{area} \times \text{time}}$



PHOTON EMITTED PER SECOND

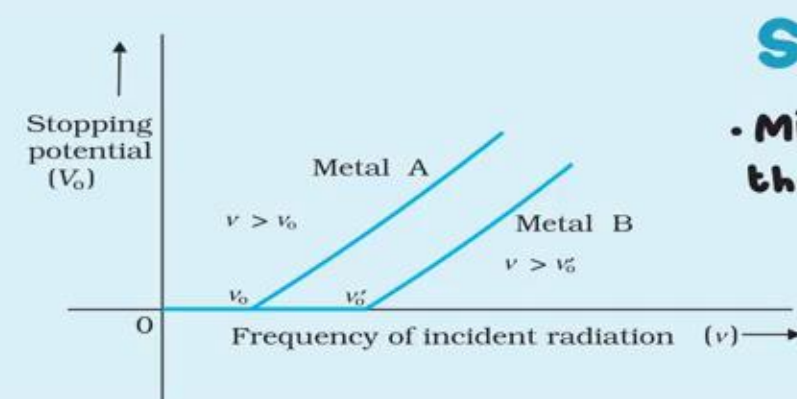
(1) $E = nh\nu$
 (2) Power, $P = nh\nu$
 $\Rightarrow = \frac{p}{h\nu} = \frac{p\lambda}{hc}$
 Number of photon per second = $\frac{\text{Power}}{\text{energy of one photon}}$

PHOTON FLUX

• Photon flux is no. of photon incident normally to a surface per seconds
 $\phi = \frac{n}{A} = \frac{P}{h\nu}$

FORCE AND RADIATION PRESSURE EXERTED BY A LIGHT BEAM

(1) $n = \frac{p\lambda}{hc}$
 (2) momentum of one photon = $\frac{h}{\lambda}$ momentum imparted per second = $n \times \frac{h}{\lambda} = \frac{P}{c}$
 $\Rightarrow \text{Force exerted} = \frac{P}{c}$
 (3) Radiation Pressure = $\frac{F}{A} = \frac{P}{CA} = \frac{I}{C}$



STOPPING POTENTIAL

• Minimum negative potential required to stop the electron of maximum K.E.

$$V_0 = \frac{K.E_{\text{max}}}{e} = \frac{h}{e} (\nu - \nu_0) \text{ Volts}$$

PHOTOELECTRIC EFFECT

- (1) It is a phenomenon of ejecting electrons by falling light of suitable frequency on a metal
- (2) Ejected electrons are called photoelectrons.
- (3) Current flowing due to the photoelectrons is called photocentric current

LAWS

- (1) No emission takes place below the threshold frequency.
- (2) Above threshold frequency, no. of photoelectrons emitted per seconds is directly proportional to intensity of radiation
- (3) The emission of photoelectrons is an instantaneous process.
- (4) Above threshold frequency, K.E (max) depends on frequency

WORK FUNCTION

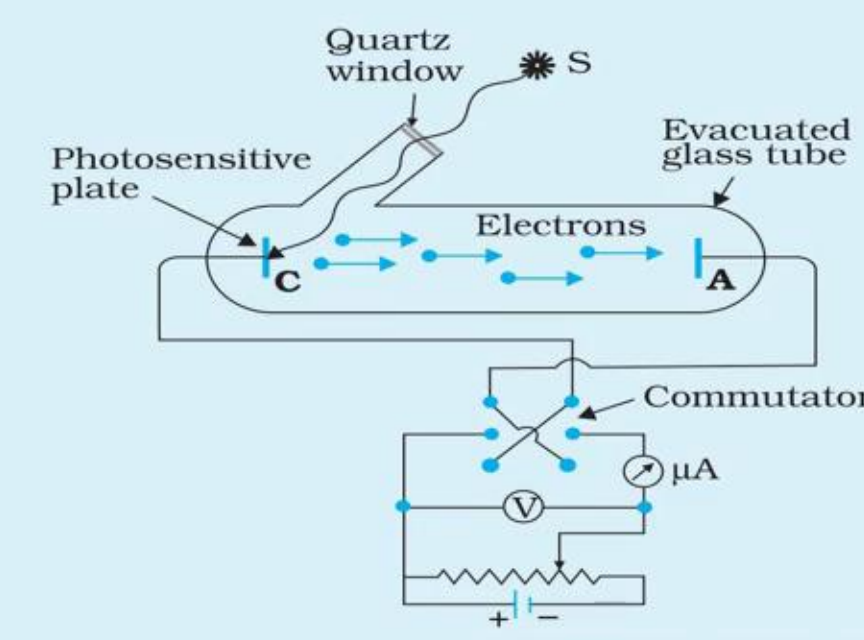
- Minimum energy required for getting a free electron away from the metal surface.
- Work function (ϕ_0) = $h\nu_0$
 $\nu_0 = \frac{\phi}{h} = \text{threshold frequency}$

EINSTEIN'S PHOTOELECTRIC EQUATION

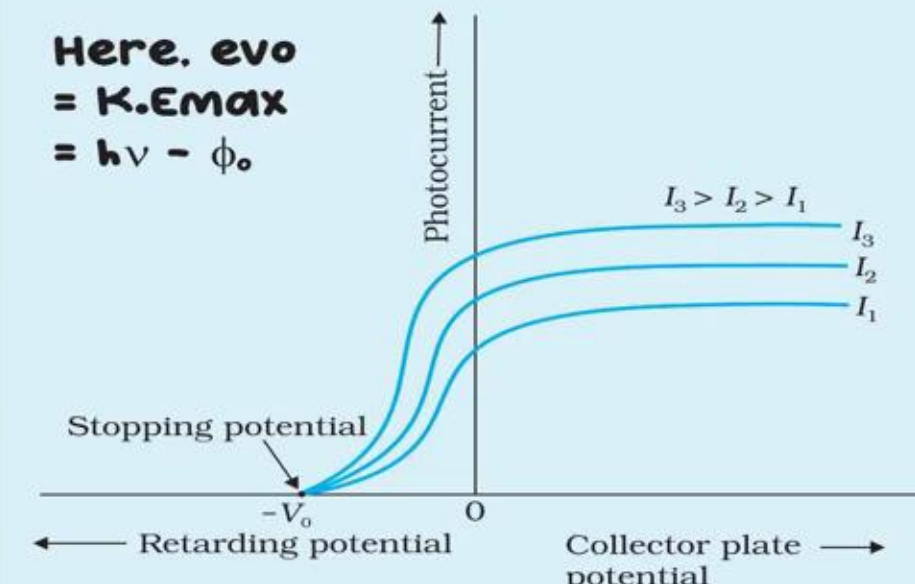
- The electron is emitted with maximum K.E
 $K.E_{\text{max}} = h\nu - \phi_0$
 $h\nu = K.E_{\text{max}} + \phi_0$
- Range of K.E.
 $0 \leq K.E_{\text{photoelectrons}} \leq h\nu - \phi_0$

EXPERIMENTAL STUDY

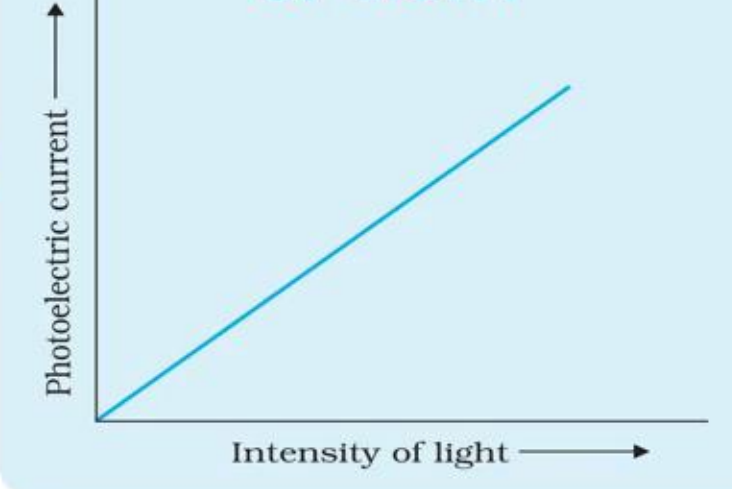
• The emission of electrons causes flow of electric current in the circuit.



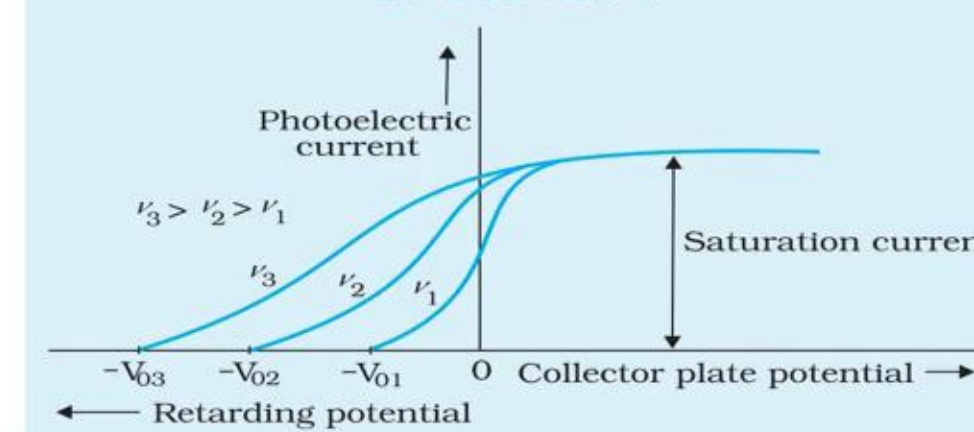
EFFECT OF POTENTIAL



EFFECT OF INTENSITY OF LIGHT



EFFECT OF FREQUENCY OF INCIDENT



DUAL NATURE OF LIGHT



de - Broglie wavelength, $\lambda = \frac{h}{mv}$ & $2\pi r = n\lambda$
 $\bullet mvr = \frac{nh}{2\pi}$
 This is Bohr quantisation Condition

PARTICLE NATURE OF LIGHT

- (1) In interaction of radiation with matter, radiation behaves as if it is made of particles called photons
- (2) $E = h\nu$ and $p = \frac{h\nu}{c}$
- (3) In a photon - particle collision, total energy and total momentum are conserved.

MATTER WAVE THEORY



- de - Broglie wavelength associated with moving particles, $\lambda = \frac{h}{p}$
- K.E of particle = $\frac{1}{2}mv^2 = \frac{p^2}{2m}$
- momentum, $p = mv = \sqrt{2m \times K.E}$

FOR UNCHARGED PARTICLES

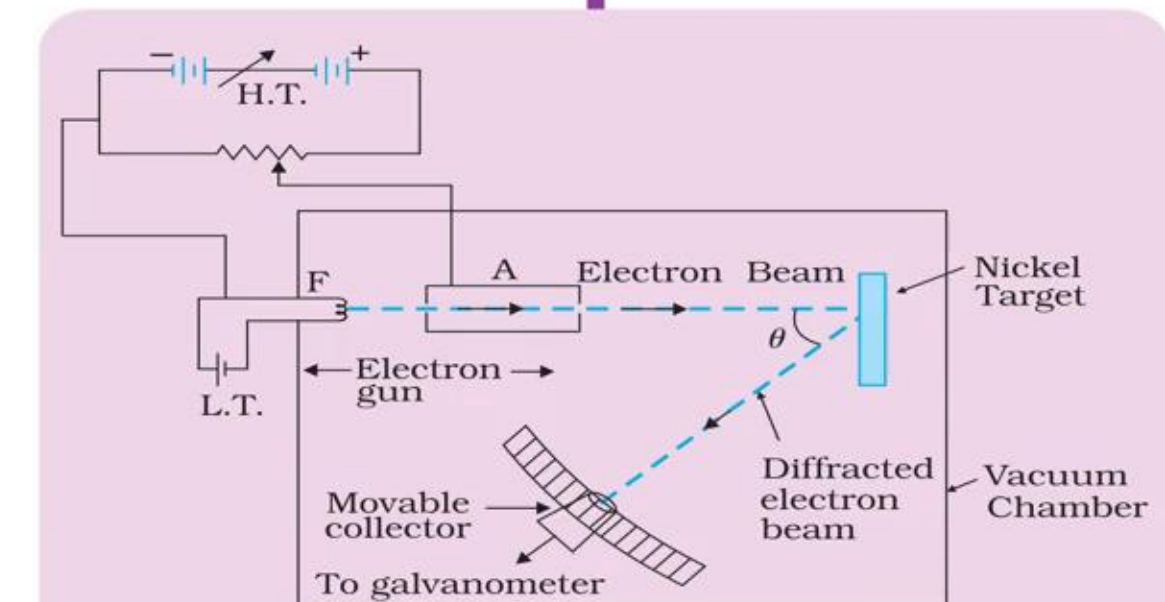
$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m \times K.E}}$$

FOR ACCELERATED CHARGED PARTICLES

$$\lambda = \frac{h}{\sqrt{2m \times K.E}}$$

V = potential difference

DAVISON-GERMER EXPERIMENT



- at $\phi = 50^\circ$ and accelerating potential = 54 V, maxima is obtained
- This experiment confirmed the wave nature of electron.

SPECIAL CASE FOR ELECTRON

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

AND
 $V = \frac{150.1}{[\lambda(\text{\AA})]^2} \text{ Volt}$

FOR GASEOUS MOLECULES

$$K.E = \frac{3}{2}KT$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2m \times \frac{3}{2}KT}}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{3mKT}}$$

INTRODUCTION

★ Wave Nature of Light Was Well Established

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- * By the end of the 19th century, light was widely accepted as a wave.
- * Maxwell's equations predicted electromagnetic waves.
- * Hertz's experiments (1887) confirmed the existence of electromagnetic waves.
- * Thus, the wave theory of light was strongly supported.

INTRODUCTION

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(i) X-rays — Roentgen (1895)

- * X-rays were discovered during experiments on gas discharge.
- * This became an important tool for understanding atomic structure.

(iii) Electron — J.J. Thomson (1897)

- * Thomson confirmed cathode rays are made of negatively charged particles (electrons).
 - * He used crossed electric and magnetic fields to determine:
 - * Speed of electrons: about 0.1–0.2 times the speed of light
 - * e/m ratio: 1.76×10^{11} C/kg
 - * The e/m ratio was the same regardless of the gas or metal used.
 - ✓ This proved electrons are universal constituents of matter.
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- * This showed the presence of rays emerging from the cathode → cathode rays.
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★ More Evidence Supporting the Existence of Electrons

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 - * Heated (thermionic emission)
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★ Millikan's Oil Drop Experiment (1913)

R.A. Millikan measured the charge of the electron accurately.

Found that: Charge (e) = 1.602×10^{-19} C

Charge appears only in integral multiples of this value → charge is quantised.

Using e and e/m , the mass of the electron was calculated.

ELECTRON EMISSION

★ Meaning of Electron Emission

- * Metals contain free electrons that are responsible for electrical conductivity.
- * These free electrons cannot normally escape from the metal surface.
- * When an electron tries to leave, the metal surface becomes positively charged and pulls the electron back.
- * An electron can come out of the metal only when it gains sufficient energy to overcome this attractive force.
- * The minimum energy required for an electron to escape from the metal surface is called the work function.

★ 2. Work Function (ϕ_0)

- * Definition: The minimum energy required by an electron to escape from the metal surface.
- * Unit: electron volt (eV) $1\text{eV} = 1.602 \times 10^{-19} \text{ J}$
- * The value of ϕ_0 depends on the metal and the nature of its surface.
- * Different metals have different work functions.
- * Example :
 - Cesium (Cs): 2.14 eV
 - Platinum (Pt): 5.65 eV — highest among the listed metals.

ELECTRON EMISSION

TABLE 11.1 WORK FUNCTIONS OF SOME METALS

Metal	Work function ϕ_0 (eV)	Metal	Work function ϕ_0 (eV)
Cs	2.14	Al	4.28
K	2.30	Hg	4.49
Na	2.75	Cu	4.65
Ca	3.20	Ag	4.70
Mo	4.17	Ni	5.15
Pb	4.25	Pt	5.65

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(iii) Photoelectric Emission

- * Electrons are emitted when light of suitable frequency falls on the metal surface.
- * The emitted electrons are called photoelectrons.
- * The minimum frequency required is called threshold frequency (ν_0).

PHOTOELECTRIC EFFECT

- * When light of suitable frequency falls on a metal surface, electrons are emitted.
- * The emitted electrons are called photoelectrons.
- * The phenomenon is called the photoelectric effect.

PHOTOELECTRIC EFFECT

(1) Hertz's observations

PHOTOELECTRIC EFFECT

(1) Hertz's observations

- * Hertz (1887) discovered the photoelectric effect accidentally during experiments on electromagnetic waves.
- * He observed that sparks across a detector loop became stronger when the emitter plate was illuminated with ultraviolet light.
- * UV light helped electrons escape from the metal surface, allowing the spark discharge to occur more easily.
- * When light shines on a metal, electrons near the surface absorb energy from incident light.
- * If the absorbed energy is greater than the work function, electrons escape from the metal surface.
- * Conclusion: Light can cause the emission of charged particles (electrons) from a metal surface.

PHOTOELECTRIC EFFECT

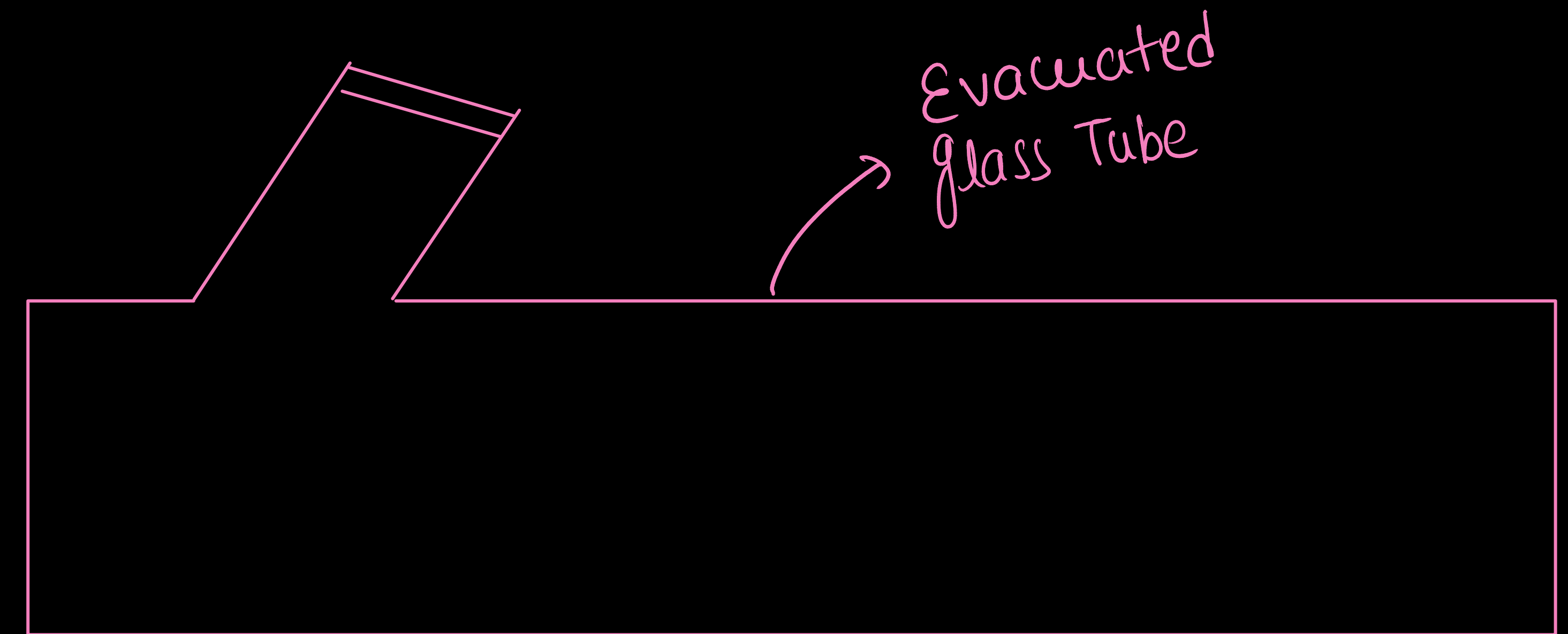
(2) Hallwachs' and Lenard's observations

Lenard's Observations :

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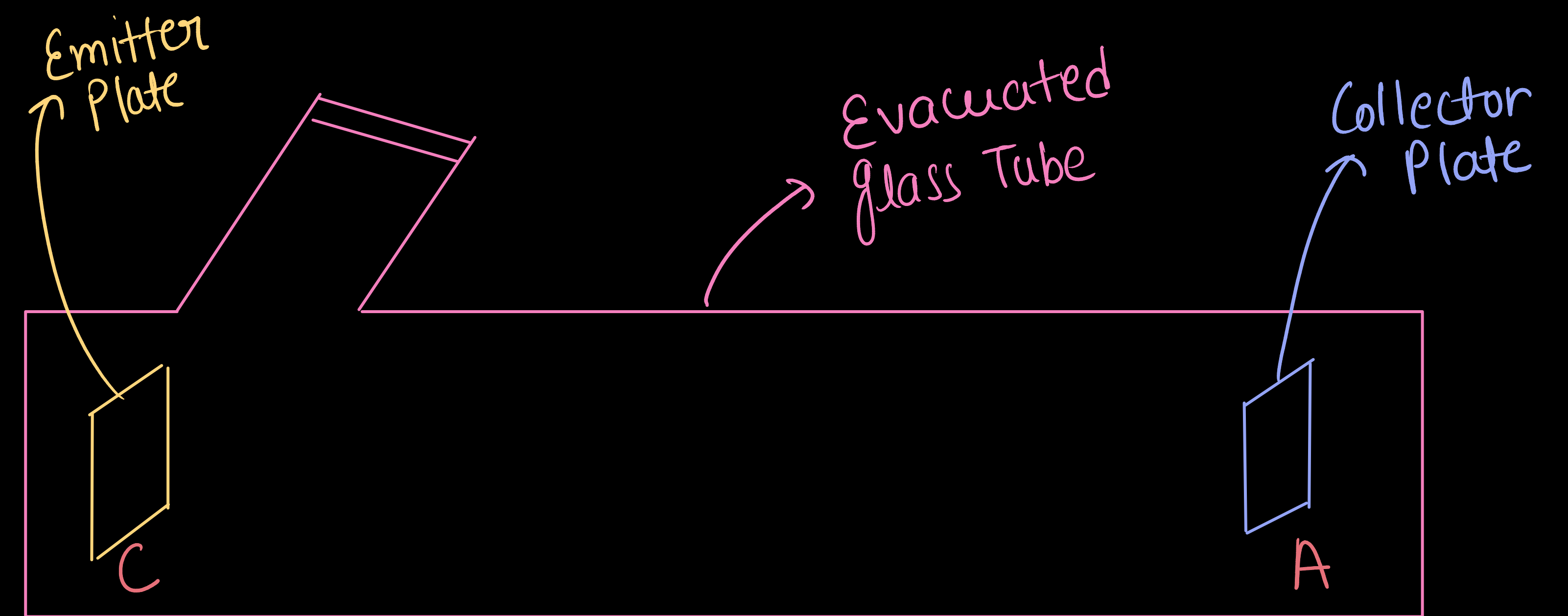
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Lenard's Observations :

* Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)



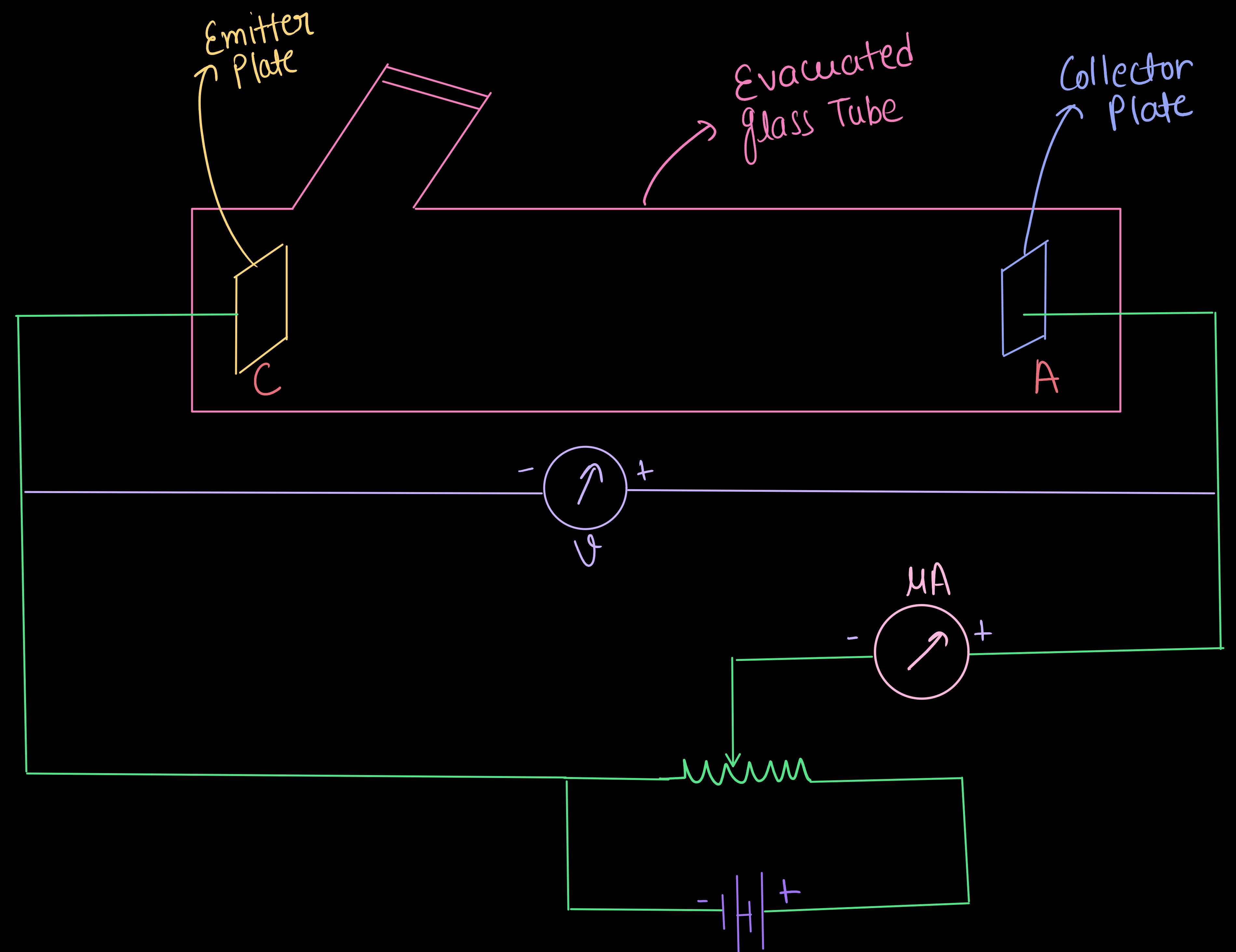
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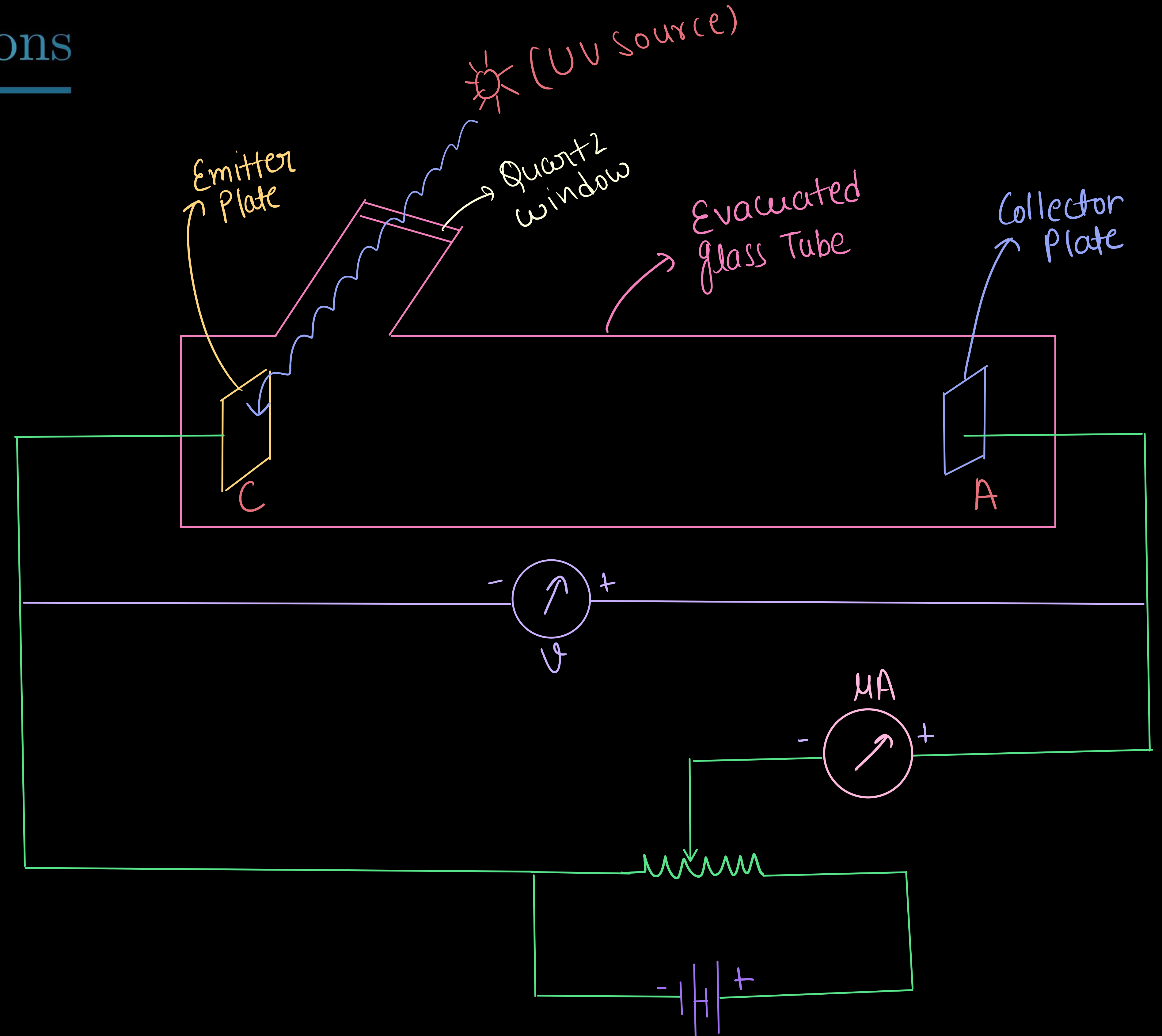
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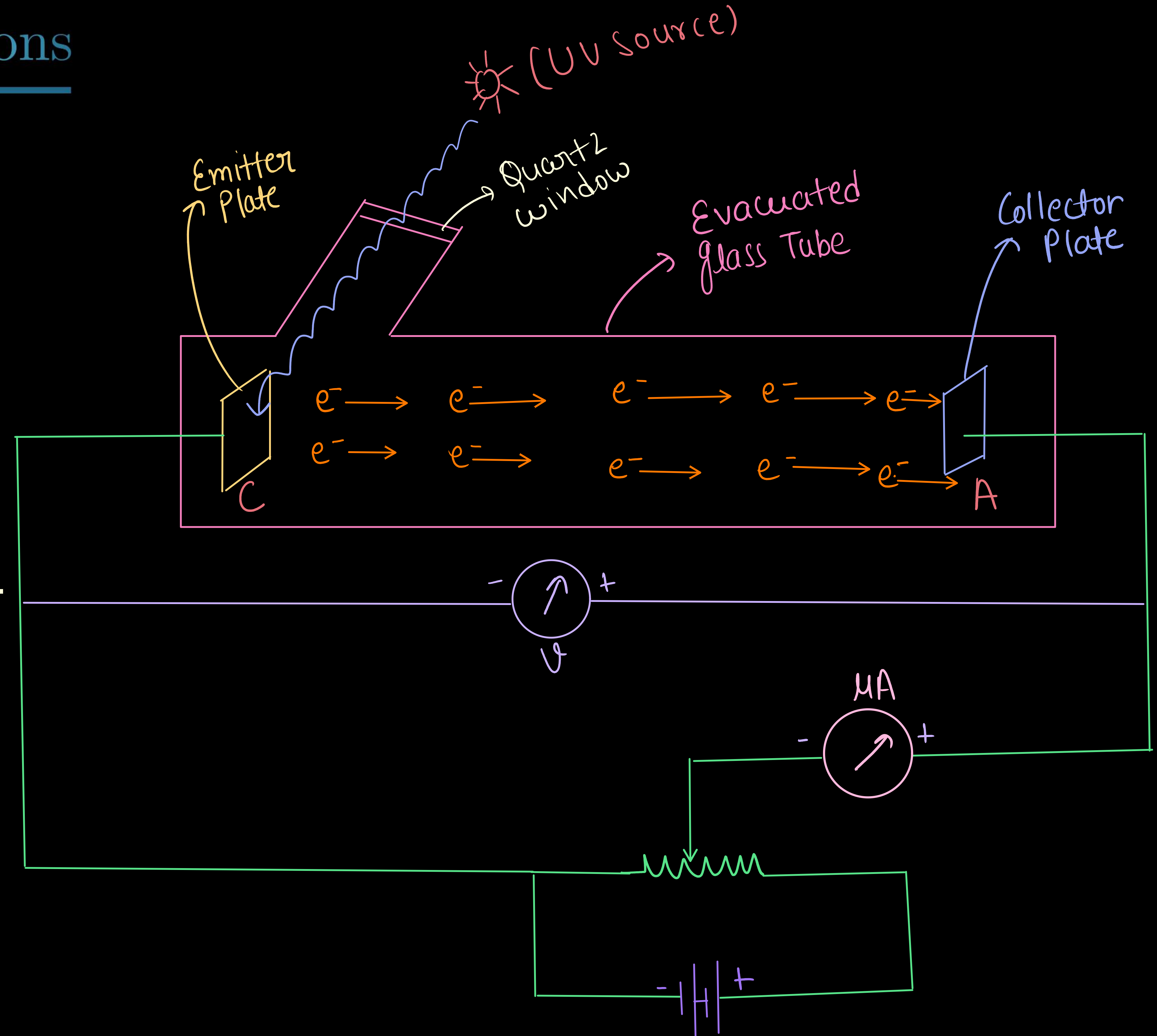
- * Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)

- * When UV light fell on plate C:

- * Electrons were emitted and attracted to plate A.

- * A current flowed in the external circuit (photoelectric current).



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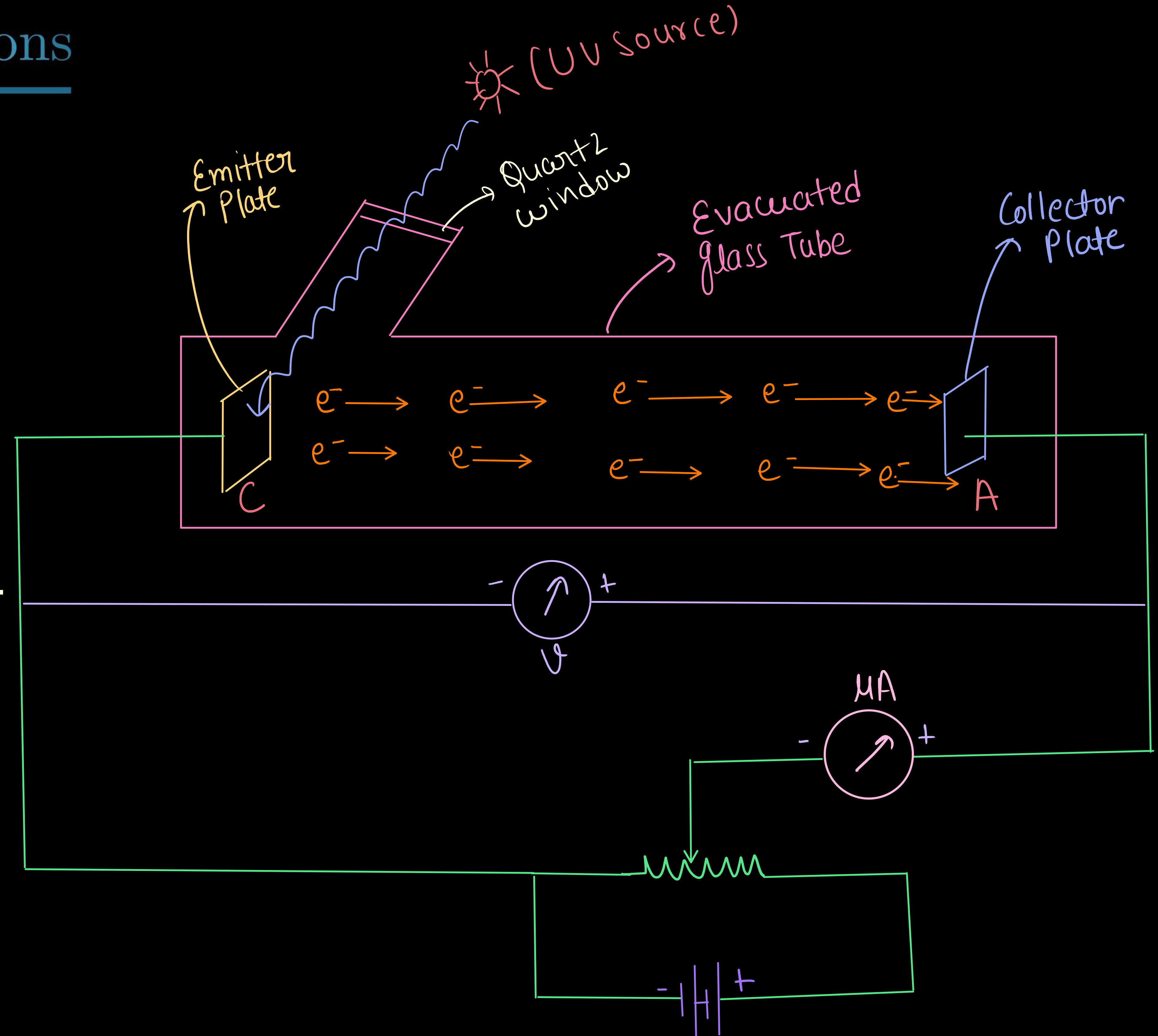
Collector plate (A)

- * When UV light fell on plate C:

- * Electrons were emitted and attracted to plate A.

- * A current flowed in the external circuit (photoelectric current).

- * Current stopped immediately when UV light was switched off.



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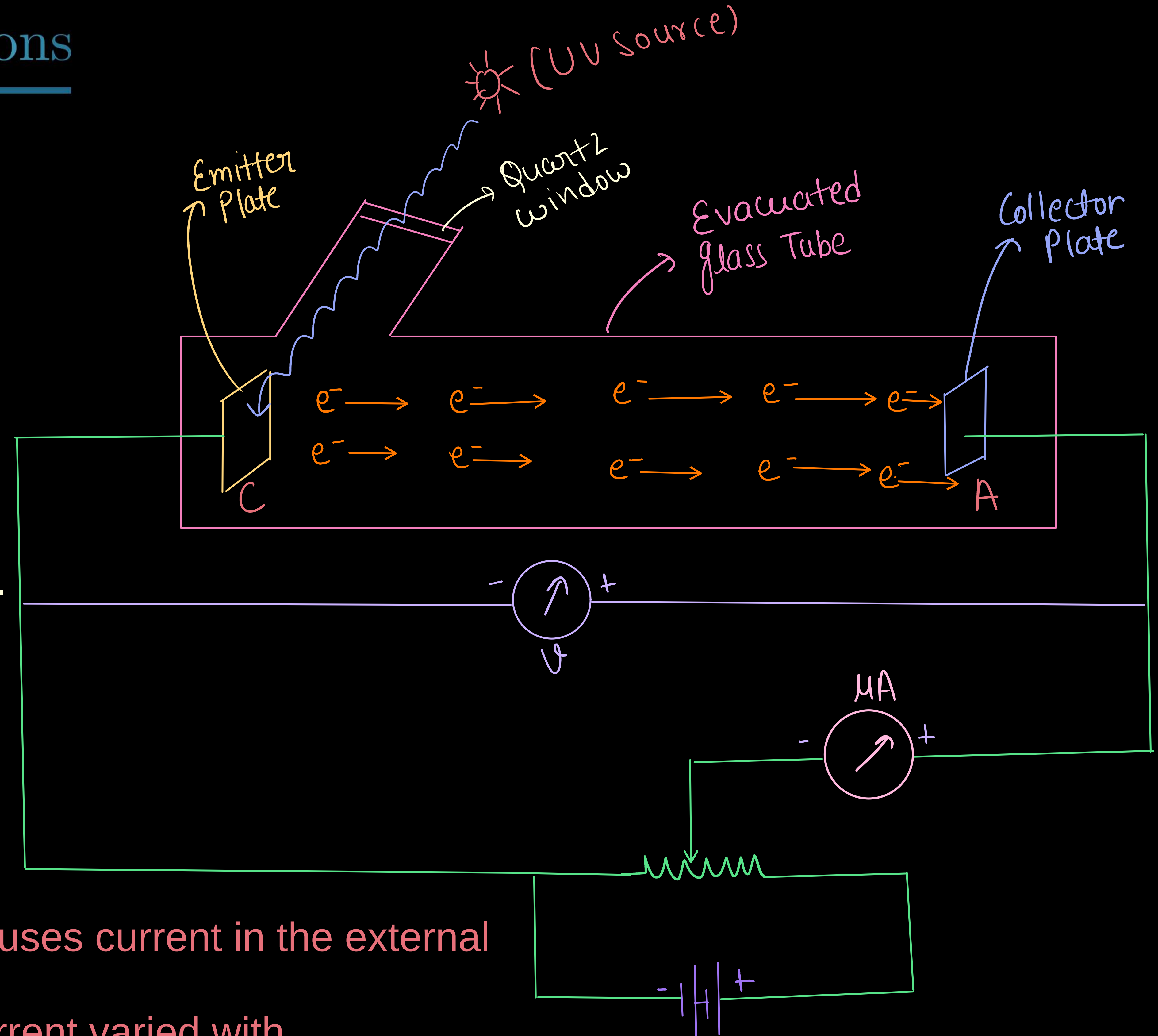
- * Electrons were emitted and attracted to plate A.

- * A current flowed in the external circuit (photoelectric current).

- * Current stopped immediately when UV light was switched off.

- * Thus, light falling on the surface of the emitter causes current in the external circuit.

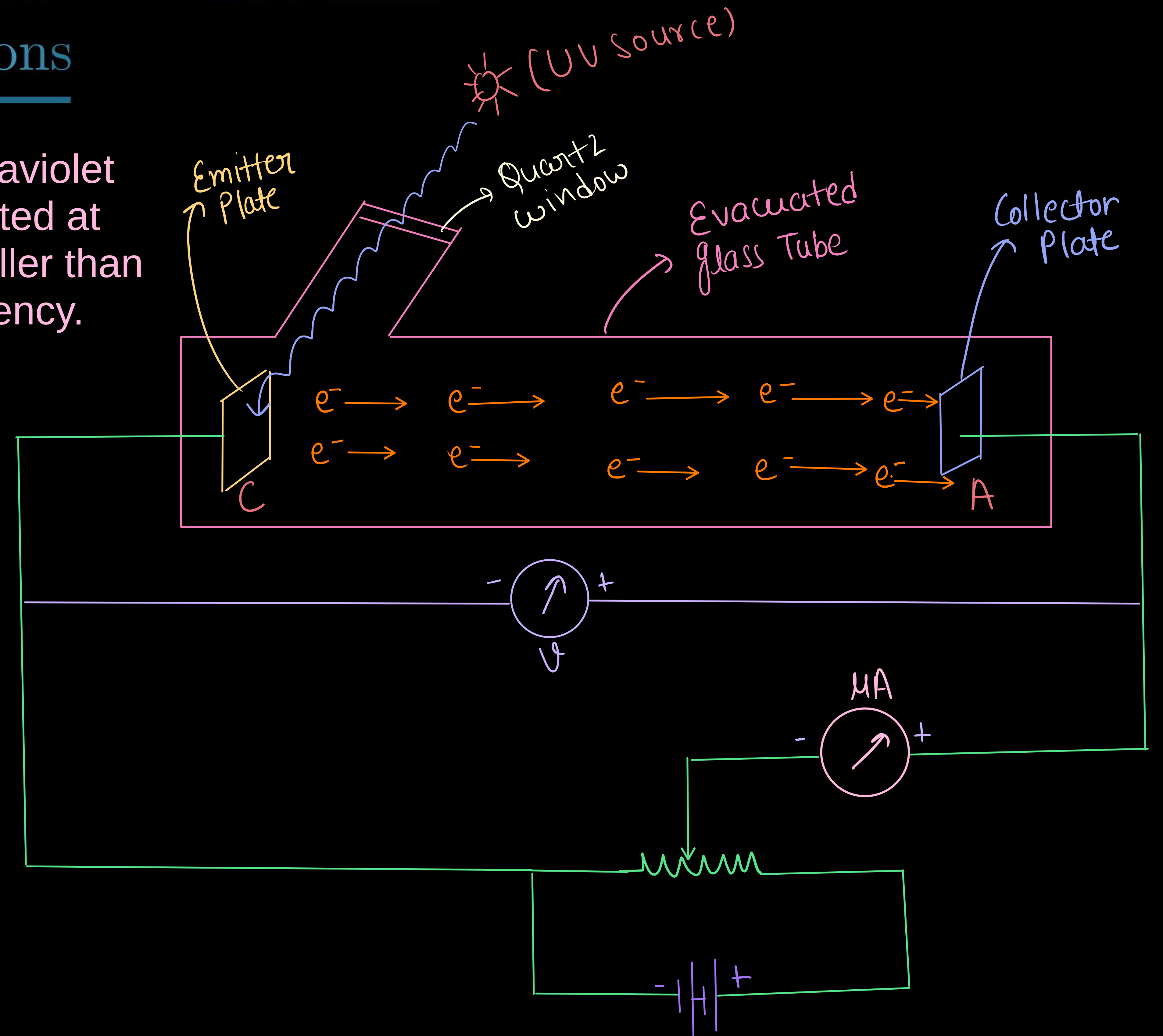
- * Hallwachs and Lenard studied how this photo current varied with collector plate potential, and with frequency and intensity of incident light.



PHOTOELECTRIC EFFECT

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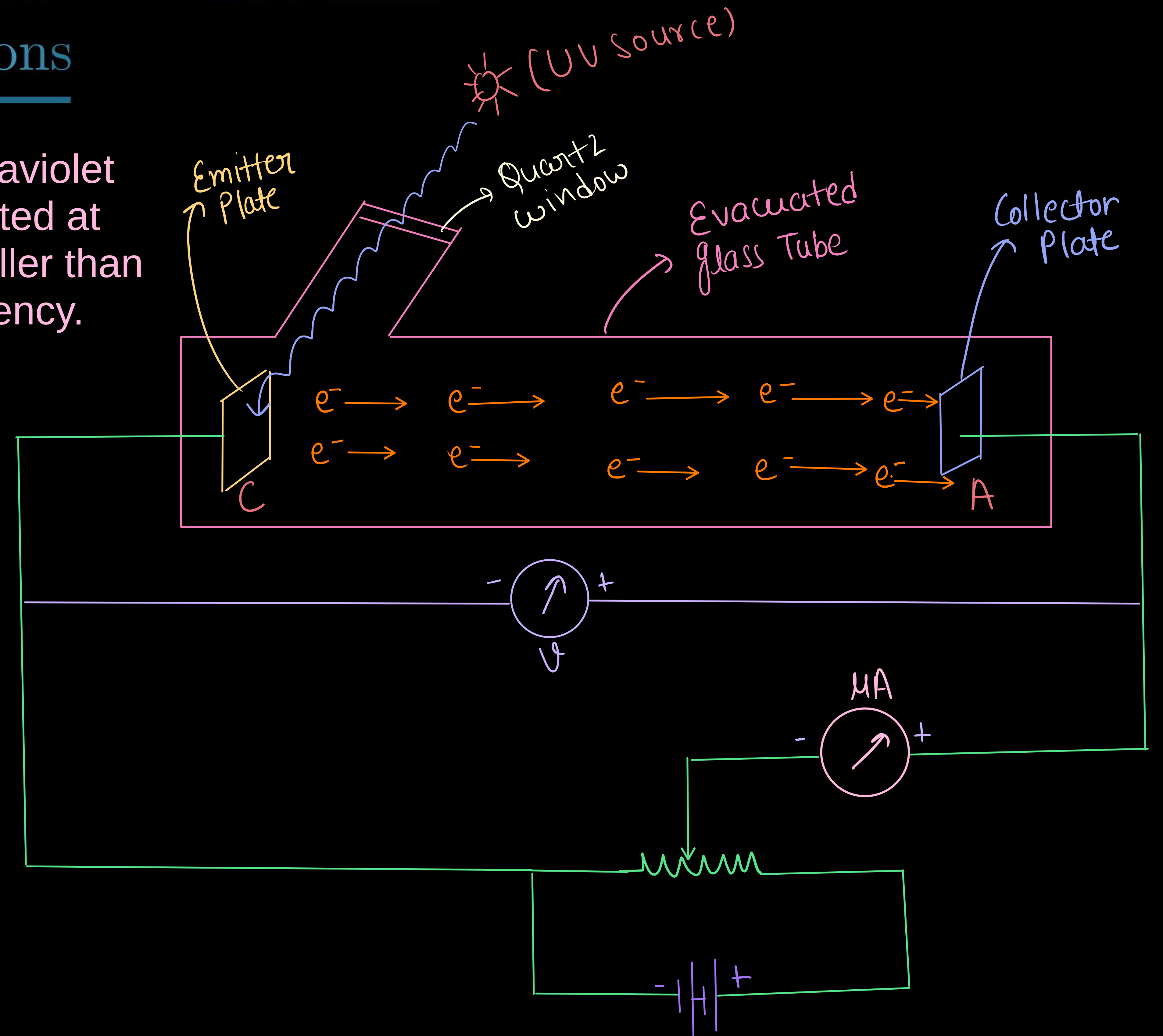
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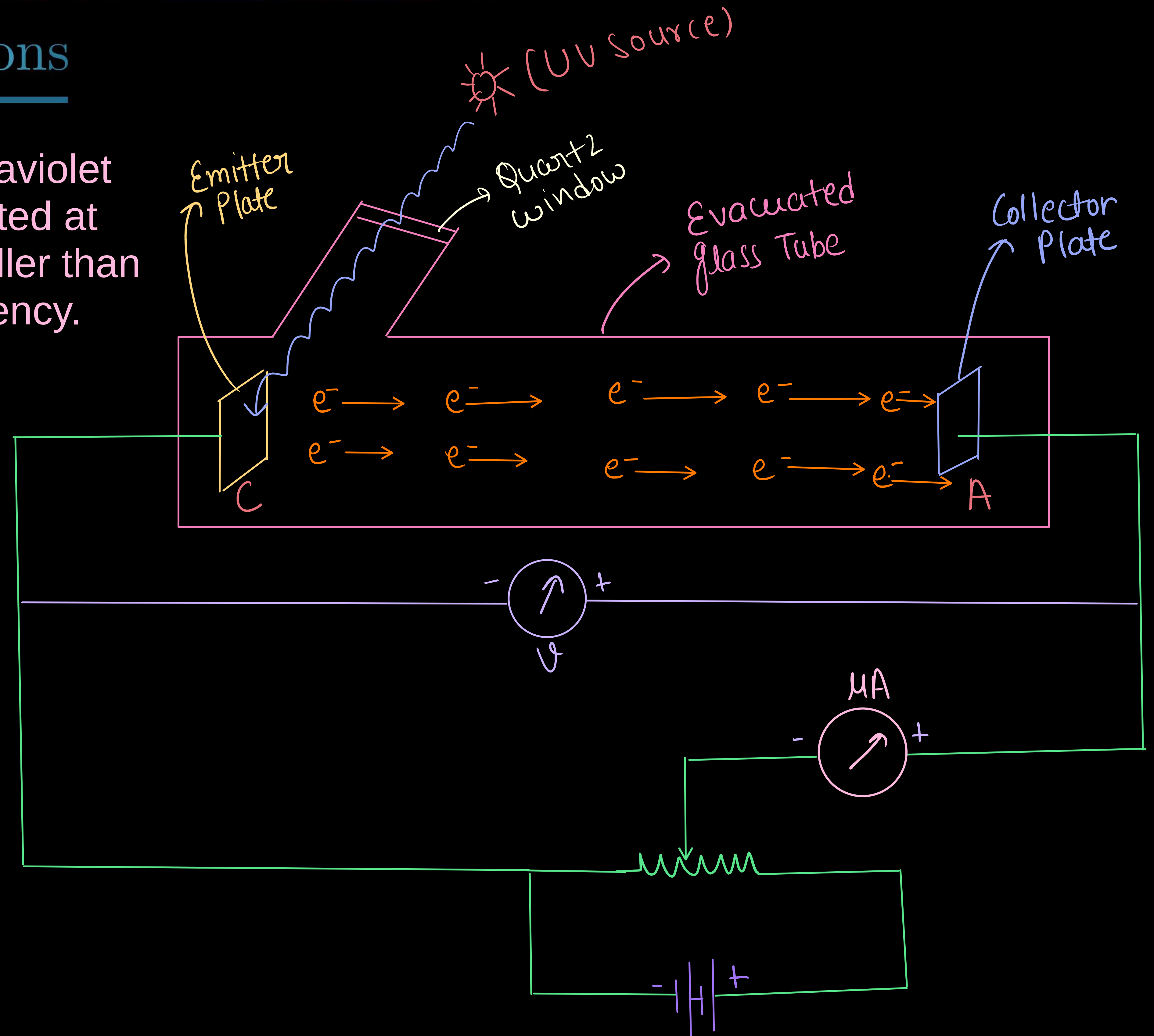
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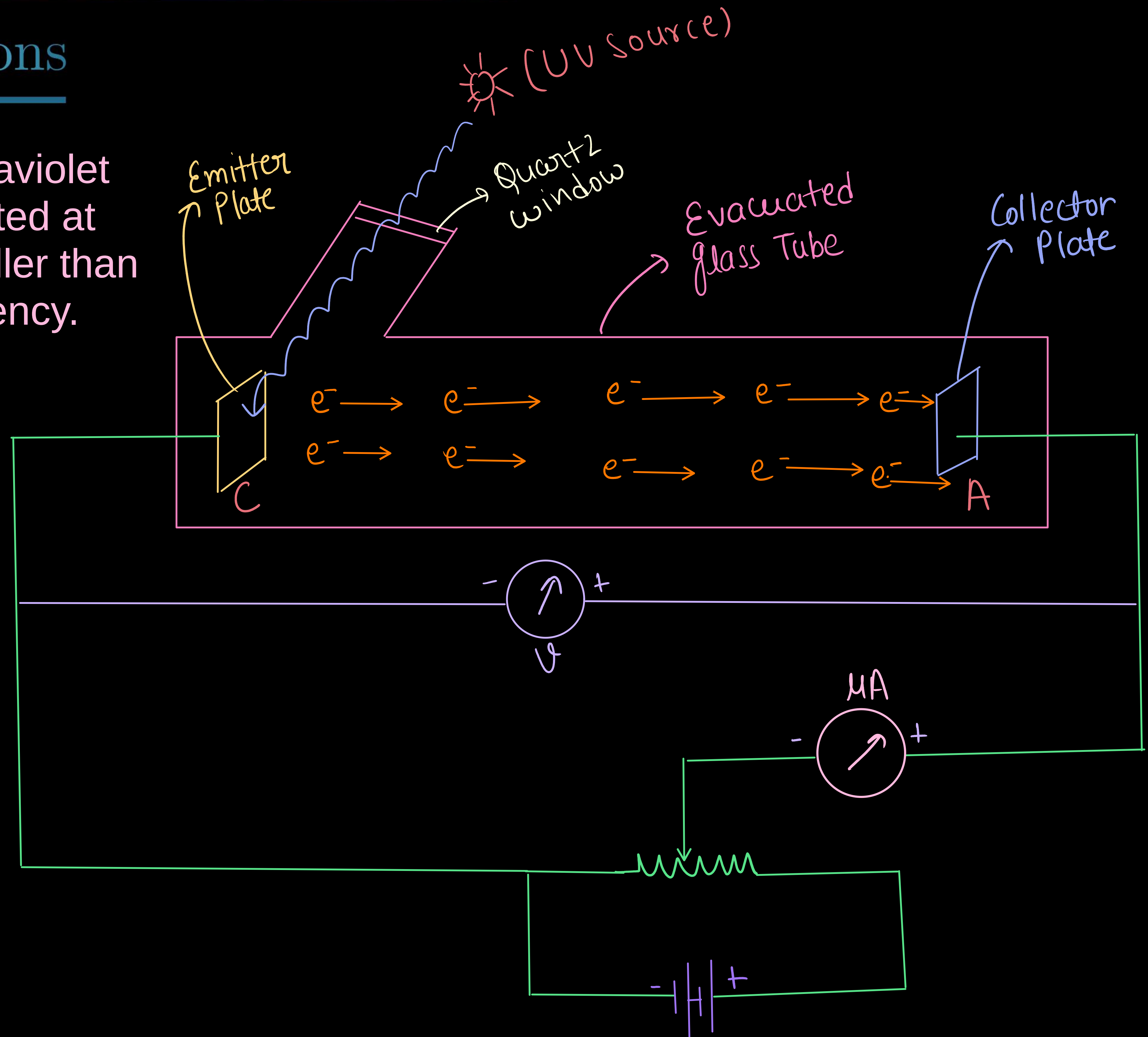
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- * It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.



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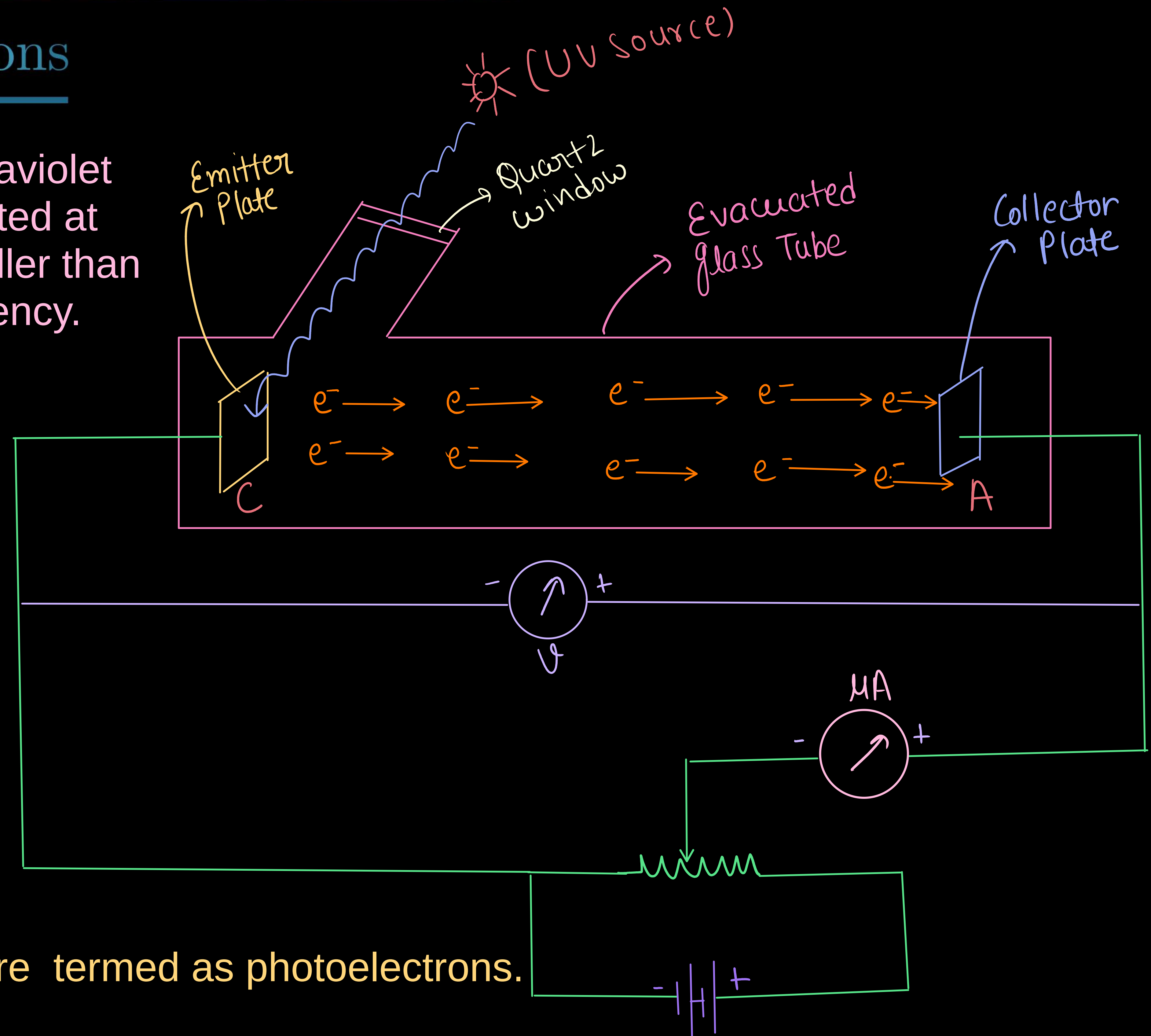
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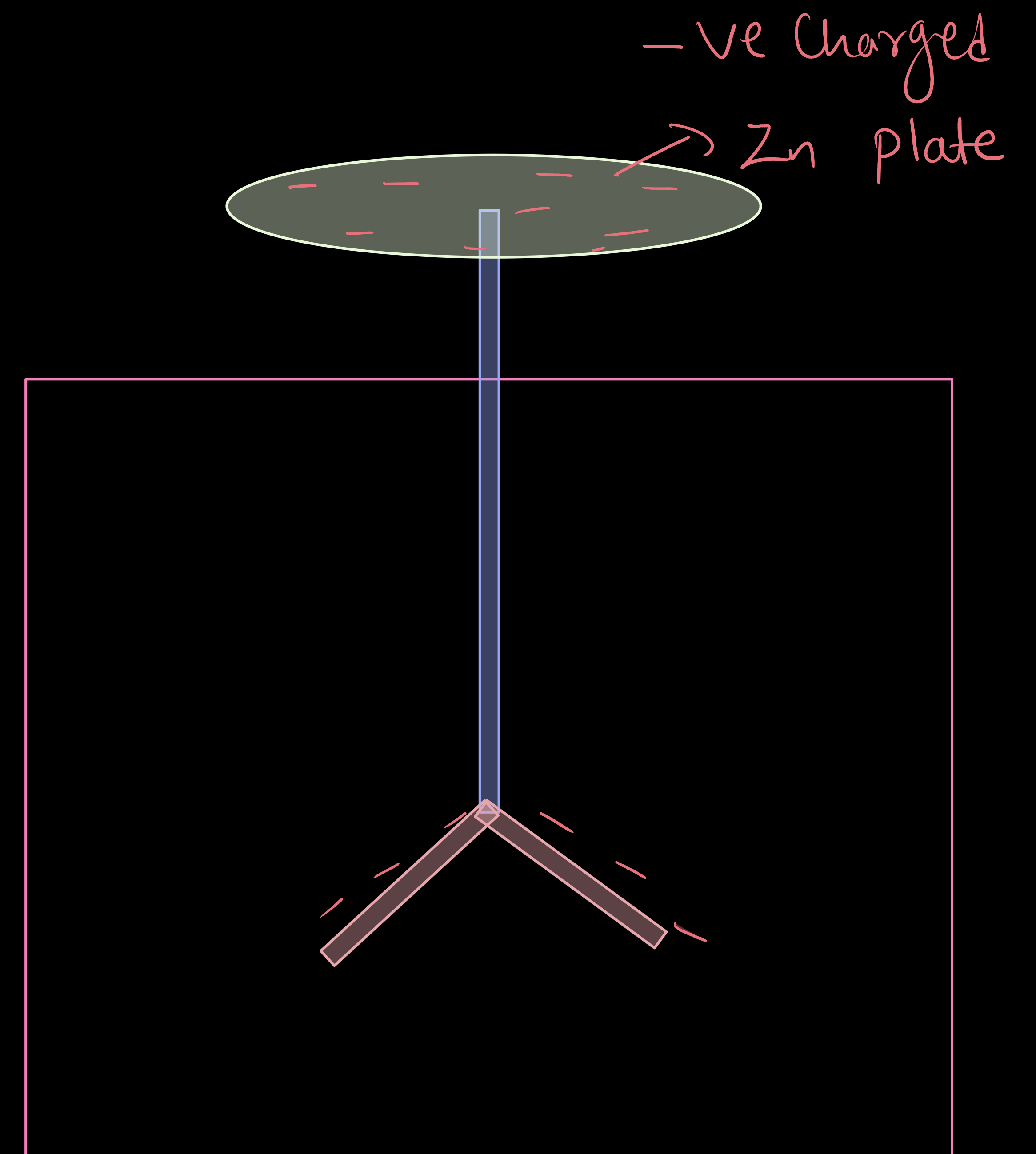
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- * It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.
- * However, some alkali metals such as lithium, sodium, potassium, caesium and rubidium were sensitive even to visible light.
- * After the discovery of electrons, these electrons were termed as photoelectrons. The phenomenon is called photoelectric effect.



PHOTOELECTRIC EFFECT

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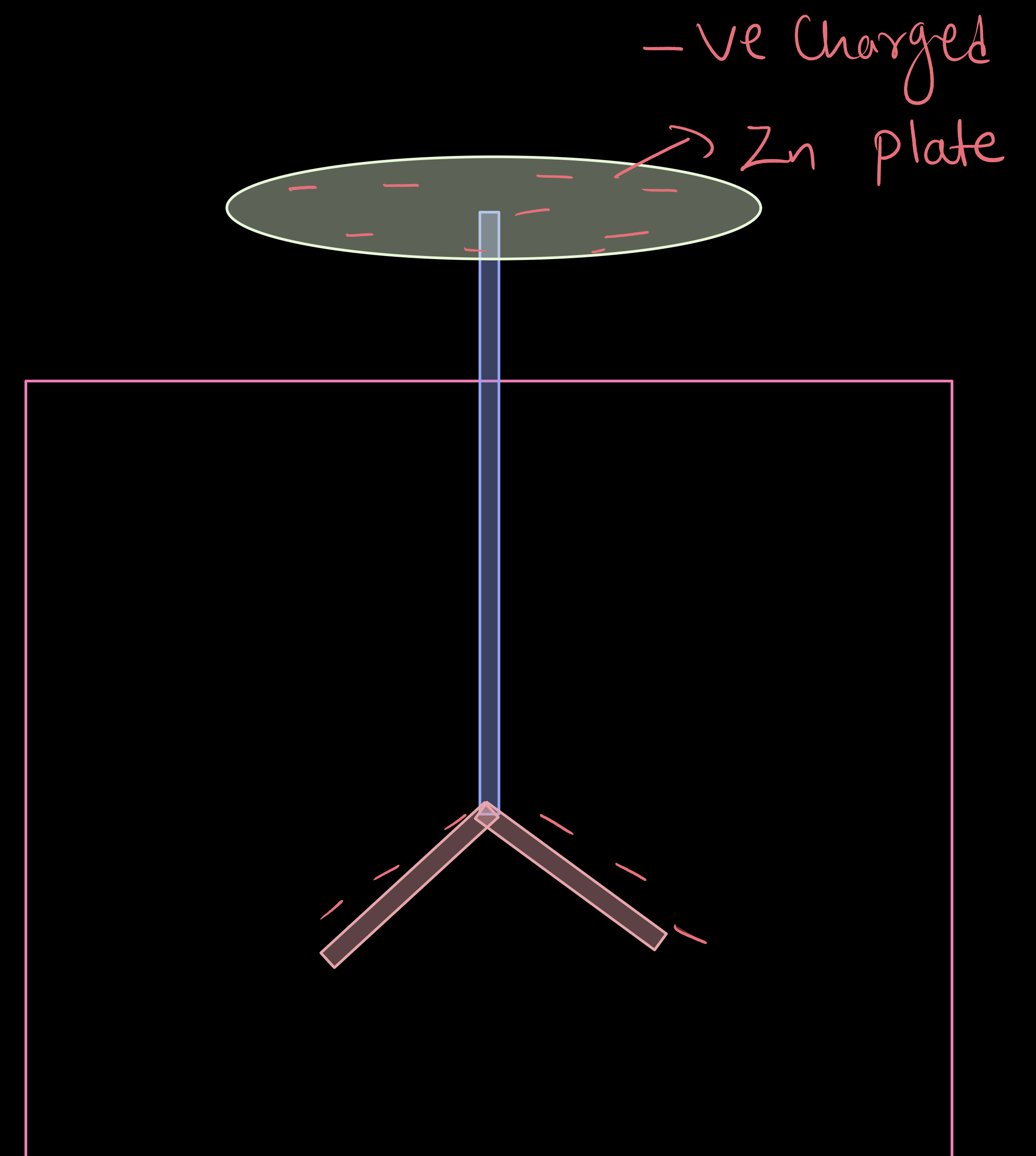
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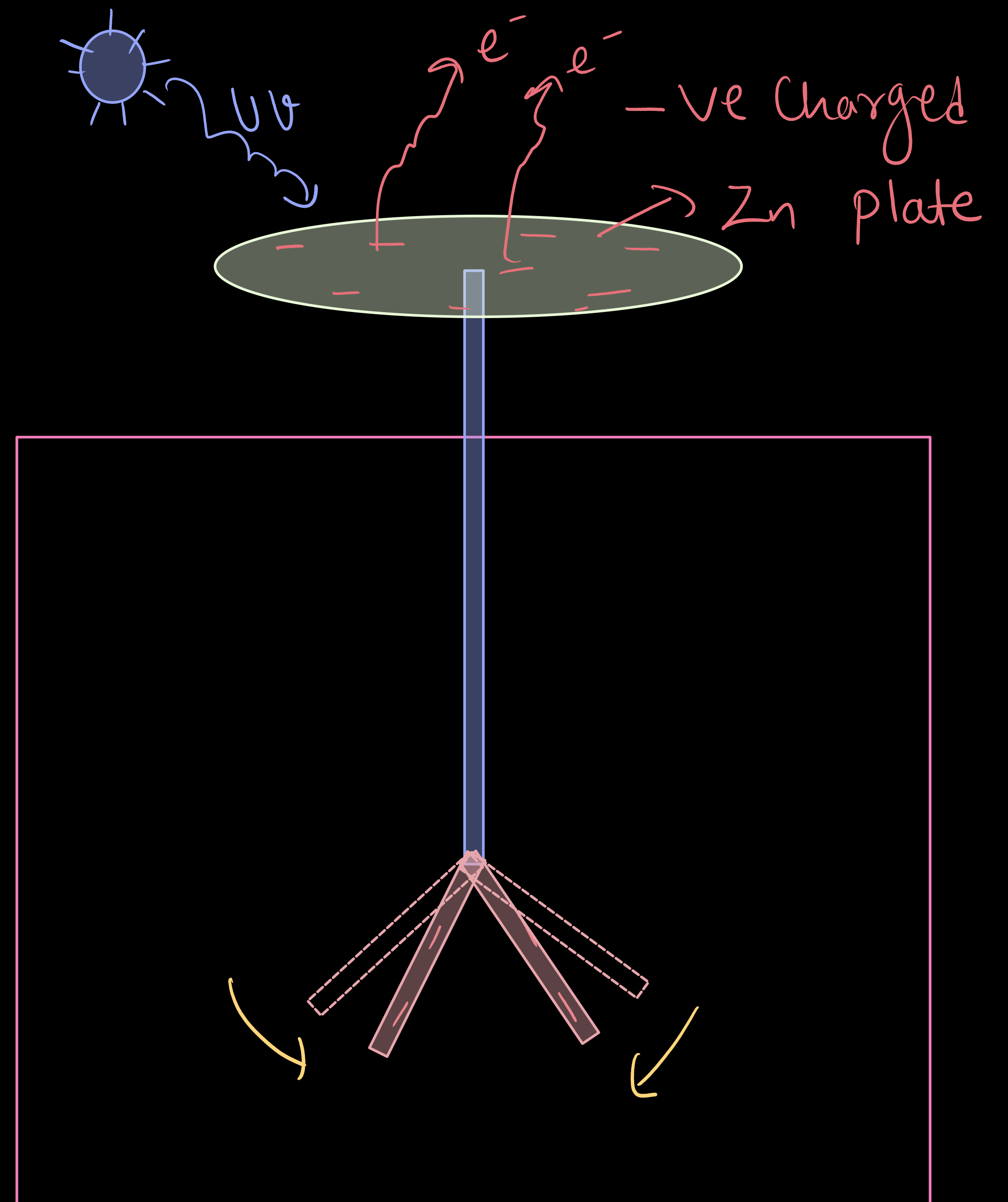
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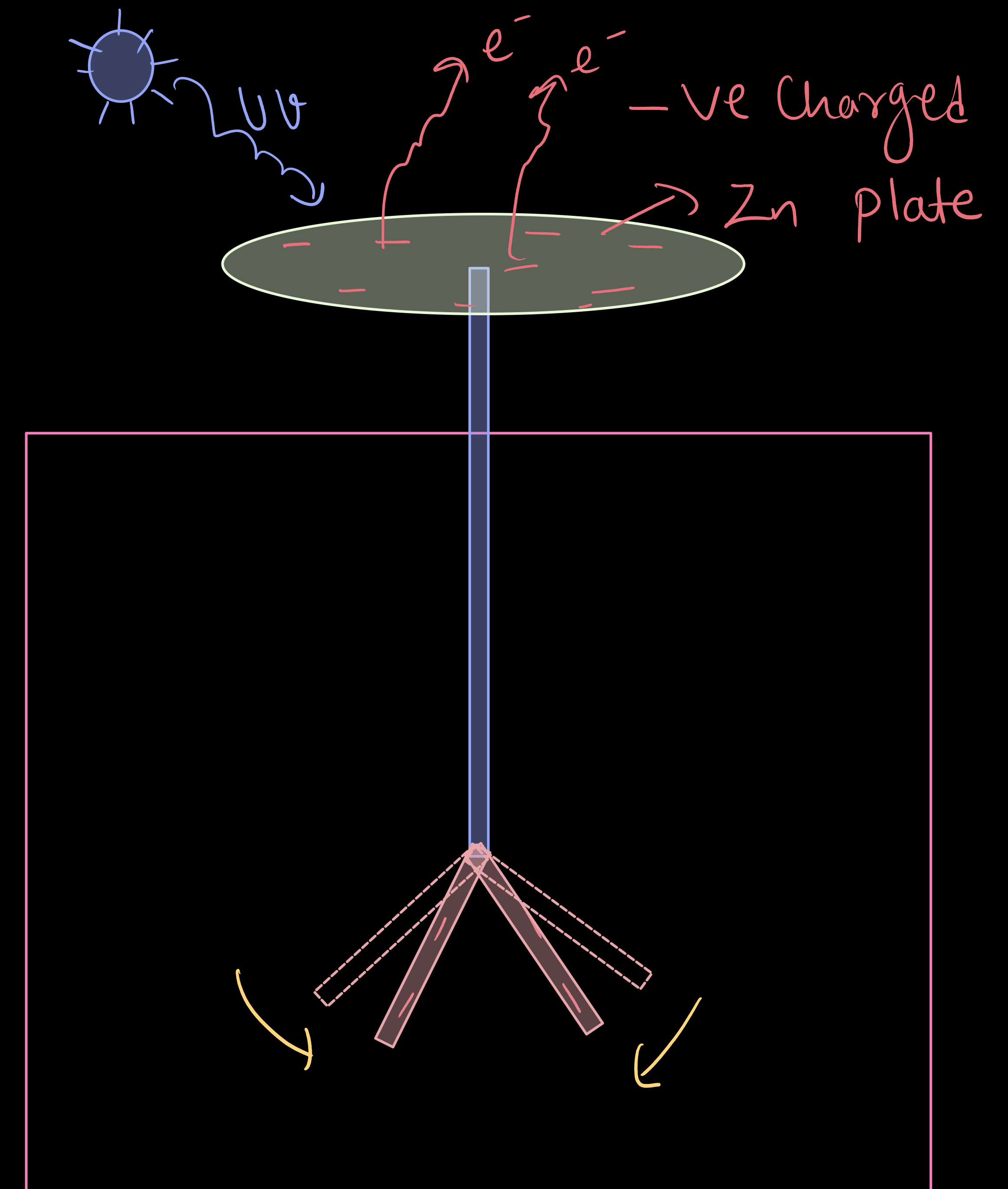
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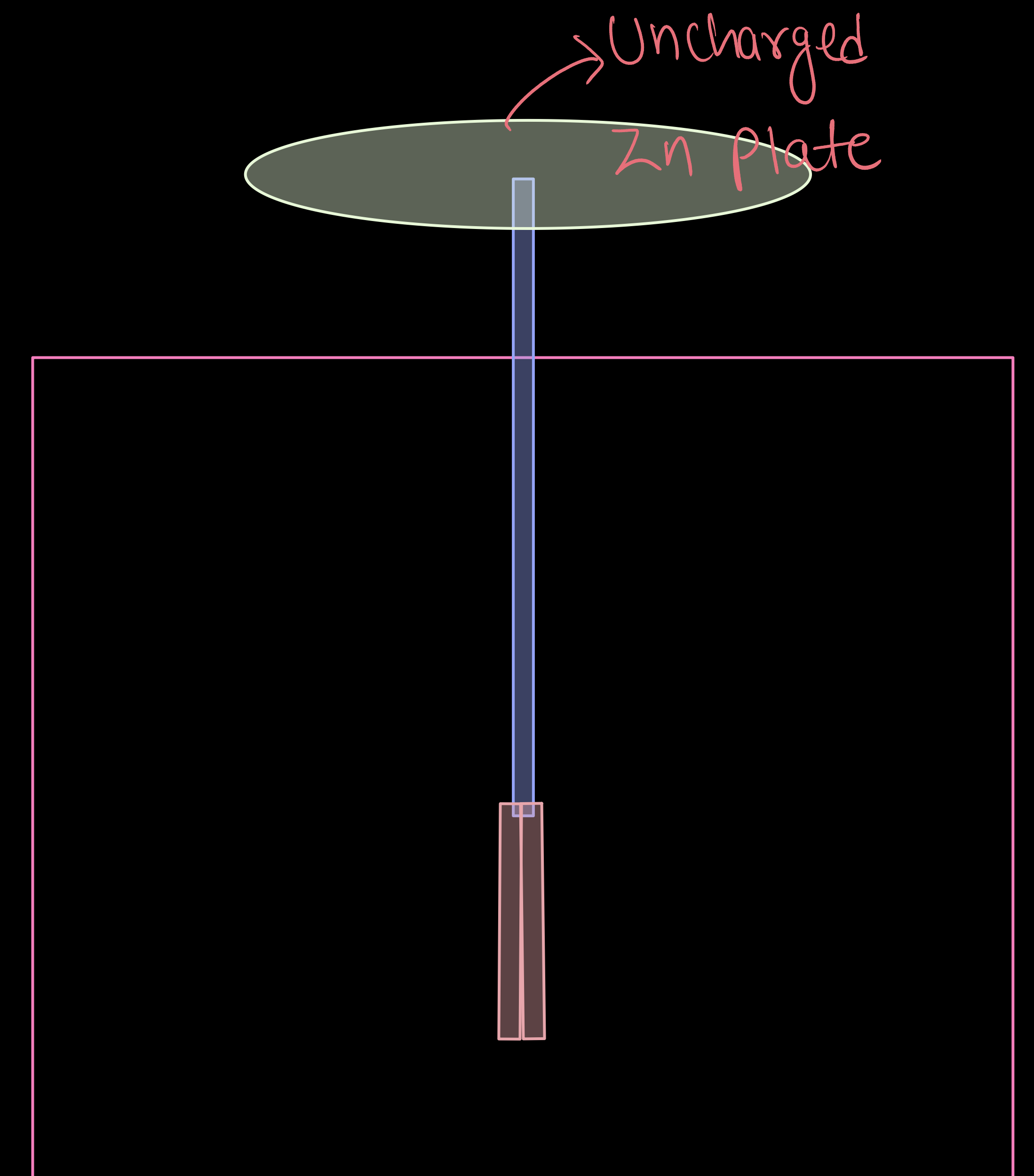


PHOTOELECTRIC EFFECT

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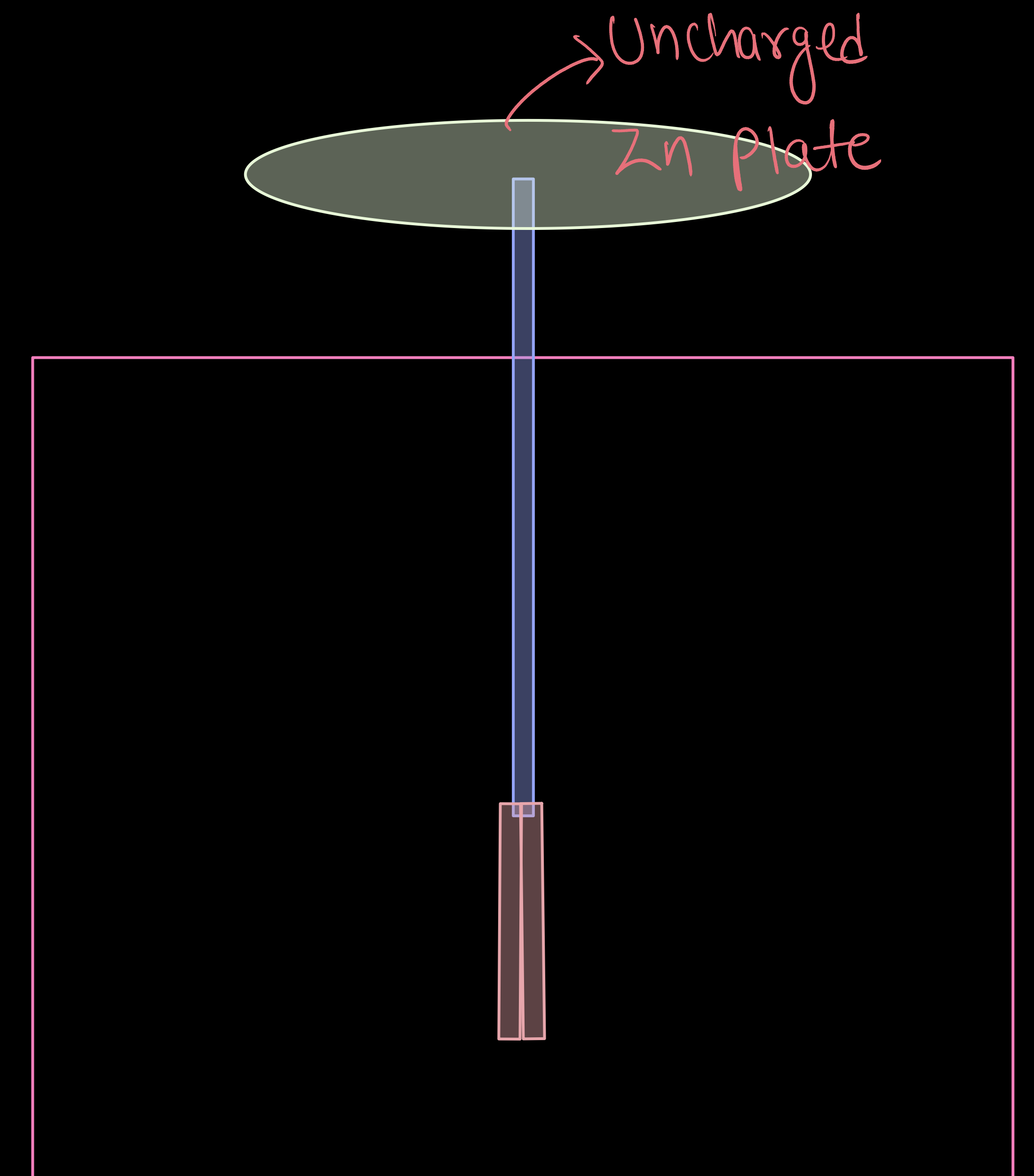
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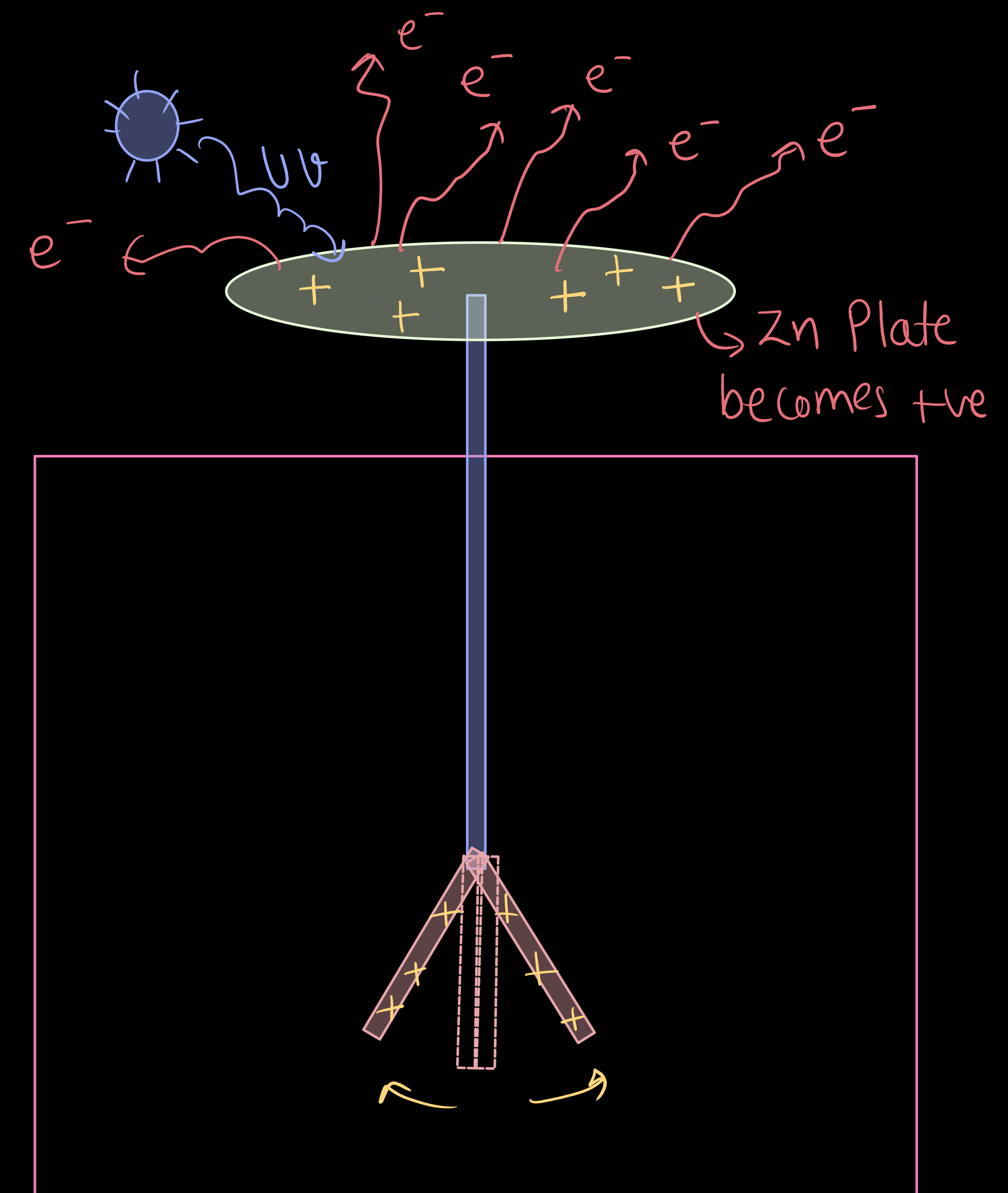


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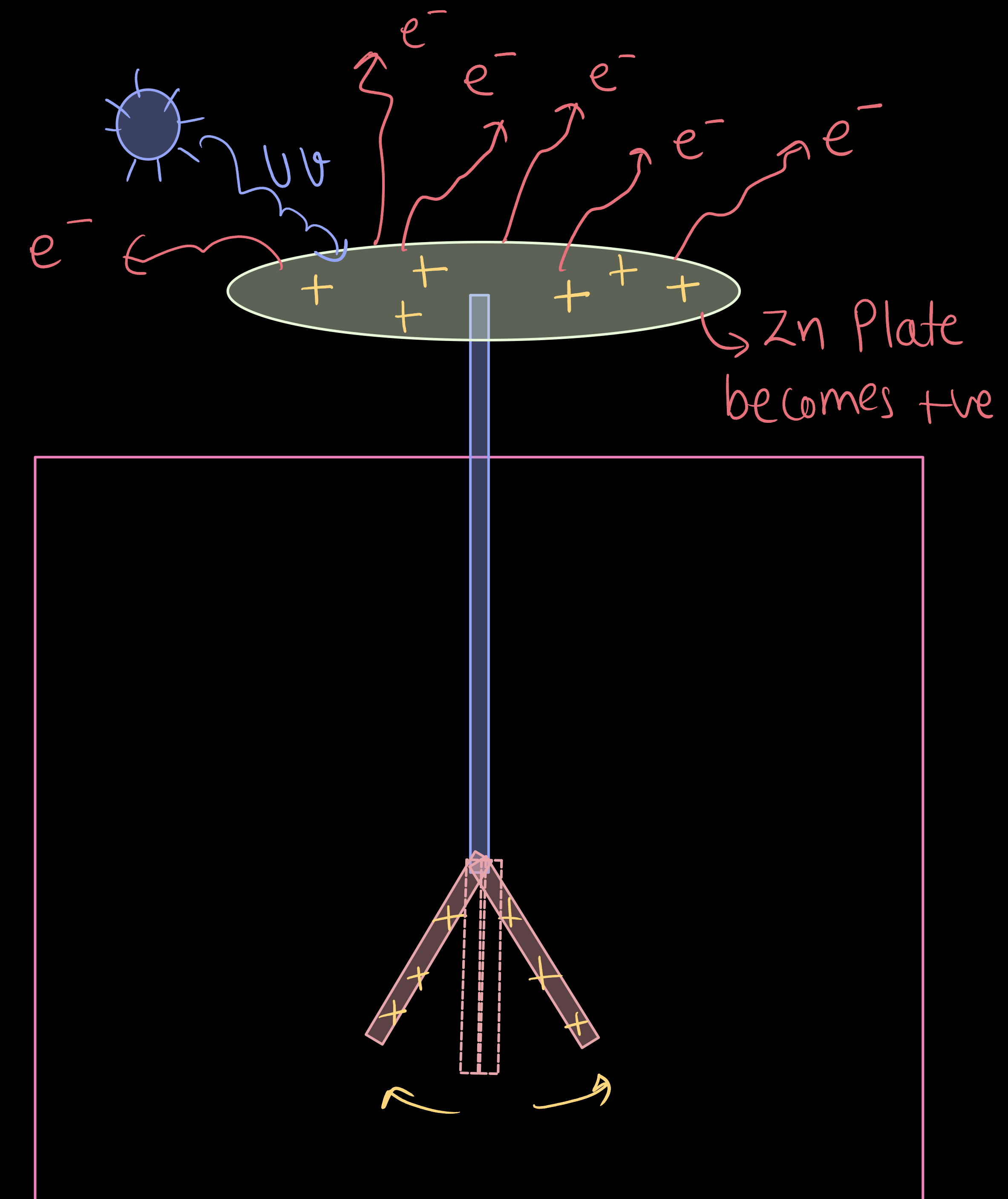
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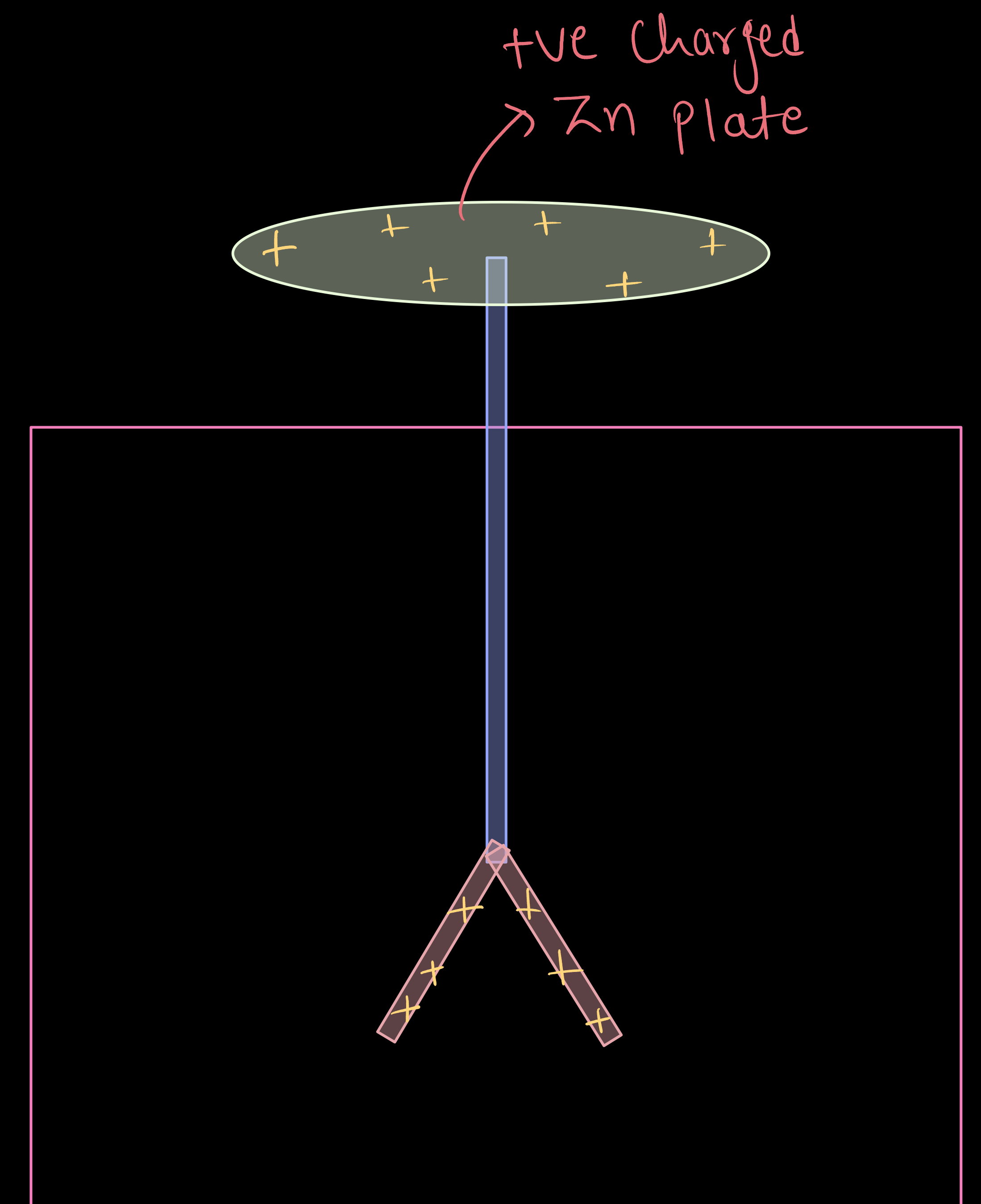
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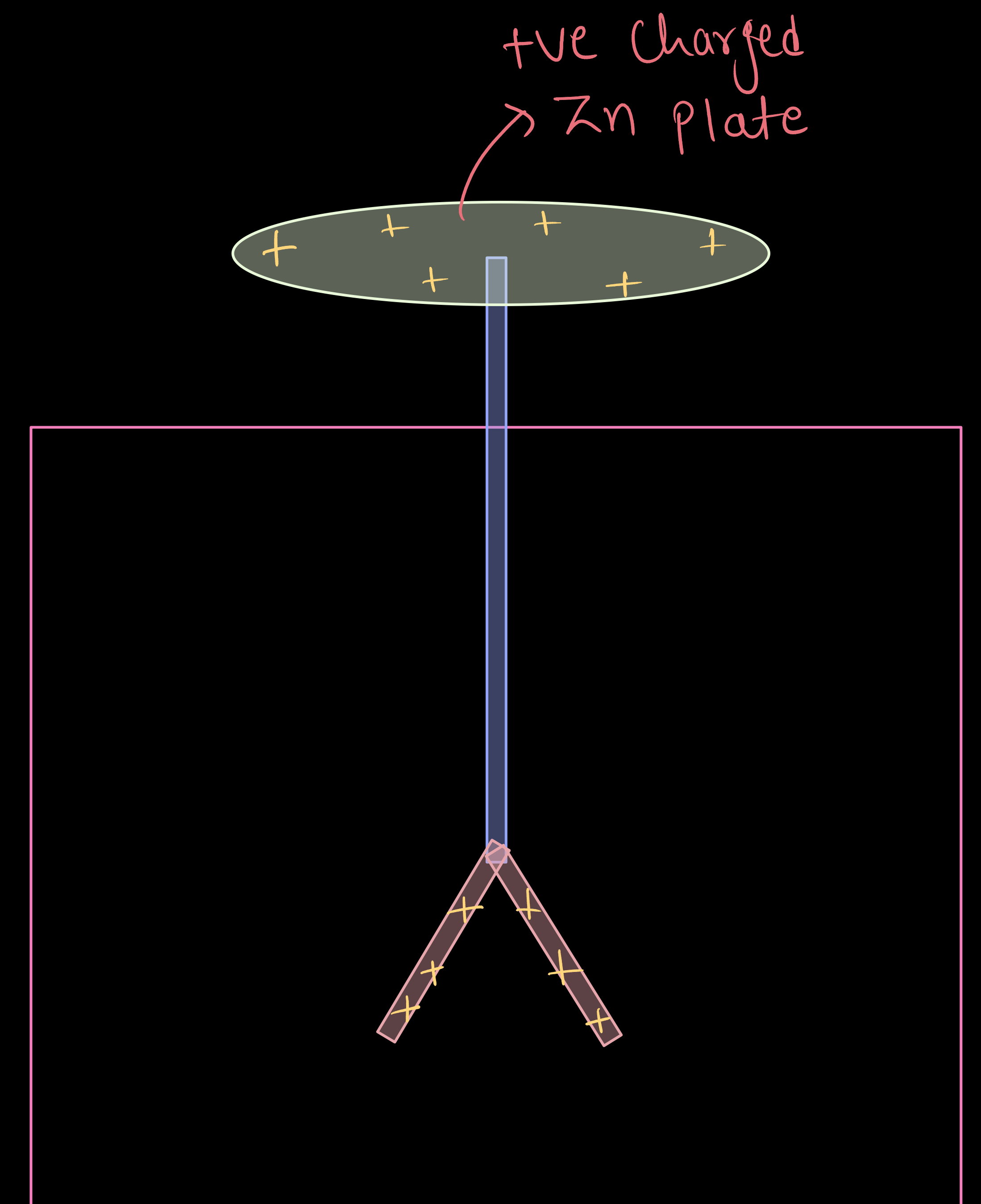
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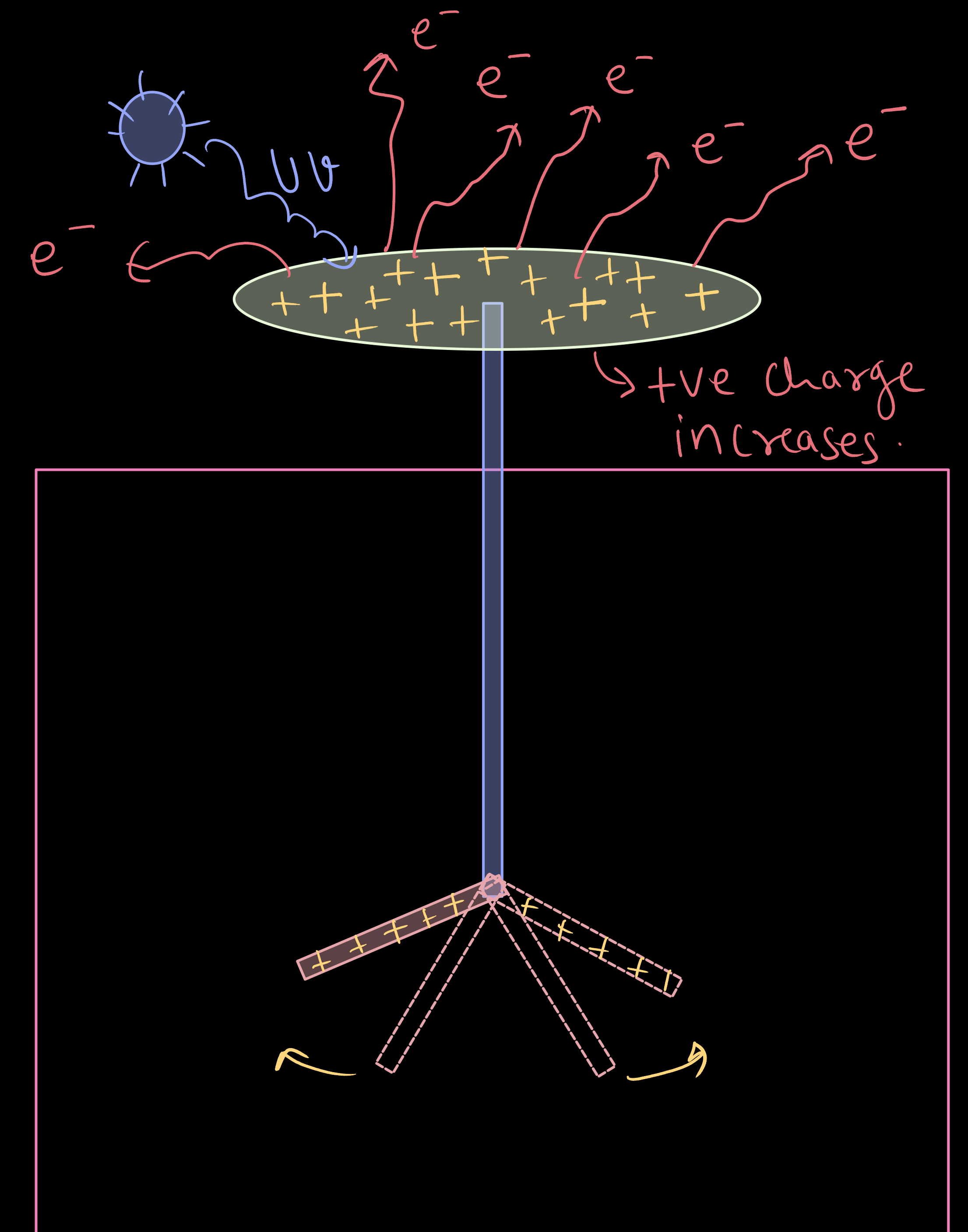
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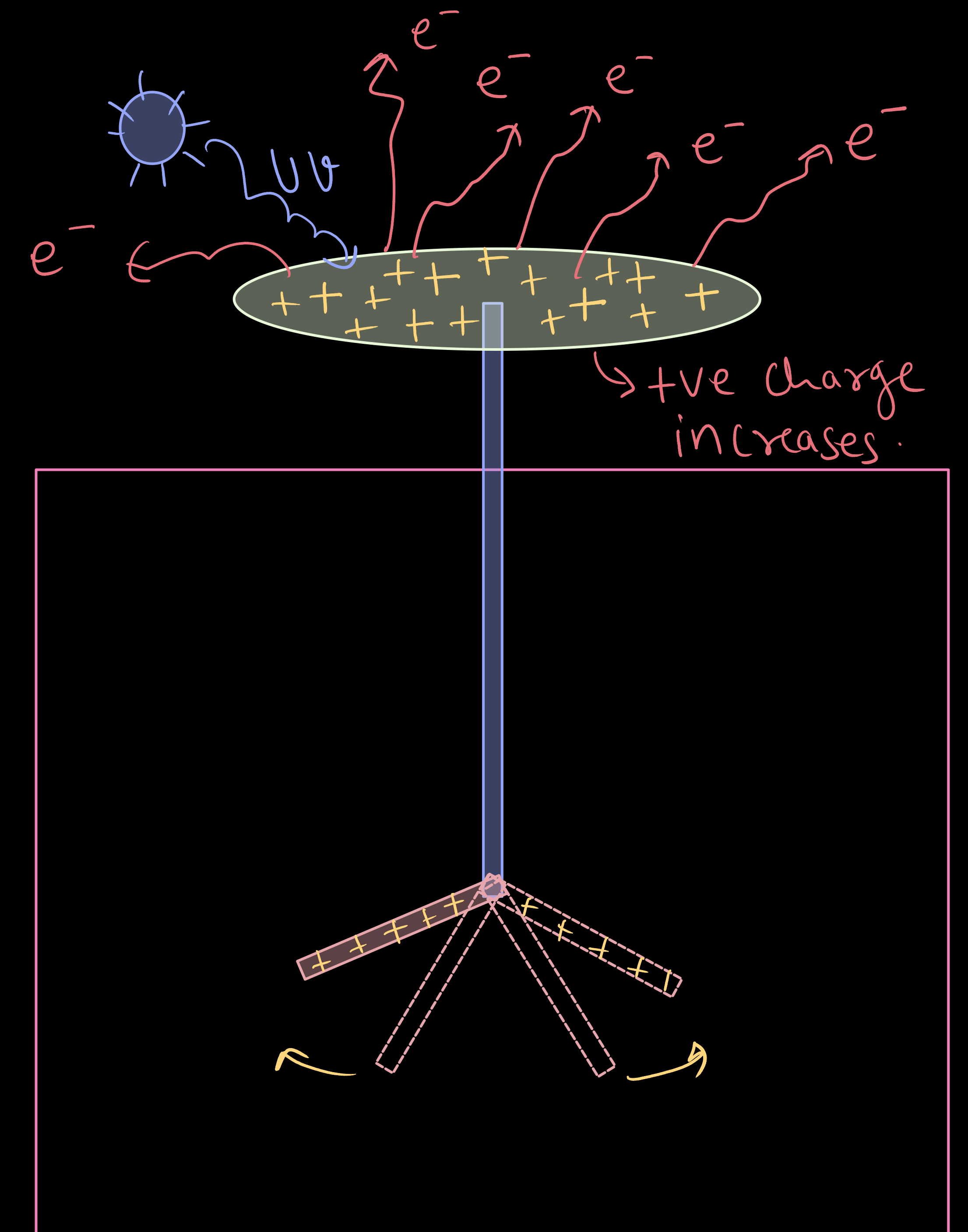
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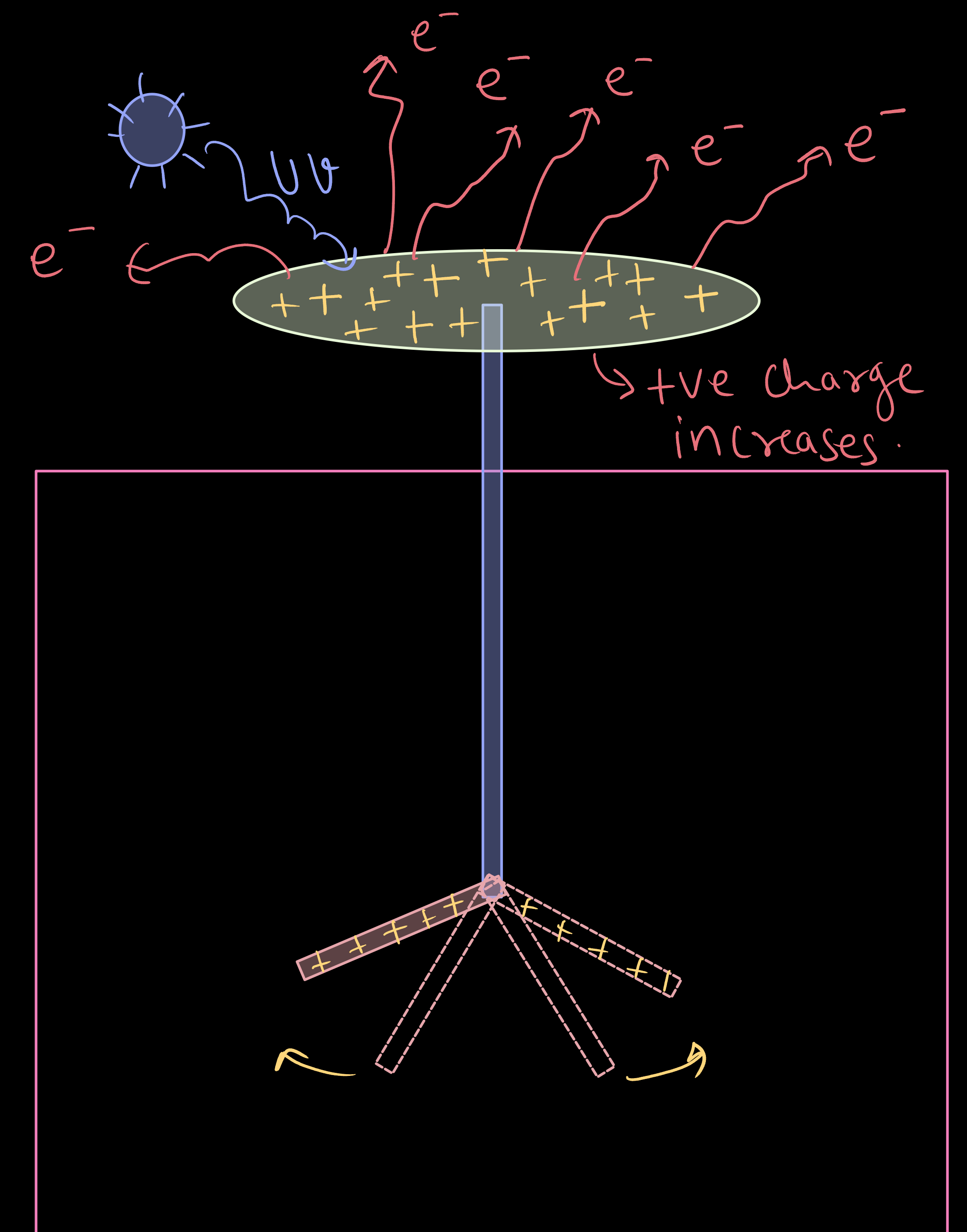
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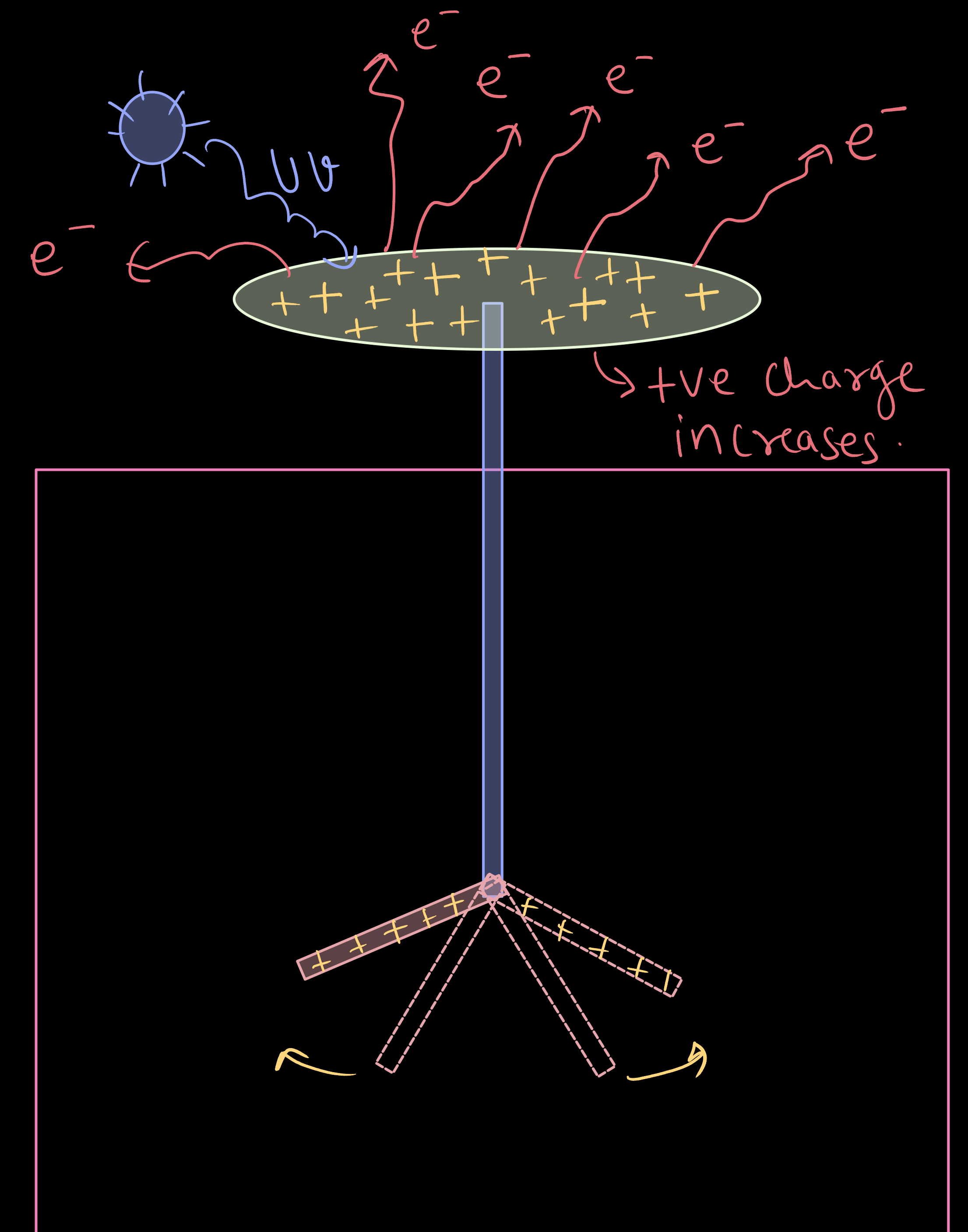
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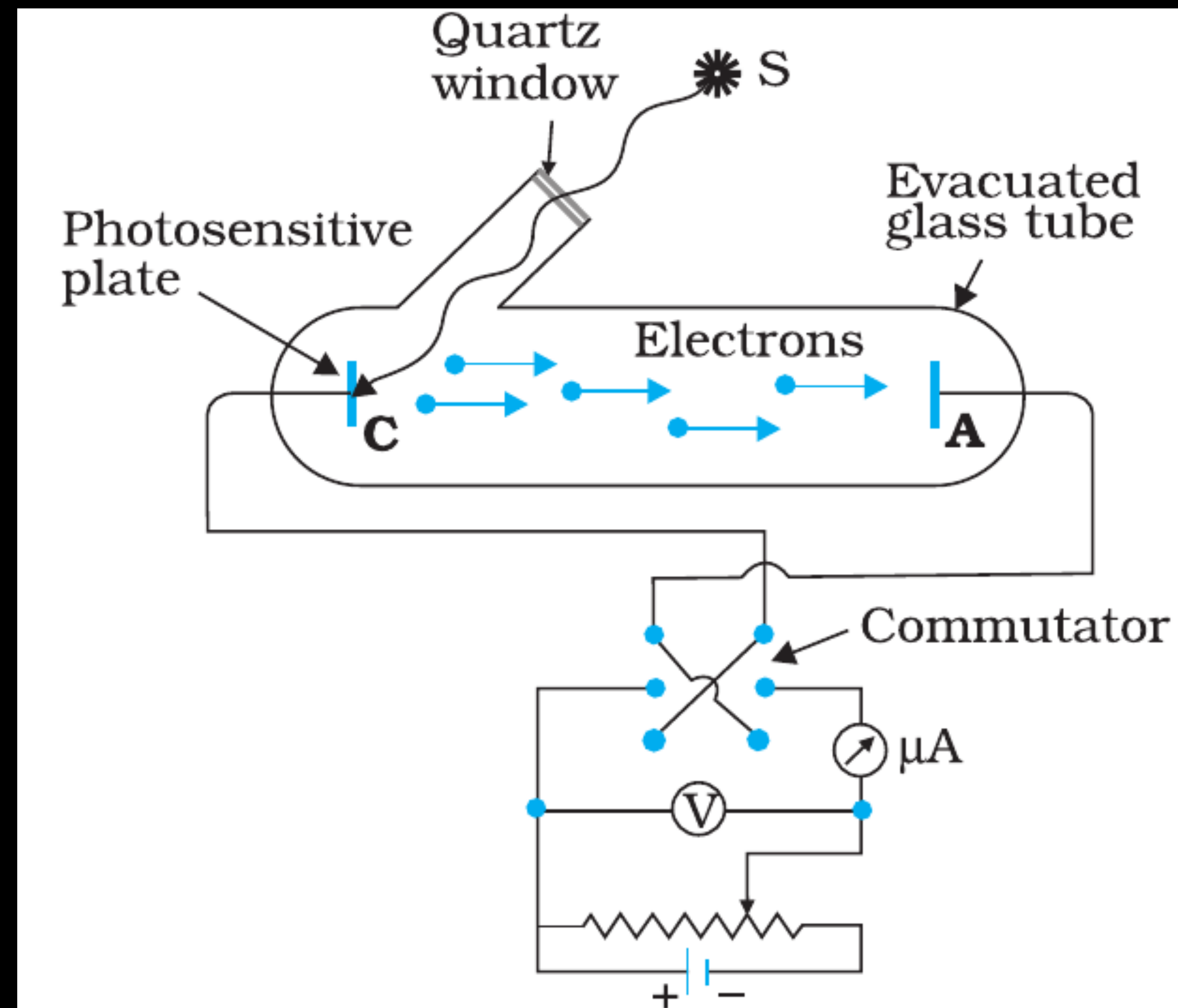


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

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- * The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.
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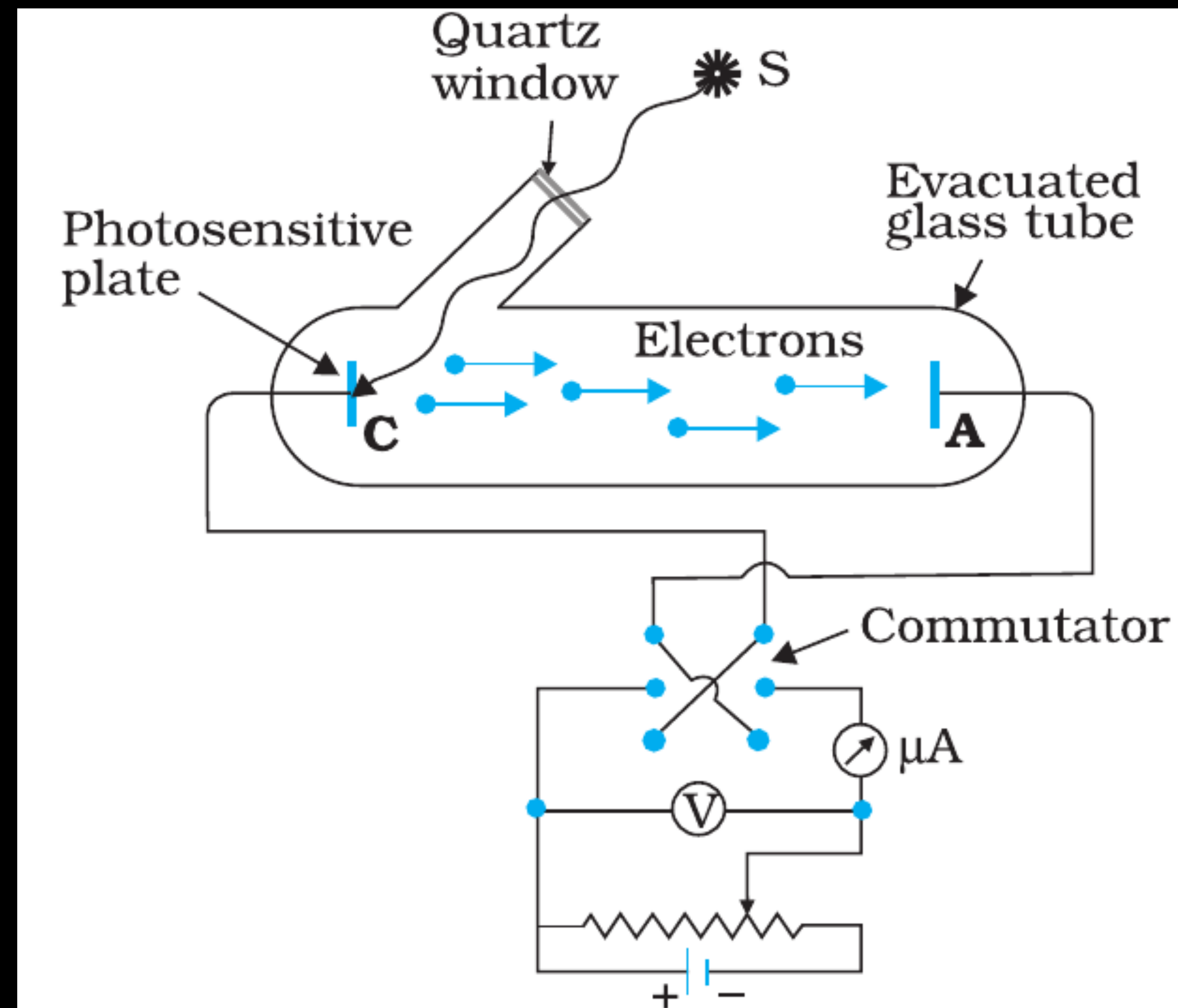


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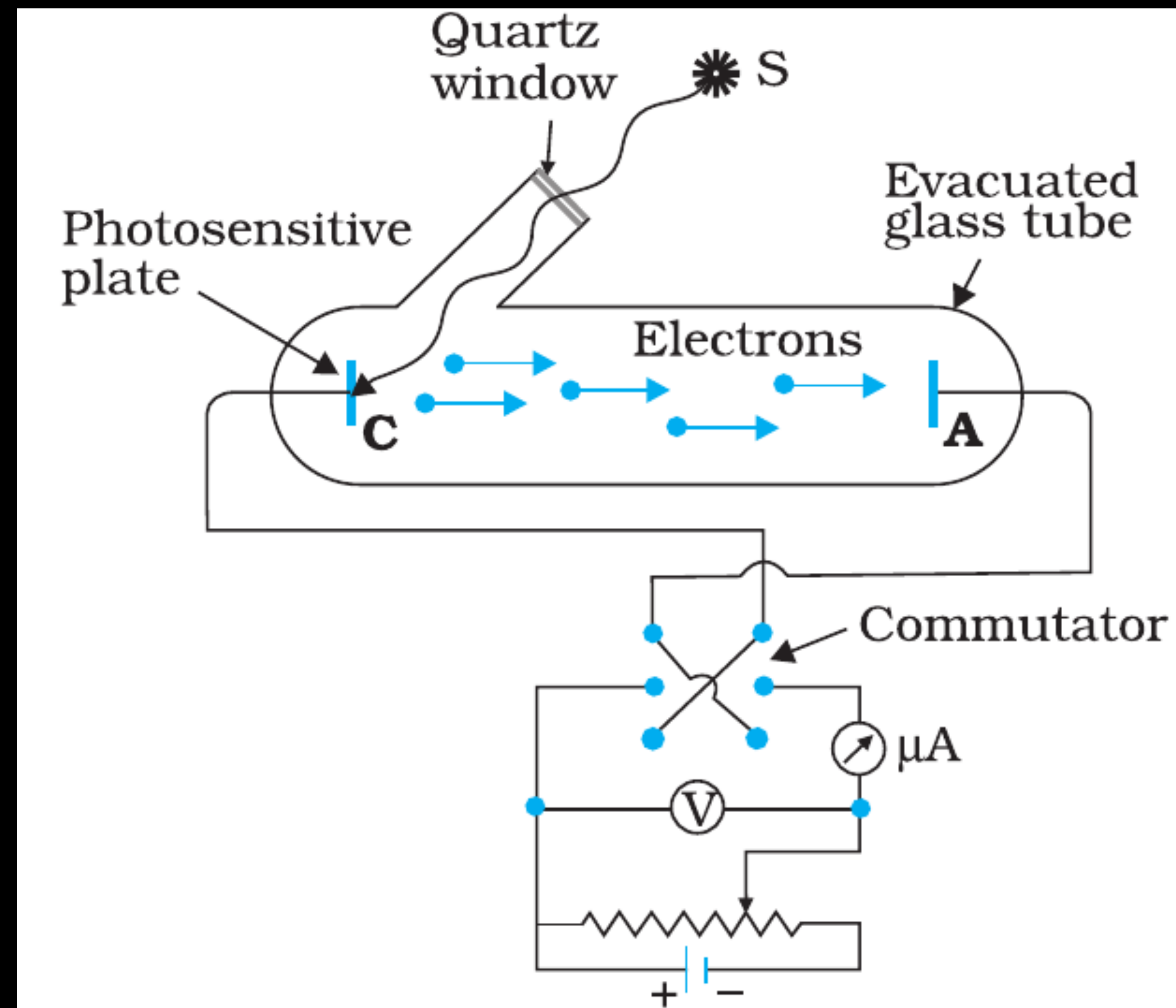


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- * The battery maintains the potential difference between the plates C and A, that can be varied. The polarity of the plates C and A can be reversed by a commutator.

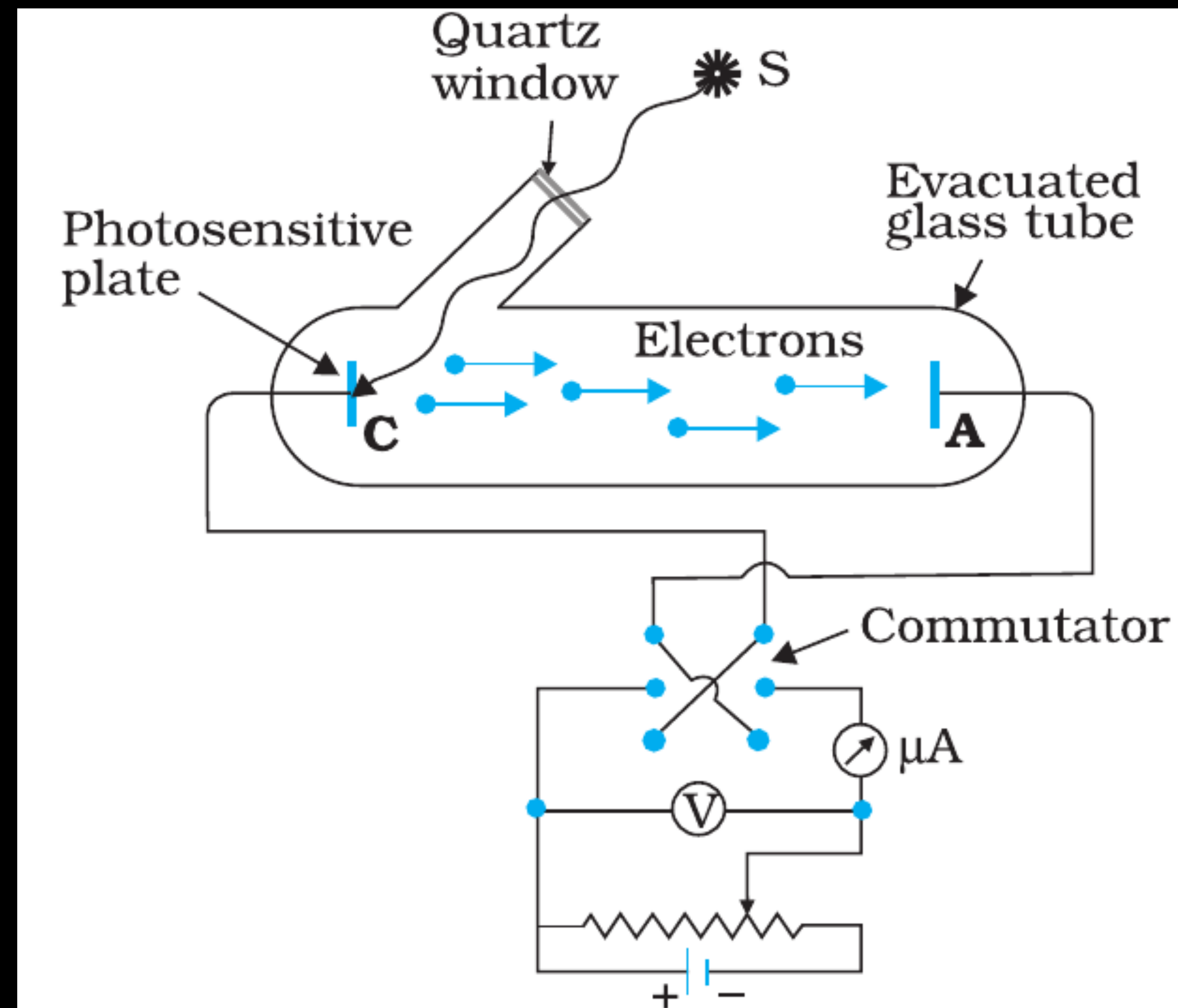


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- * The potential difference between the emitter and collector plates is measured by a voltmeter (V) whereas the resulting photo current flowing in the circuit is measured by a microammeter (μA).

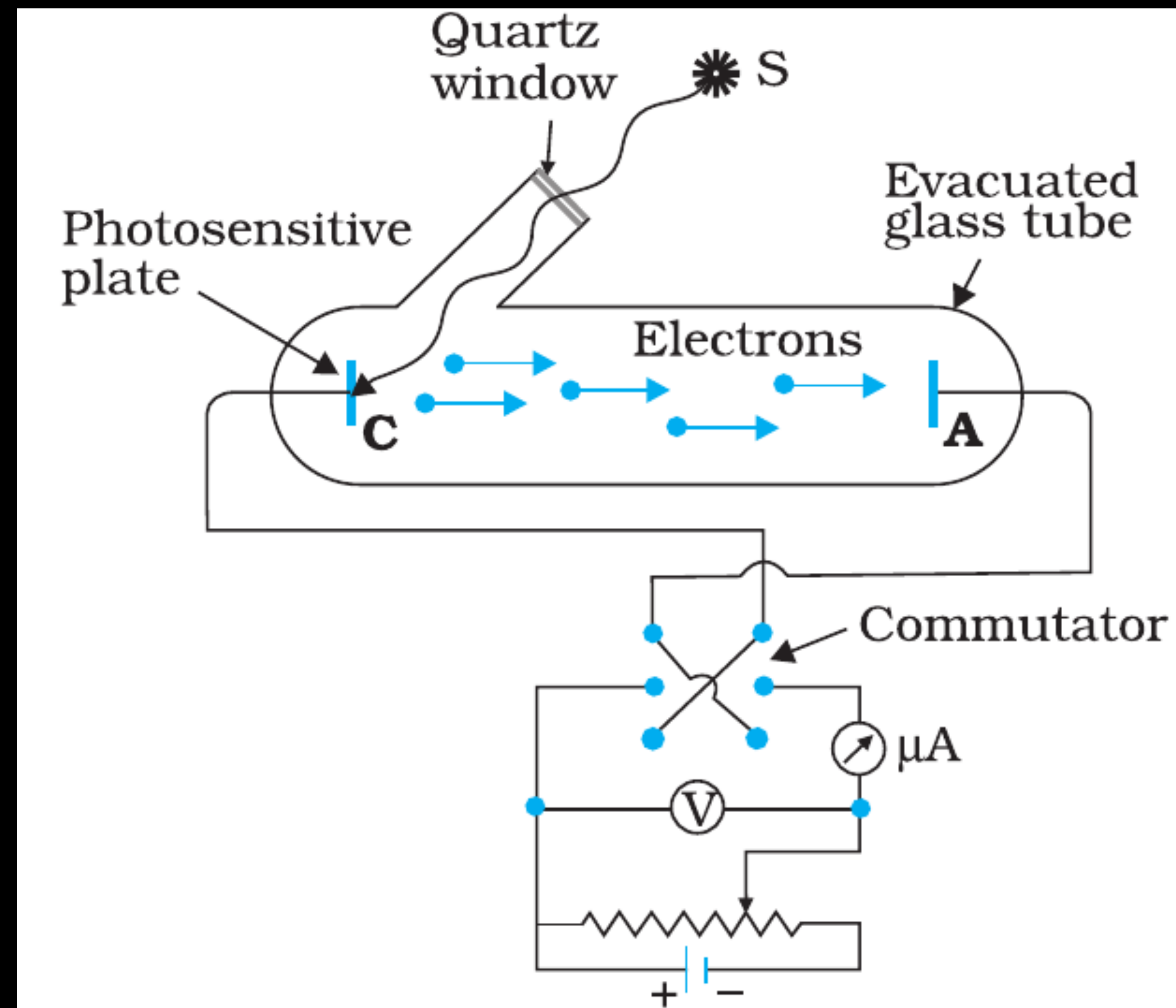


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- * The potential difference between the emitter and collector plates is measured by a voltmeter (V) whereas the resulting photo current flowing in the circuit is measured by a microammeter (μA).
- * We can use the experimental arrangement of to study the variation of photocurrent with (a) intensity of radiation, (b) frequency of incident radiation, (c) the potential difference between the plates A and C, and (d) the nature of the material of plate C

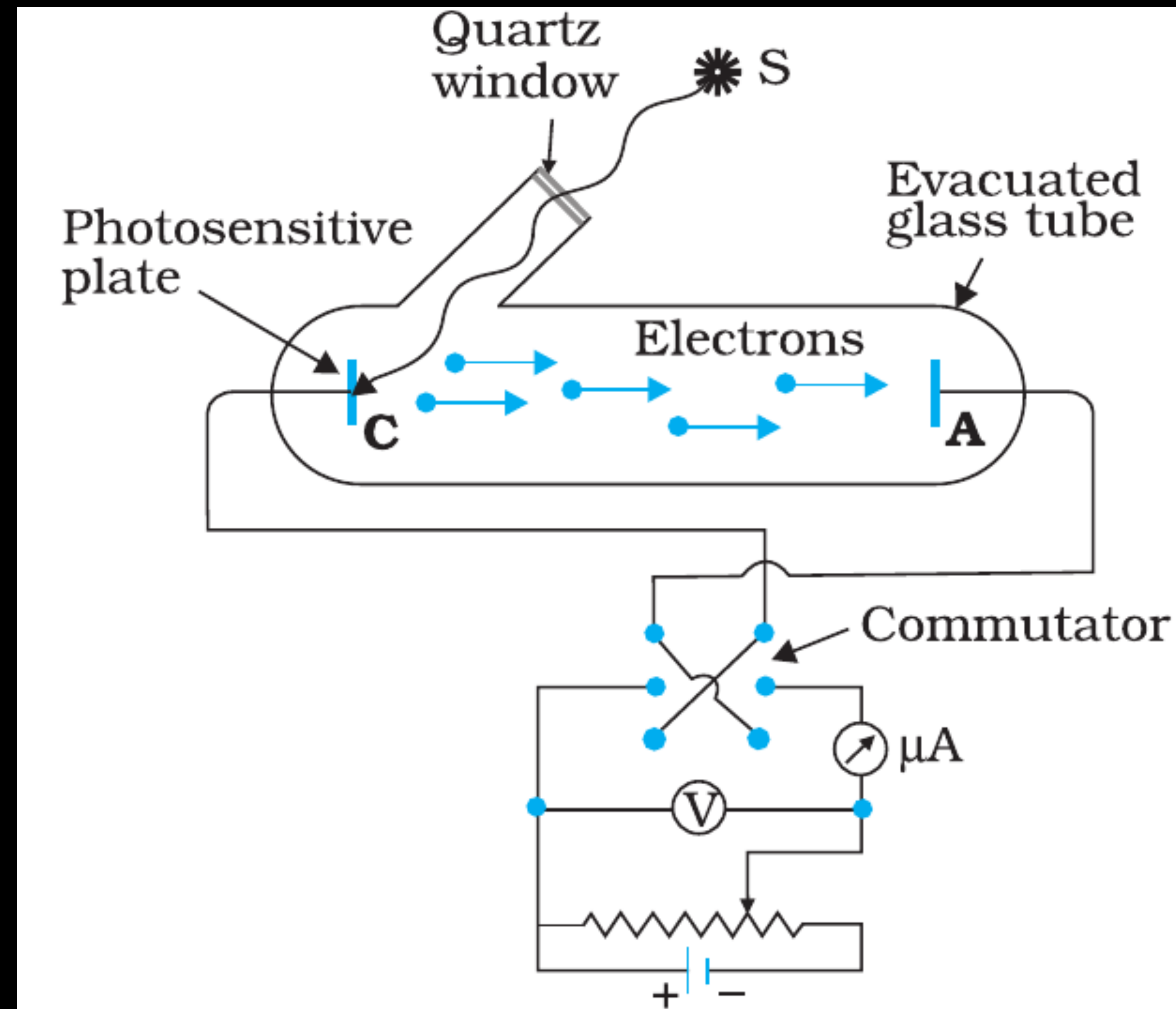


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(1) Effect of intensity of light on photocurrent

(Keep frequency and potential constant, vary only intensity.)

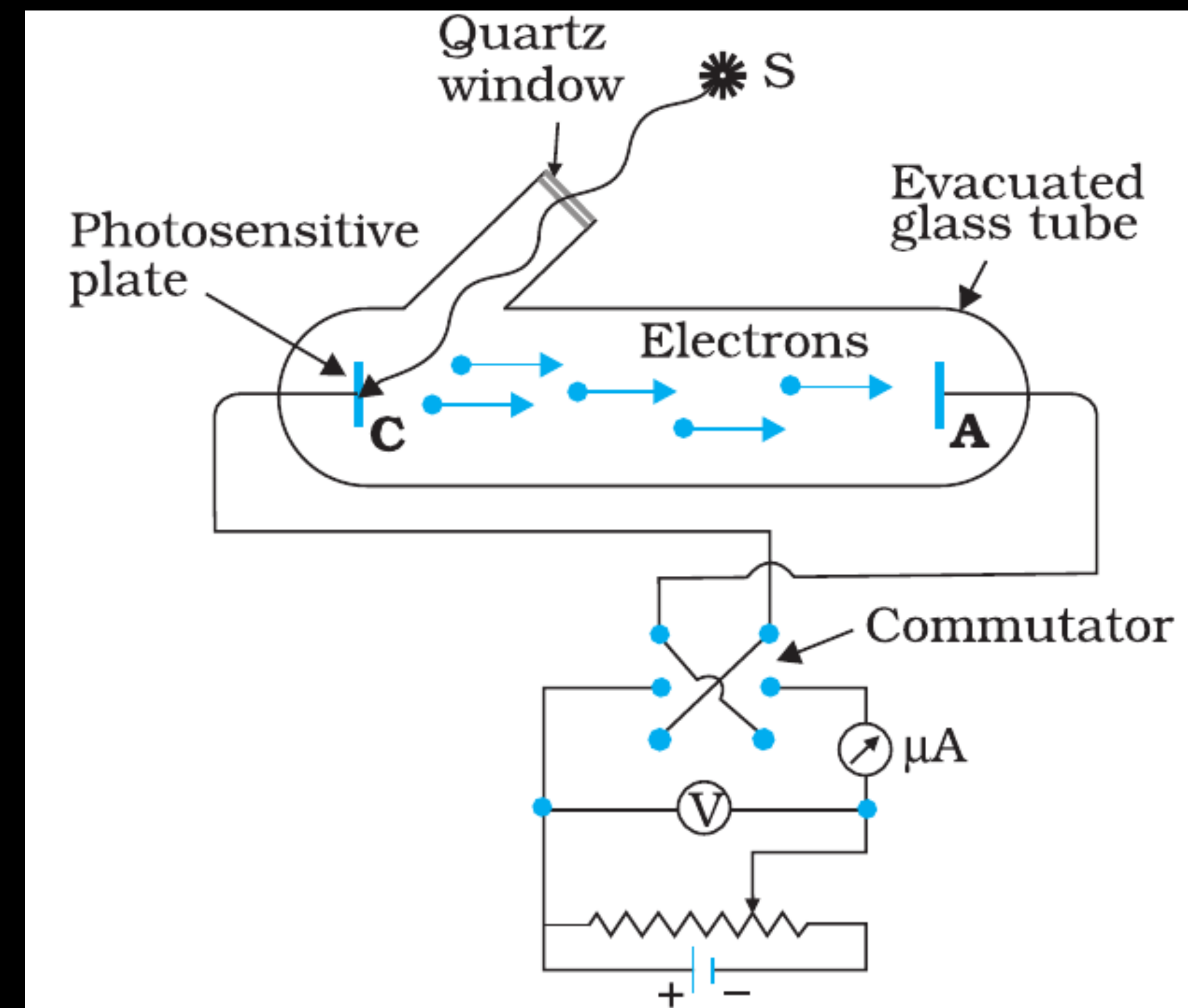


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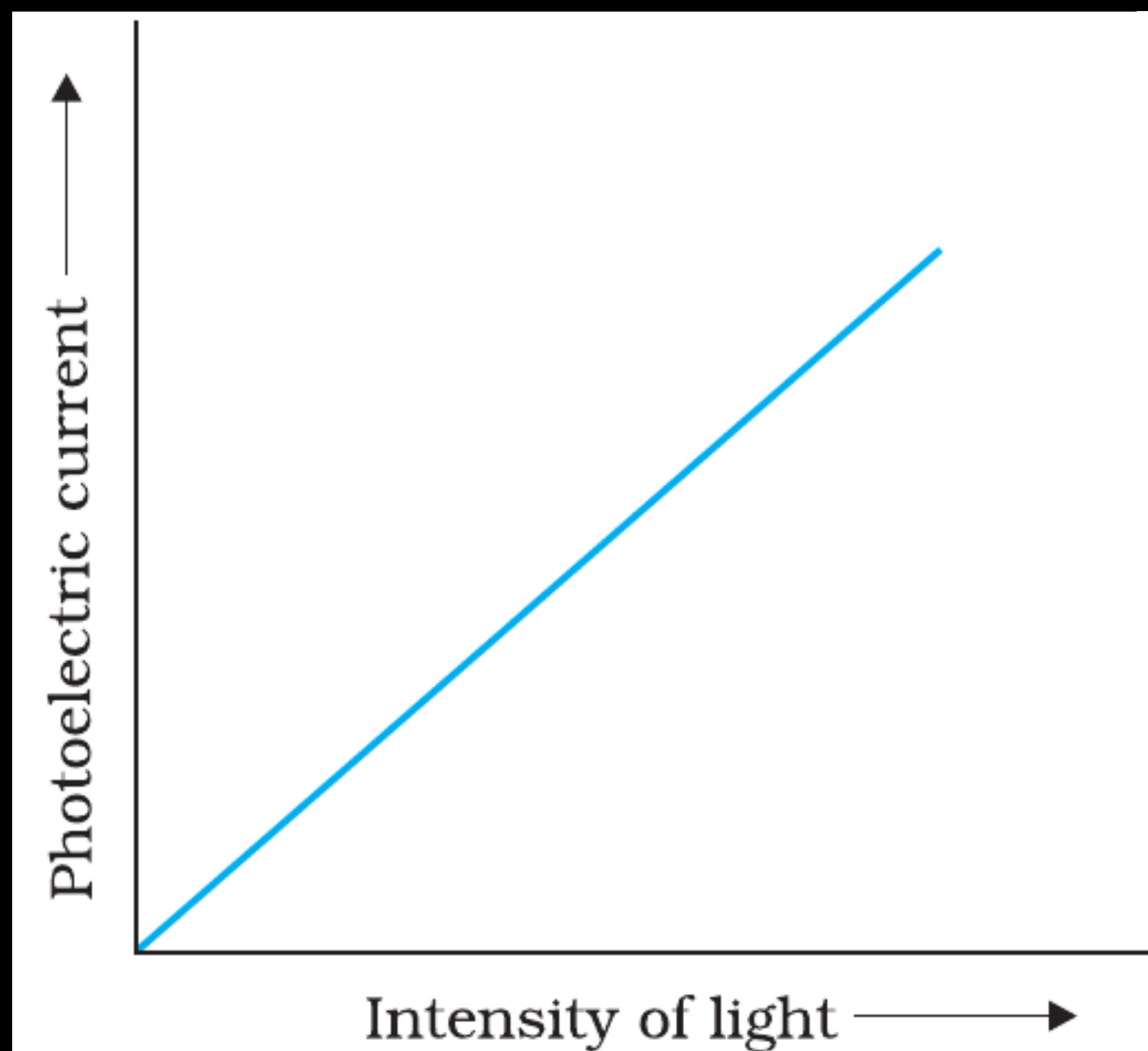


FIGURE 11.2 Variation of Photoelectric current with intensity of light.

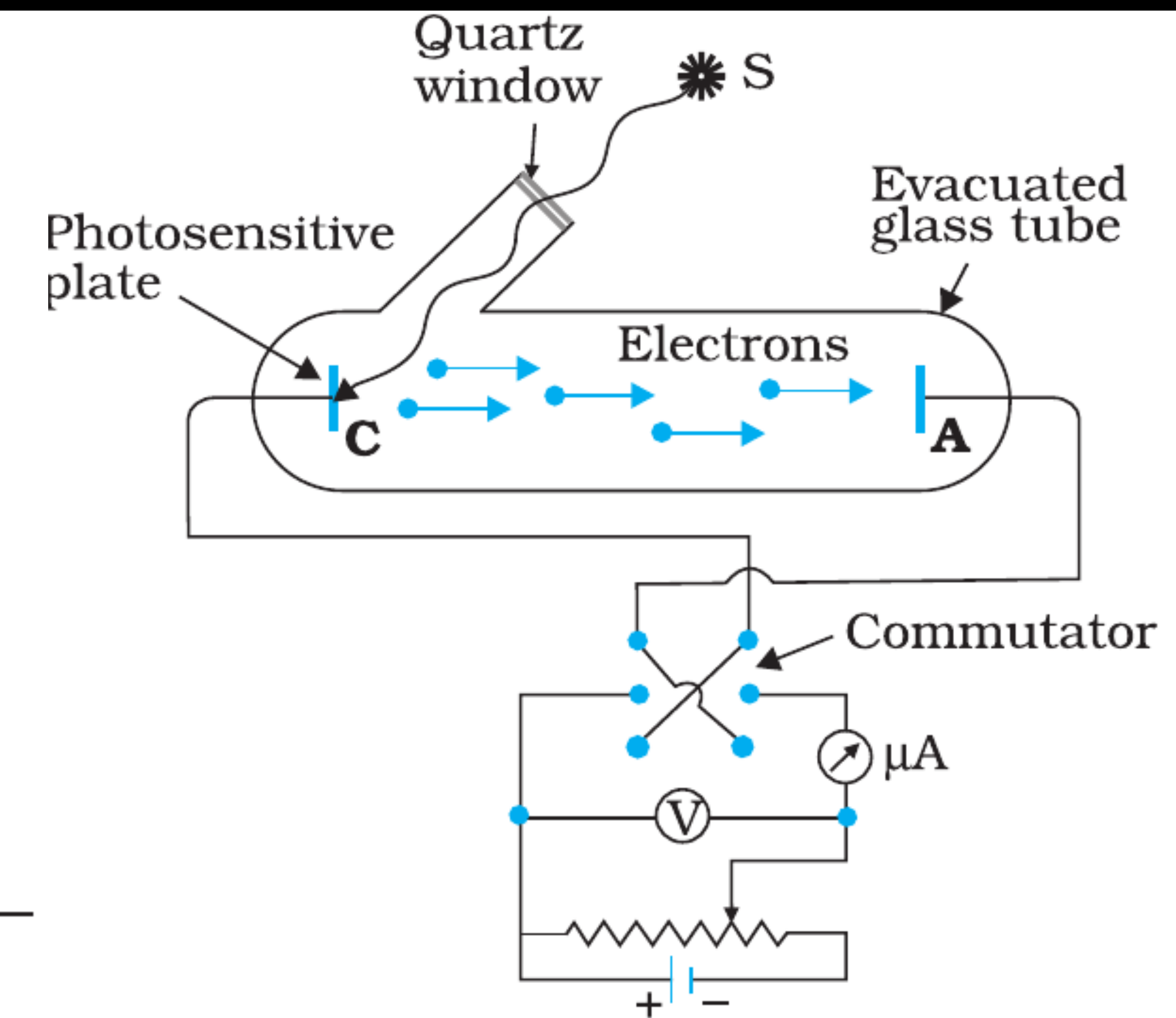


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* The number of photoelectrons emitted per second (or photocurrent) is directly proportional to the intensity of incident radiation.

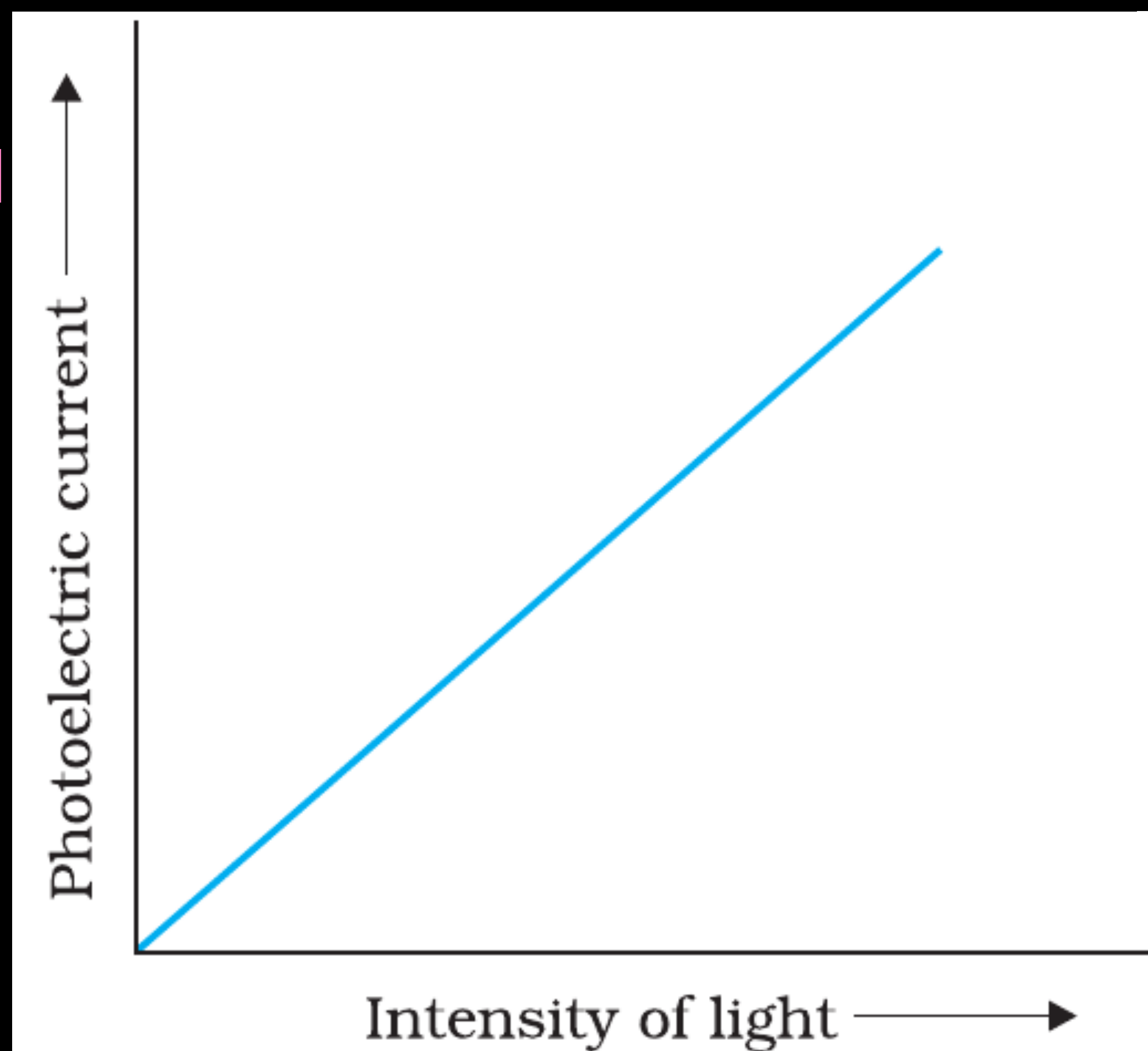


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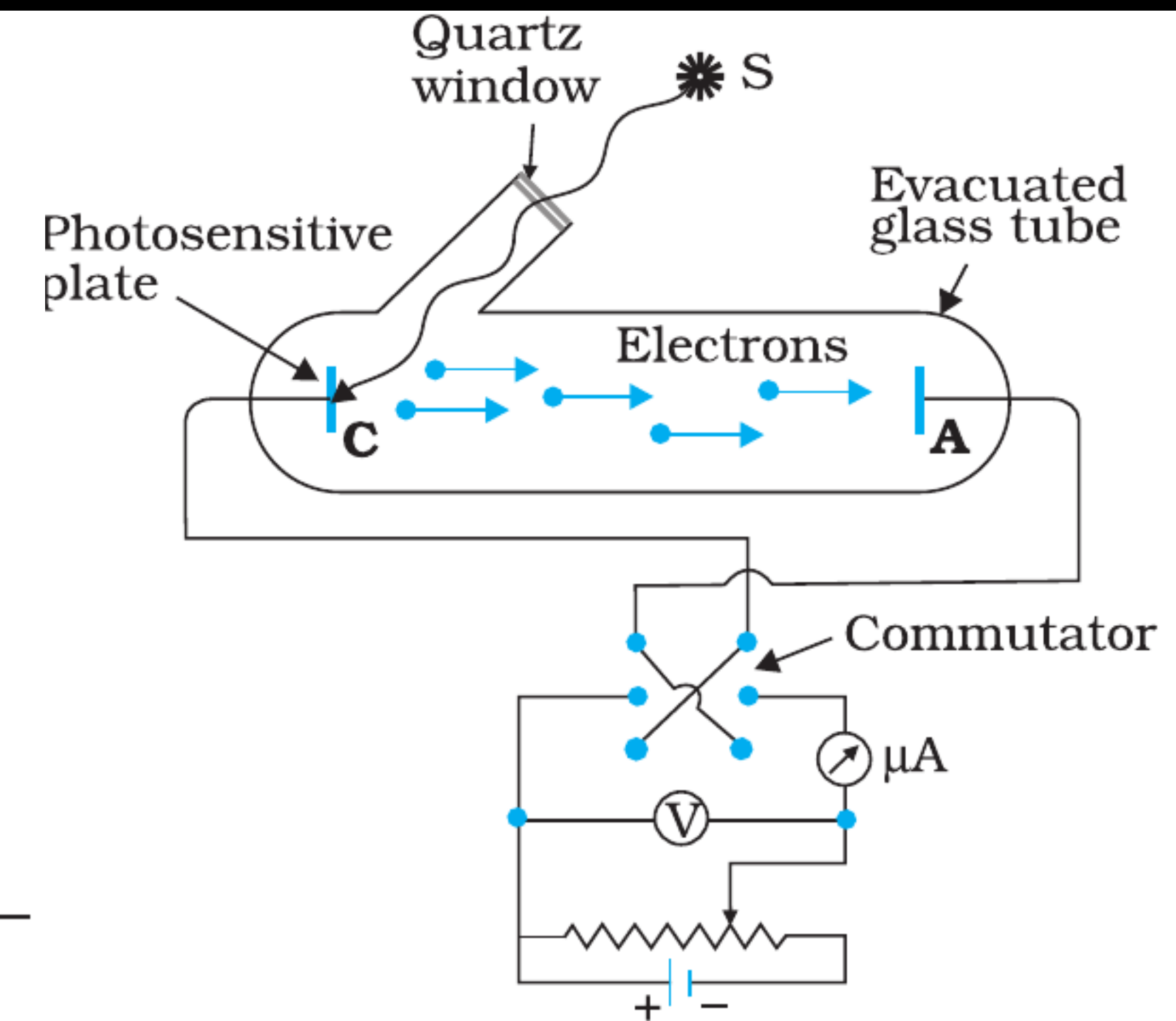


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EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

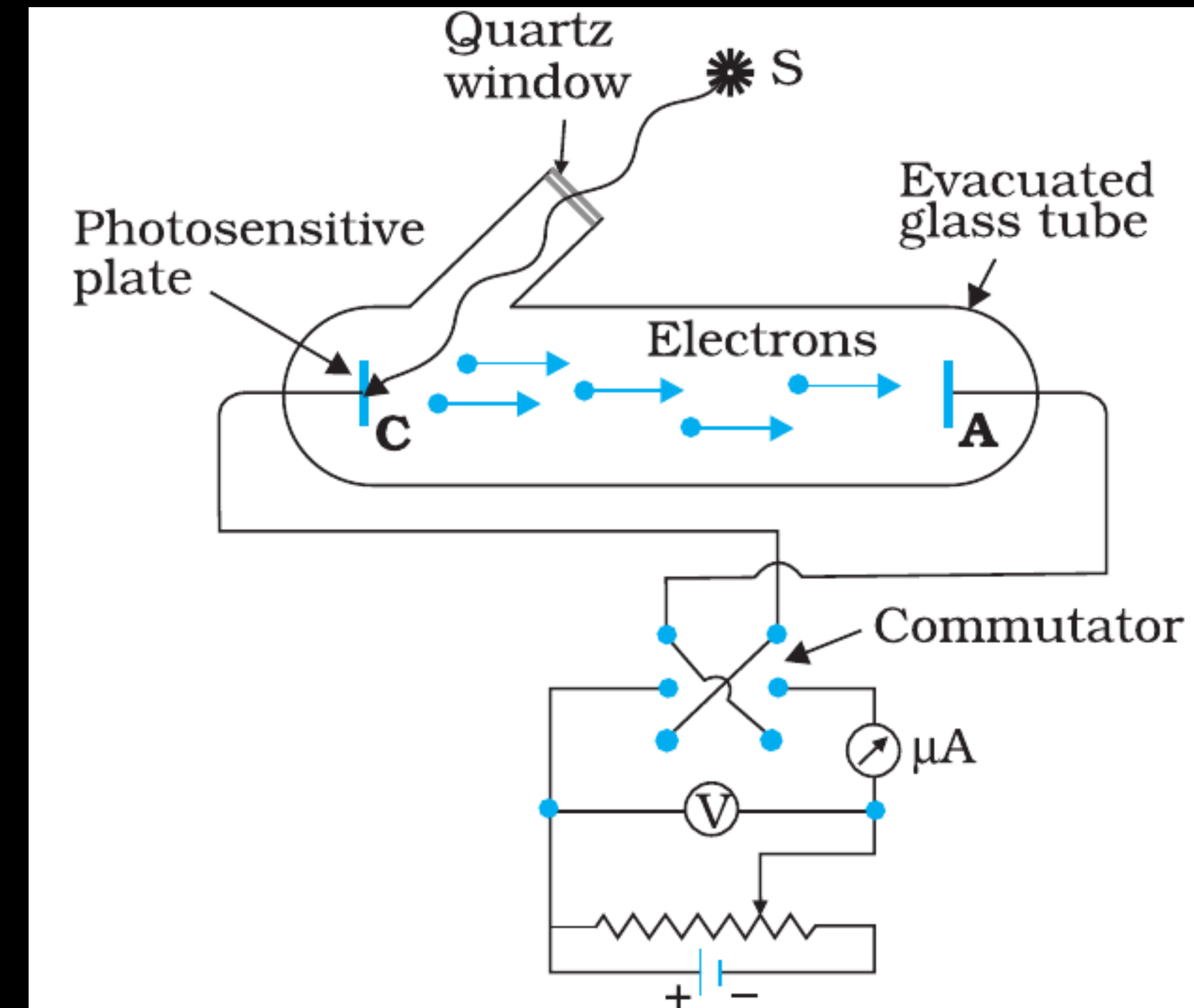


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

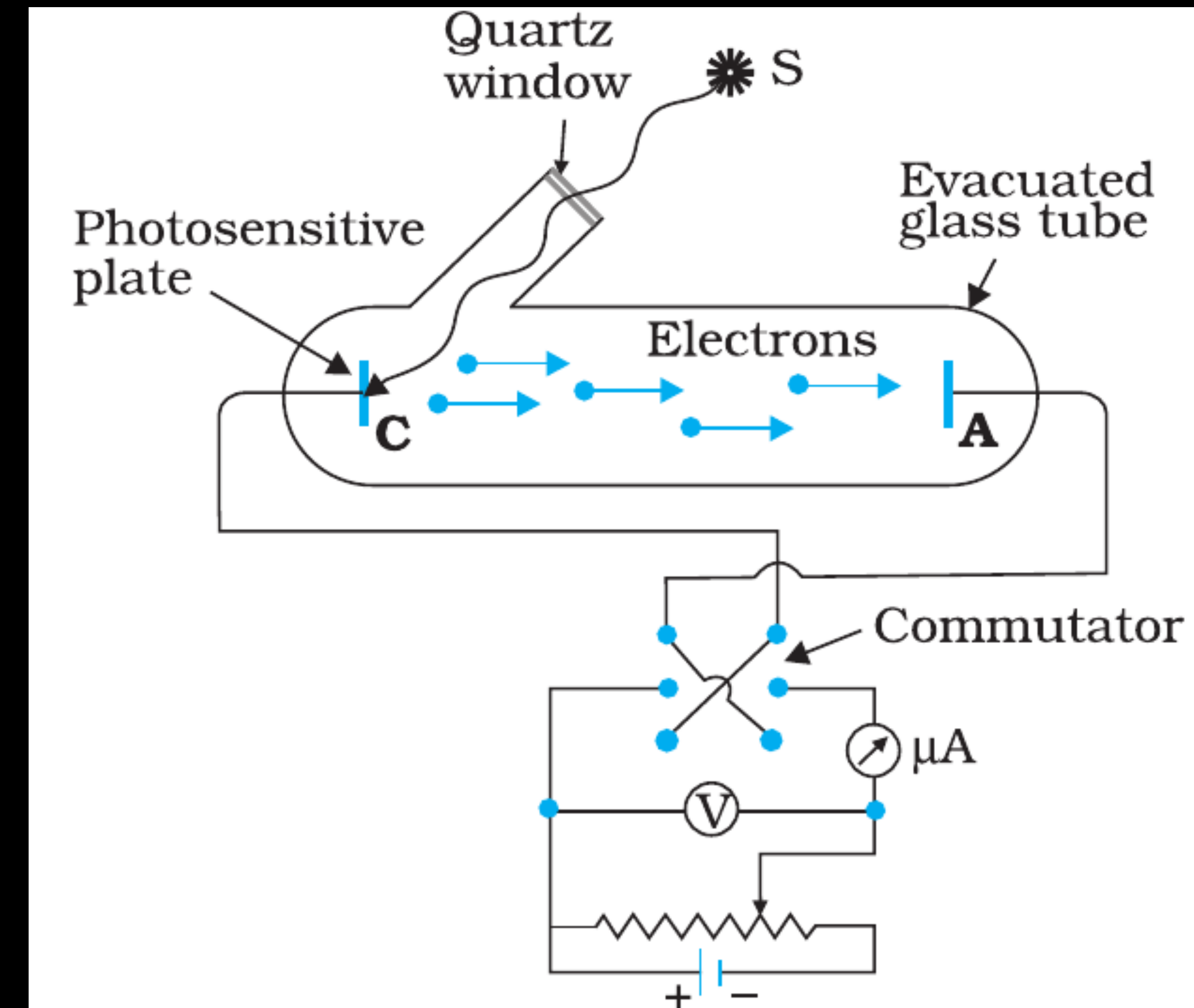


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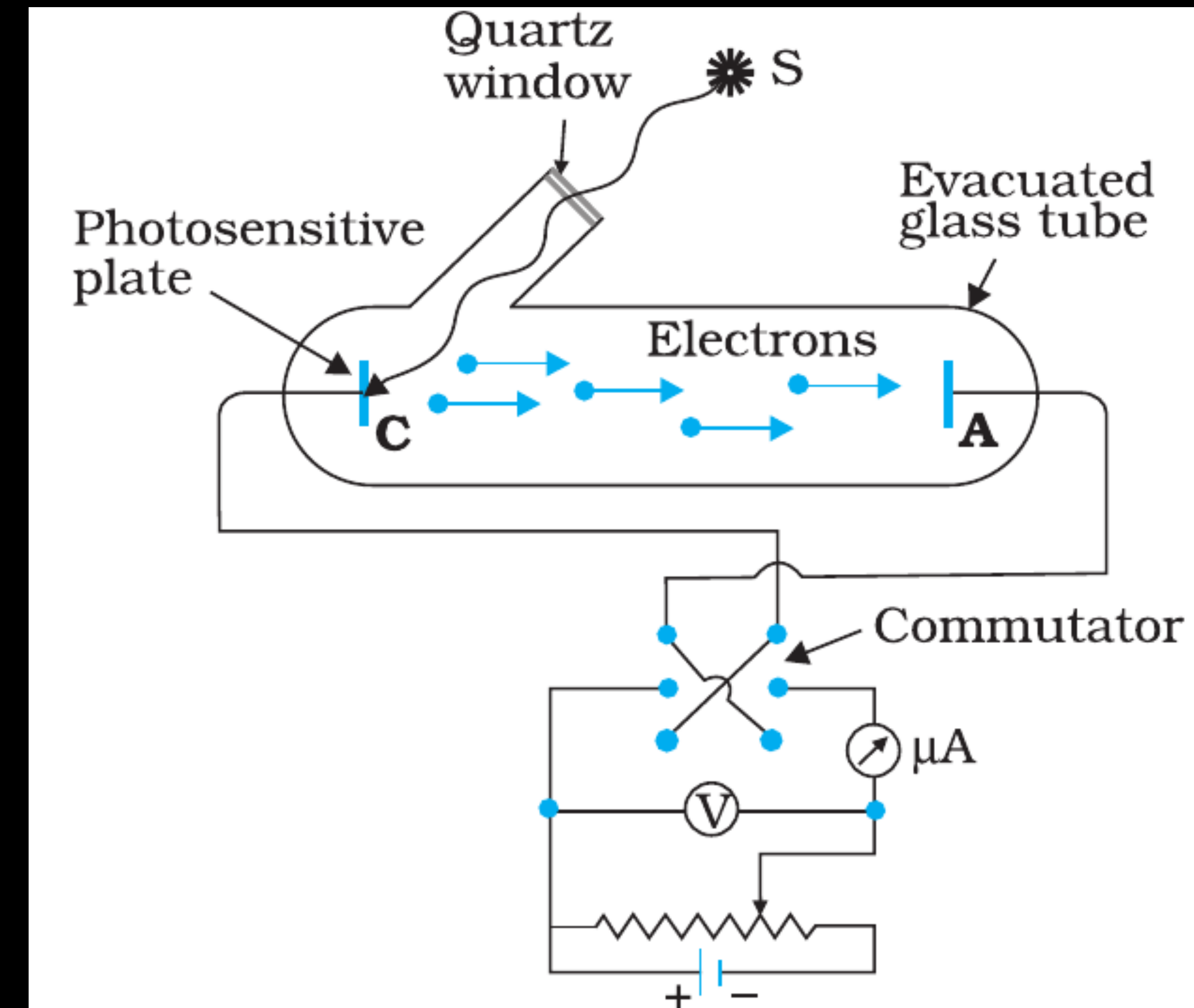


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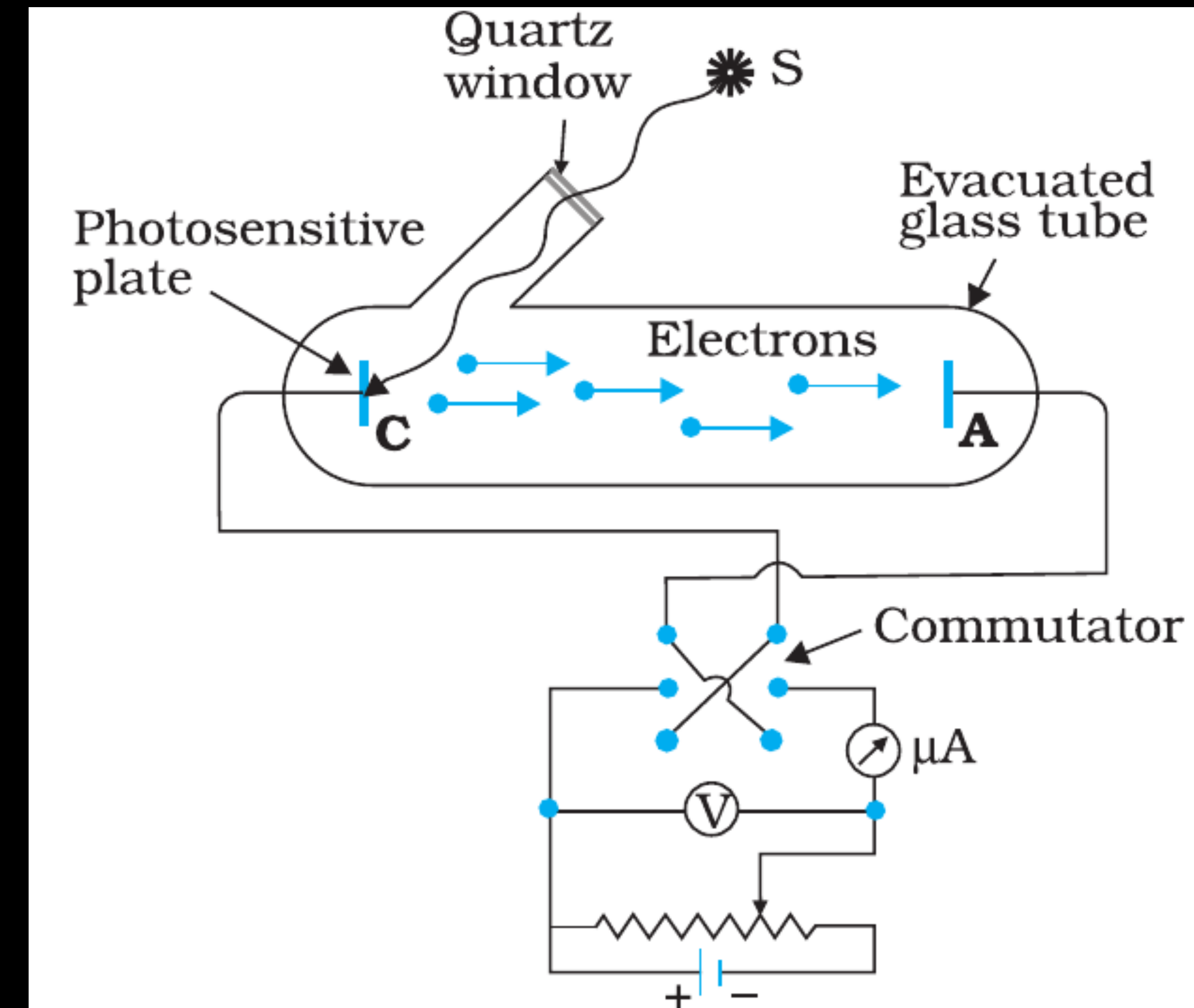


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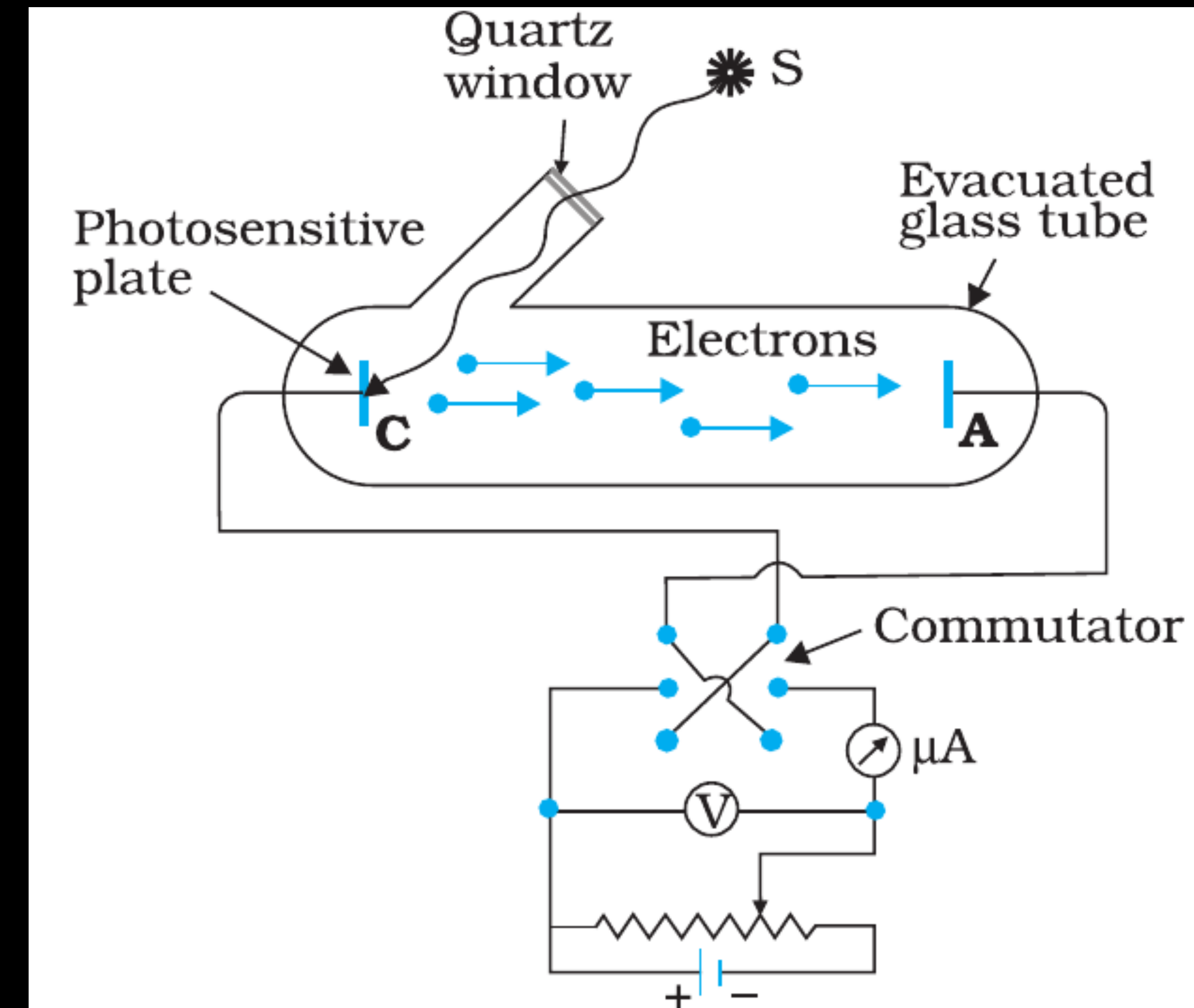


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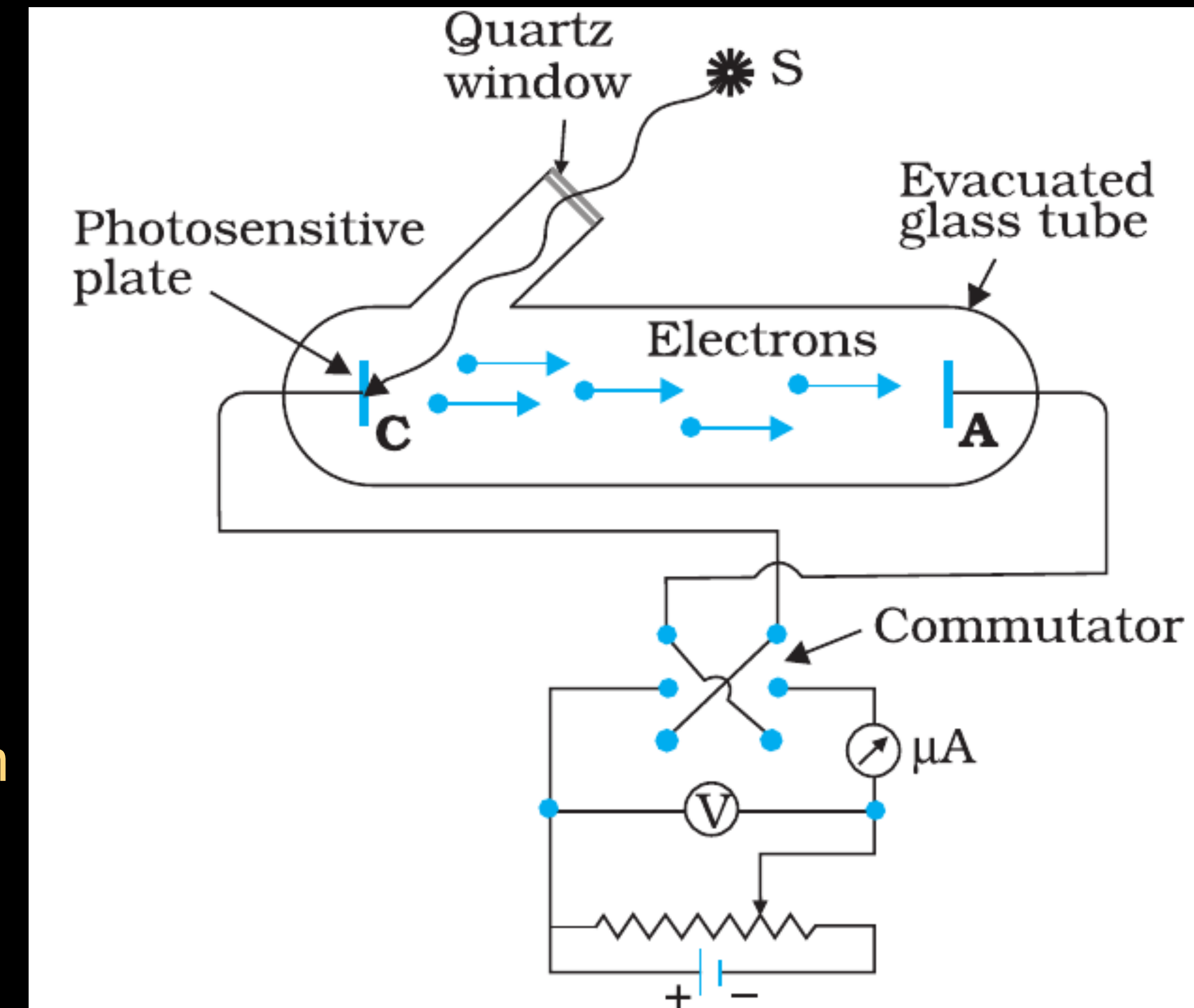


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(ii) Negative Potential (Retarding Potential) and make it increasingly negative gradually.

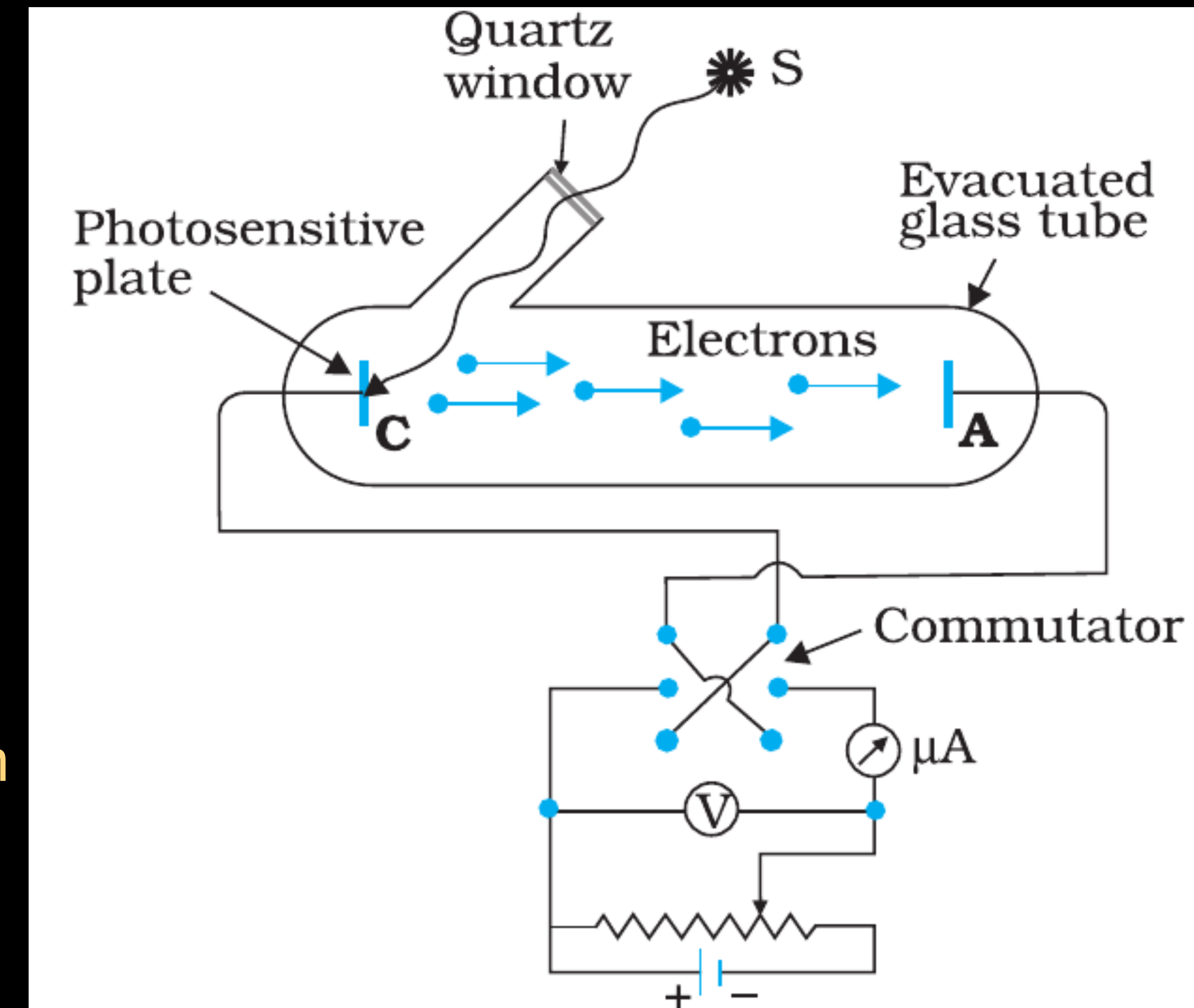


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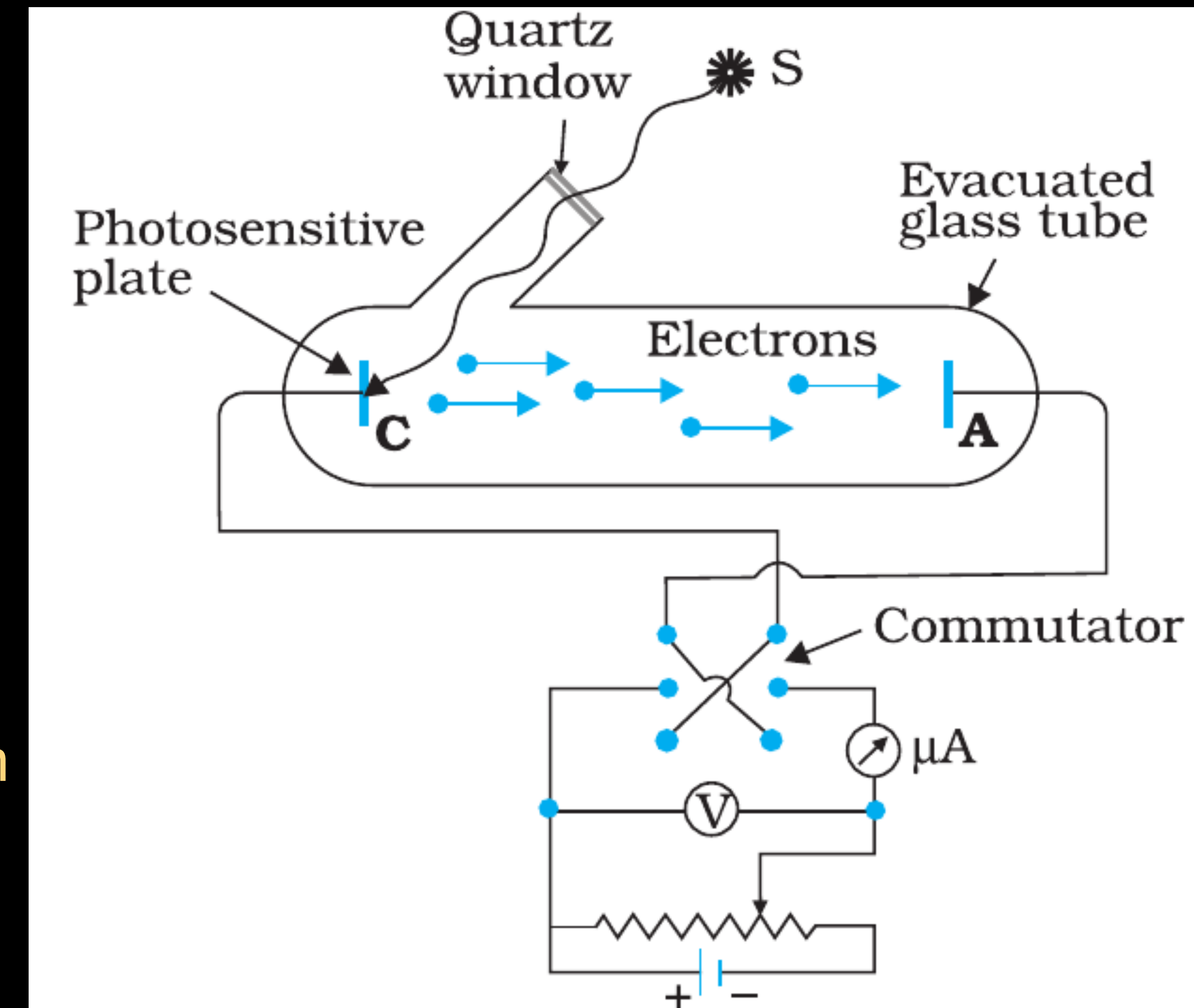


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- * When the polarity is reversed, the electrons are repelled and only the sufficiently energetic electrons are able to reach the collector A.
- * At a certain negative voltage (V_0) called stopping potential: Even the most energetic electrons cannot reach A. Photocurrent becomes zero.

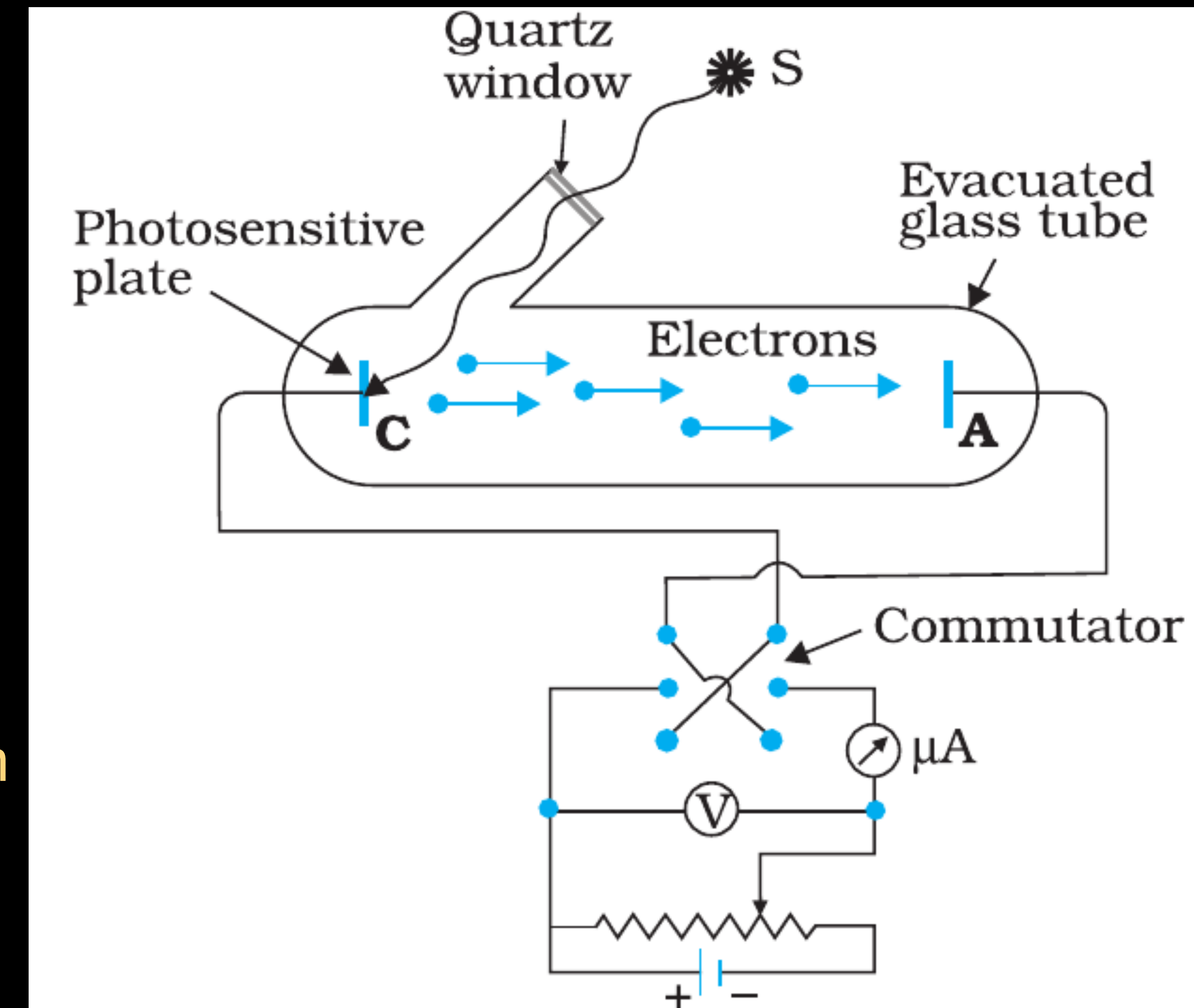


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EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stopping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

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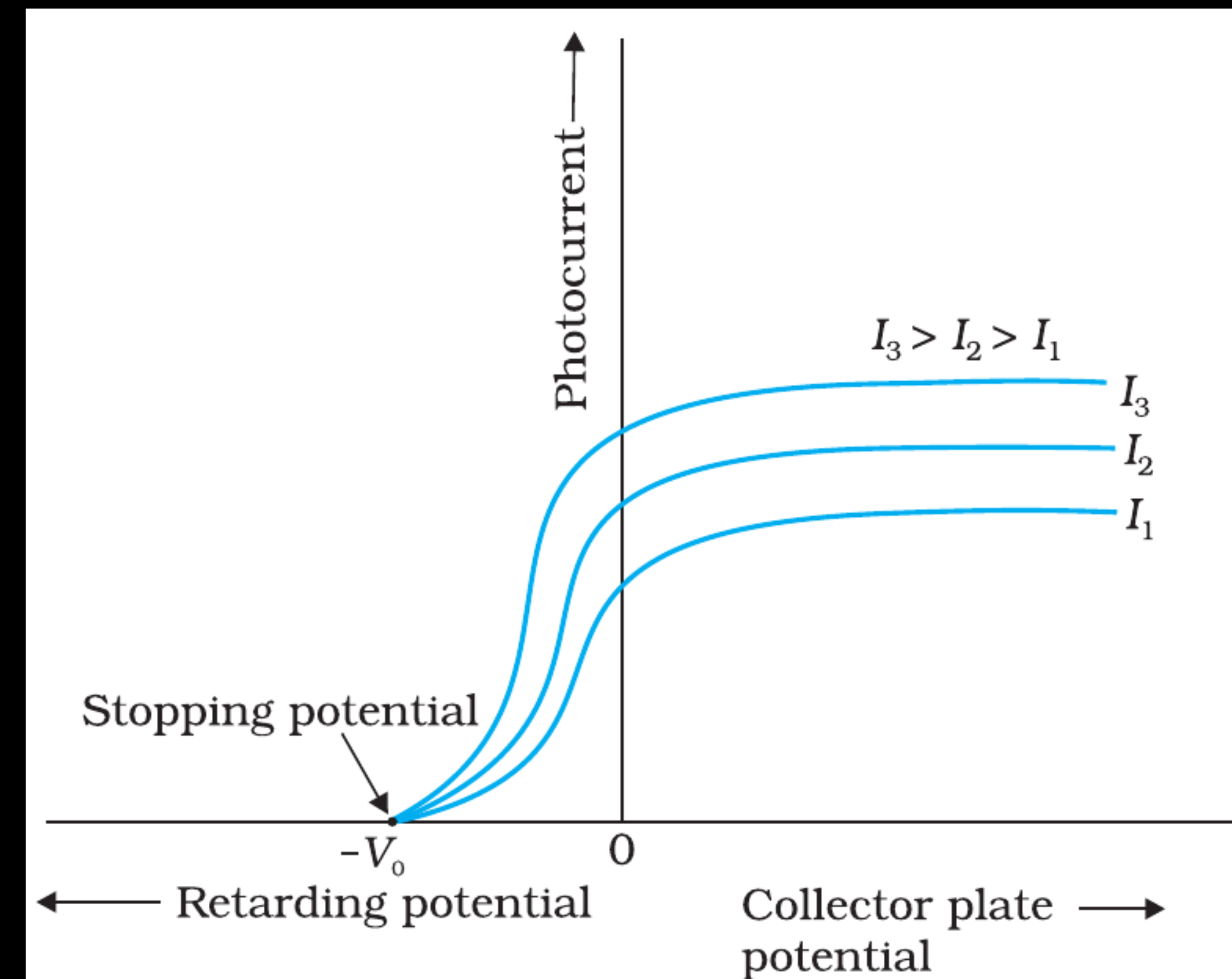


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

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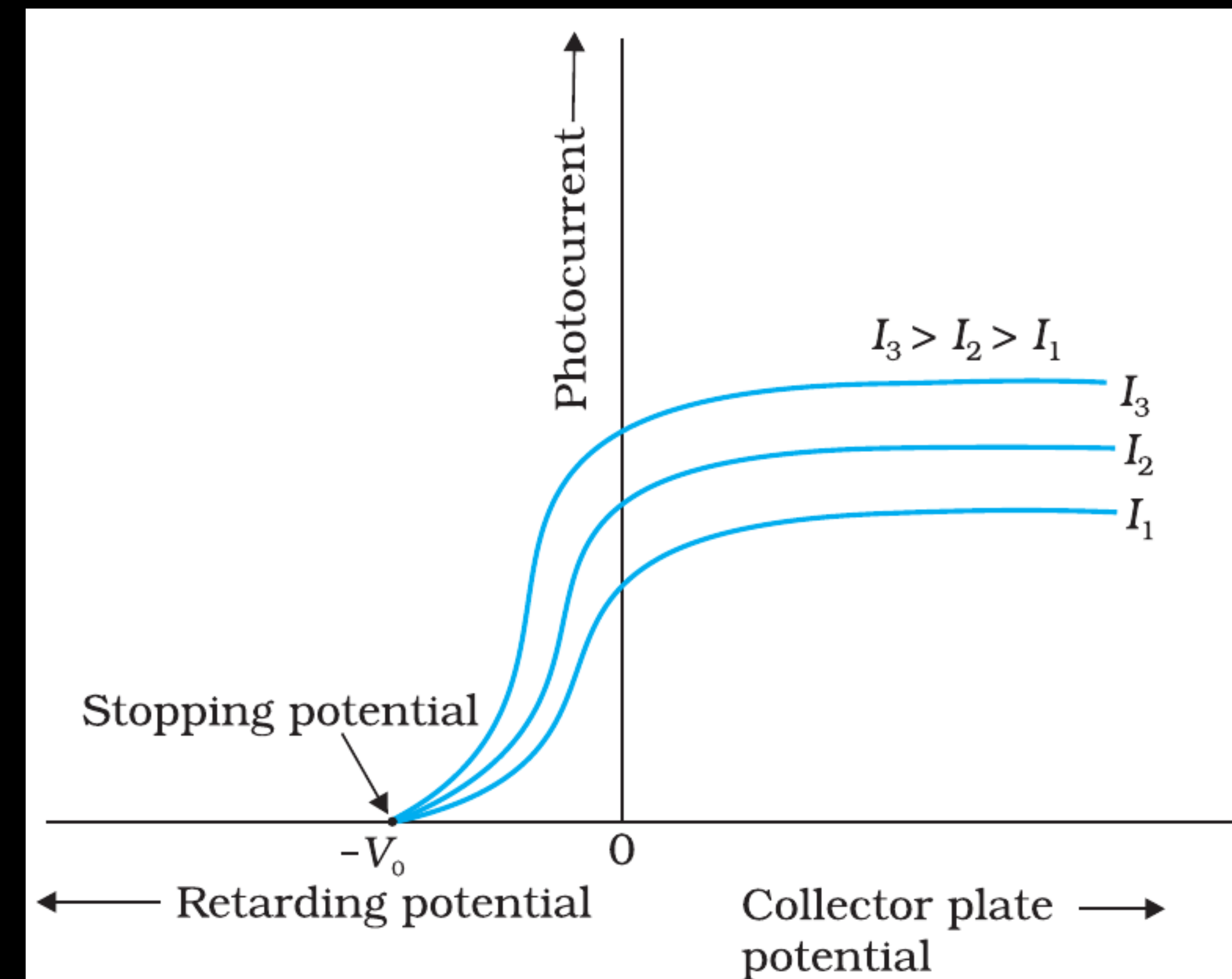


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- * All the photoelectrons emitted from the metal do not have the same energy.
- * Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy ($K_{\max}$), so that

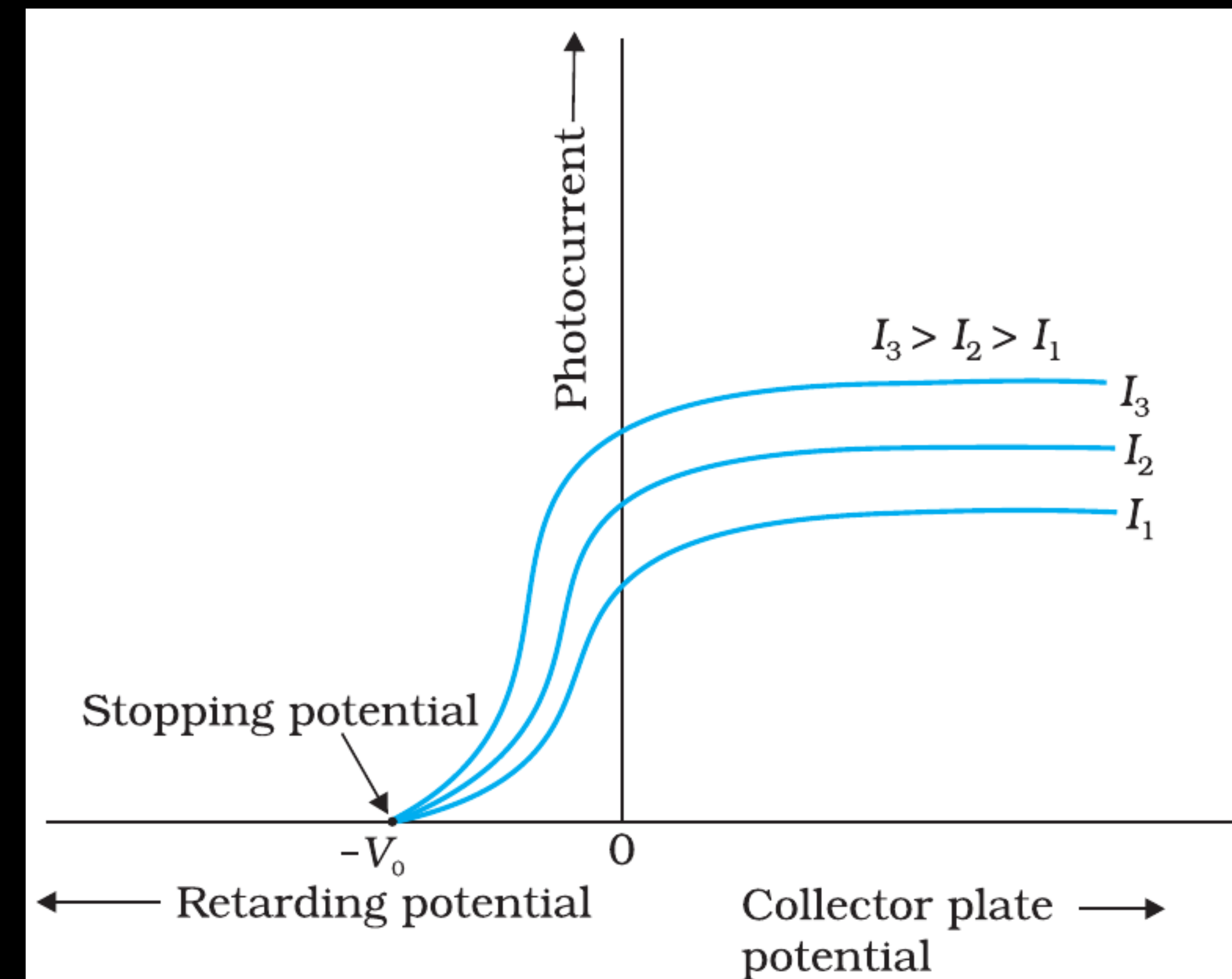


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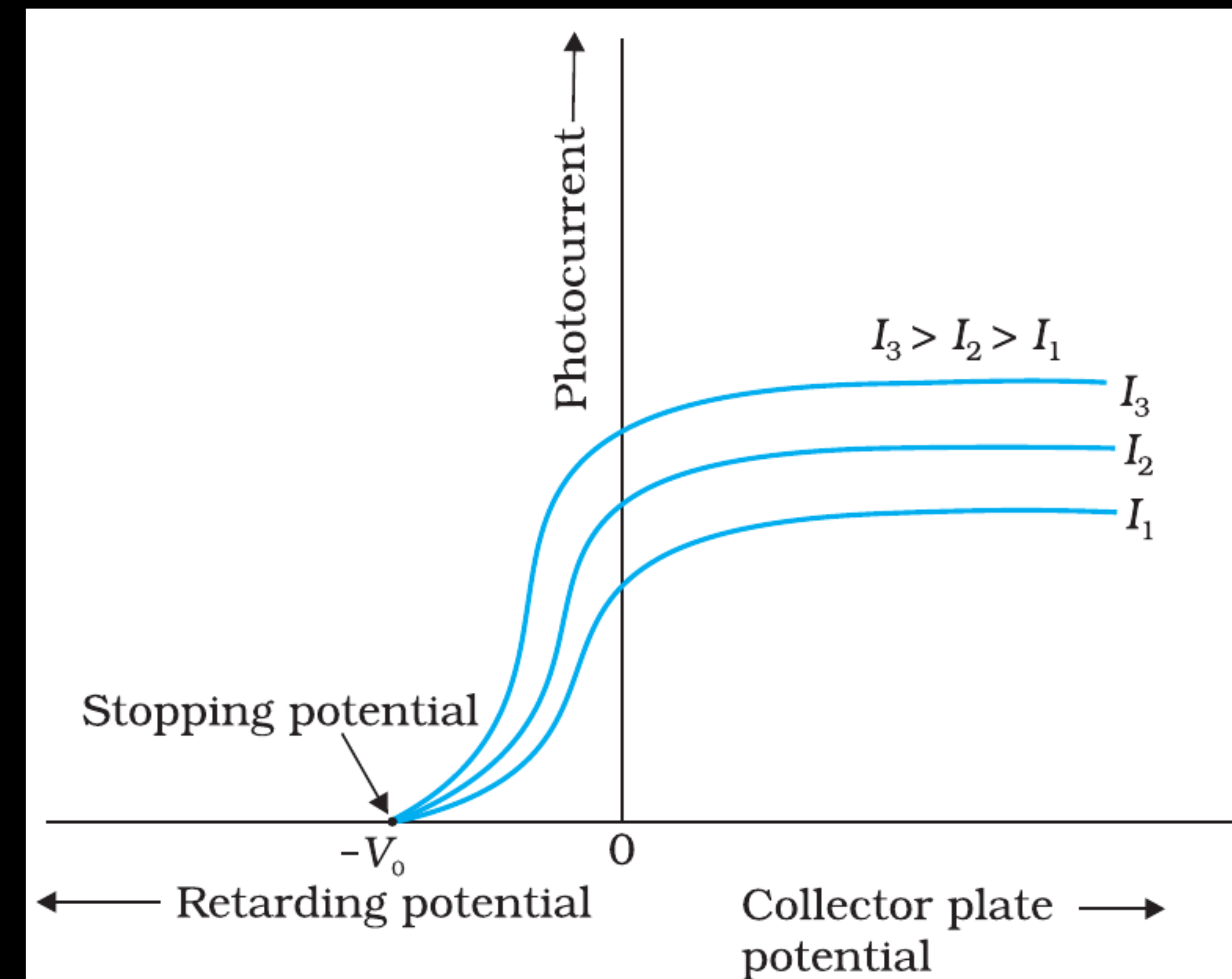


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- * For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.

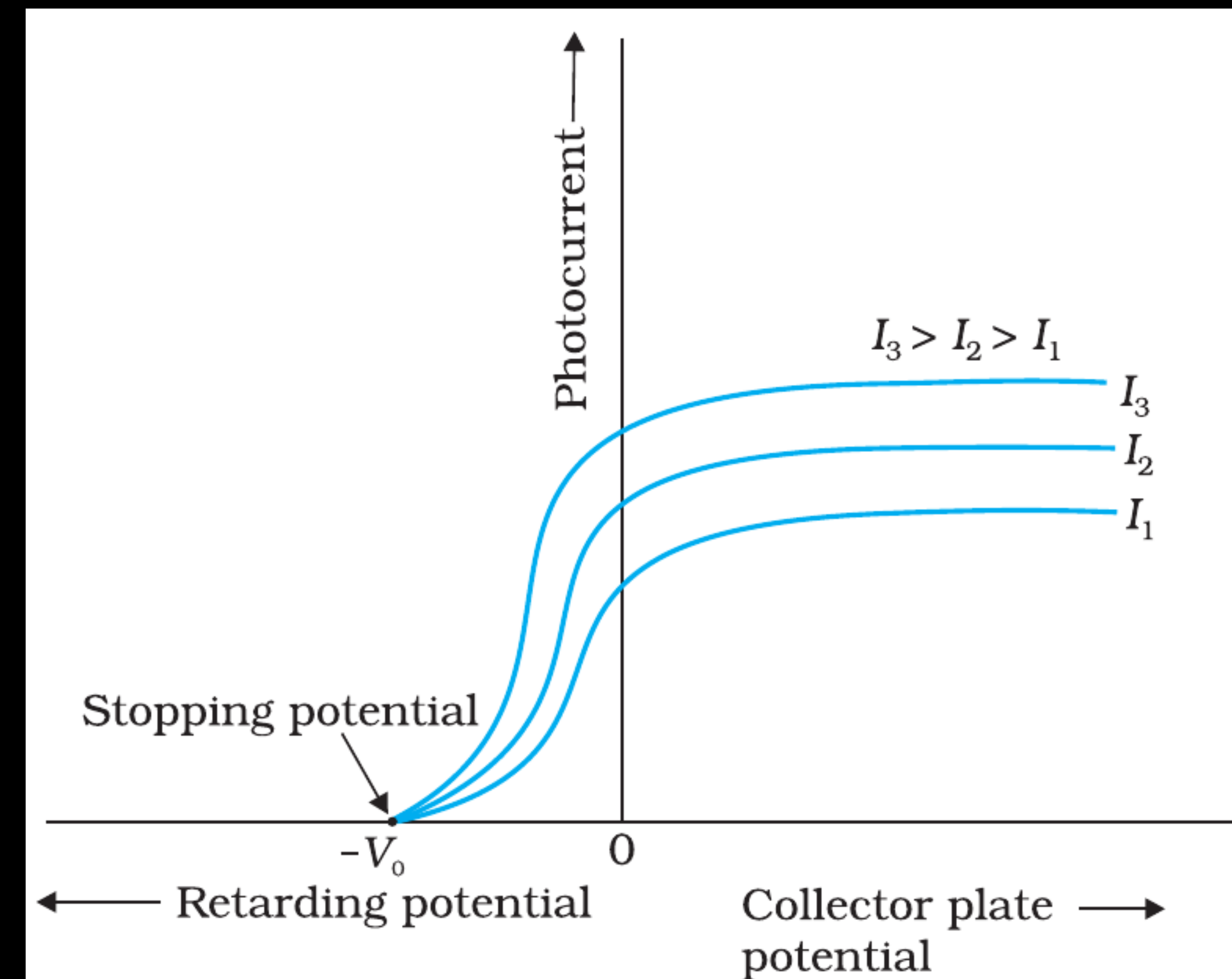


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- * For a given frequency of the incident radiation, the stopping potential is independent of its intensity.

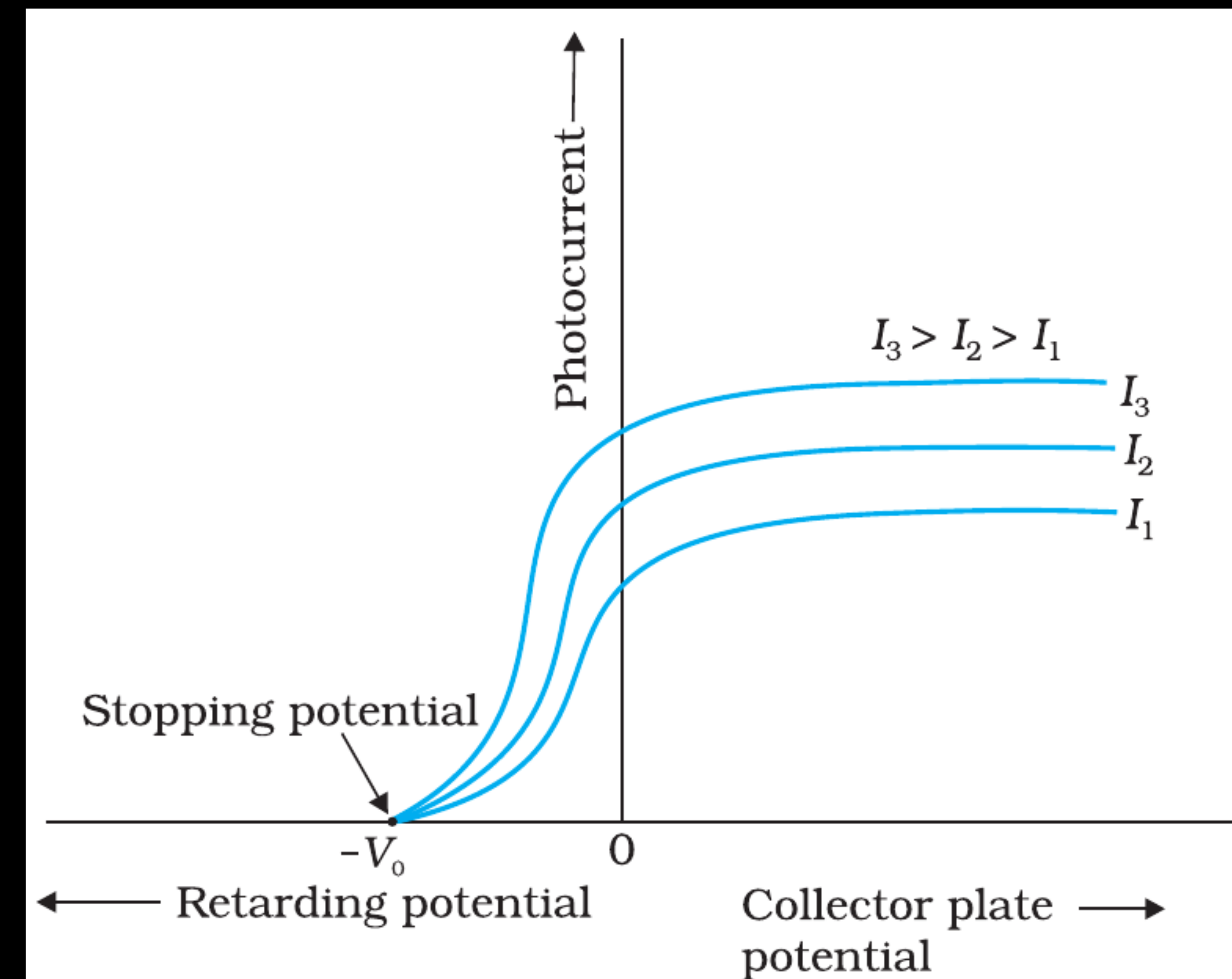


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$$K_{\max} = eV_0$$

- * For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.
- * For a given frequency of the incident radiation, the stopping potential is independent of its intensity.
- * In other words, the maximum kinetic energy of photoelectrons depends on the light source and the emitter plate material, but is independent of intensity of incident radiation.

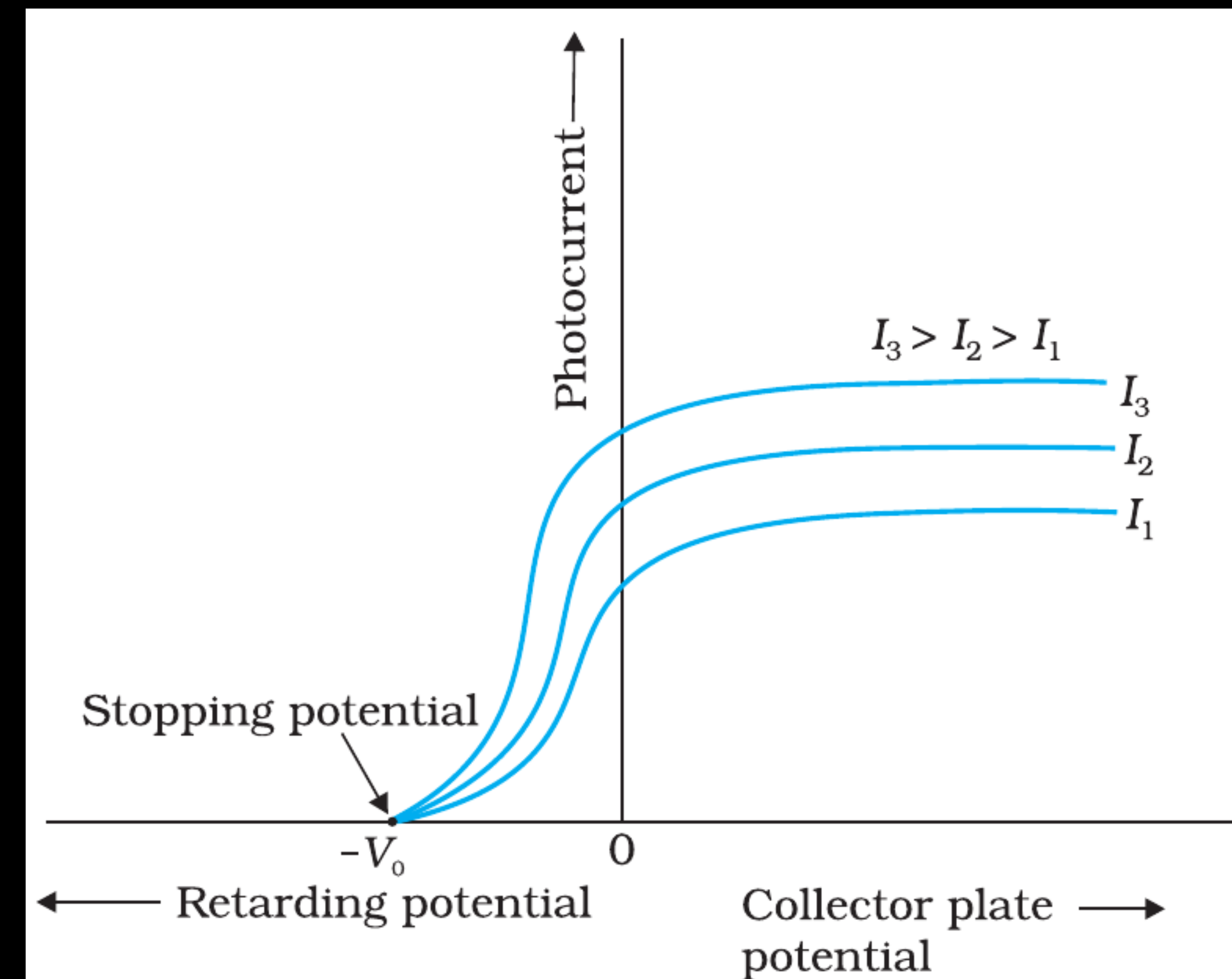


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.

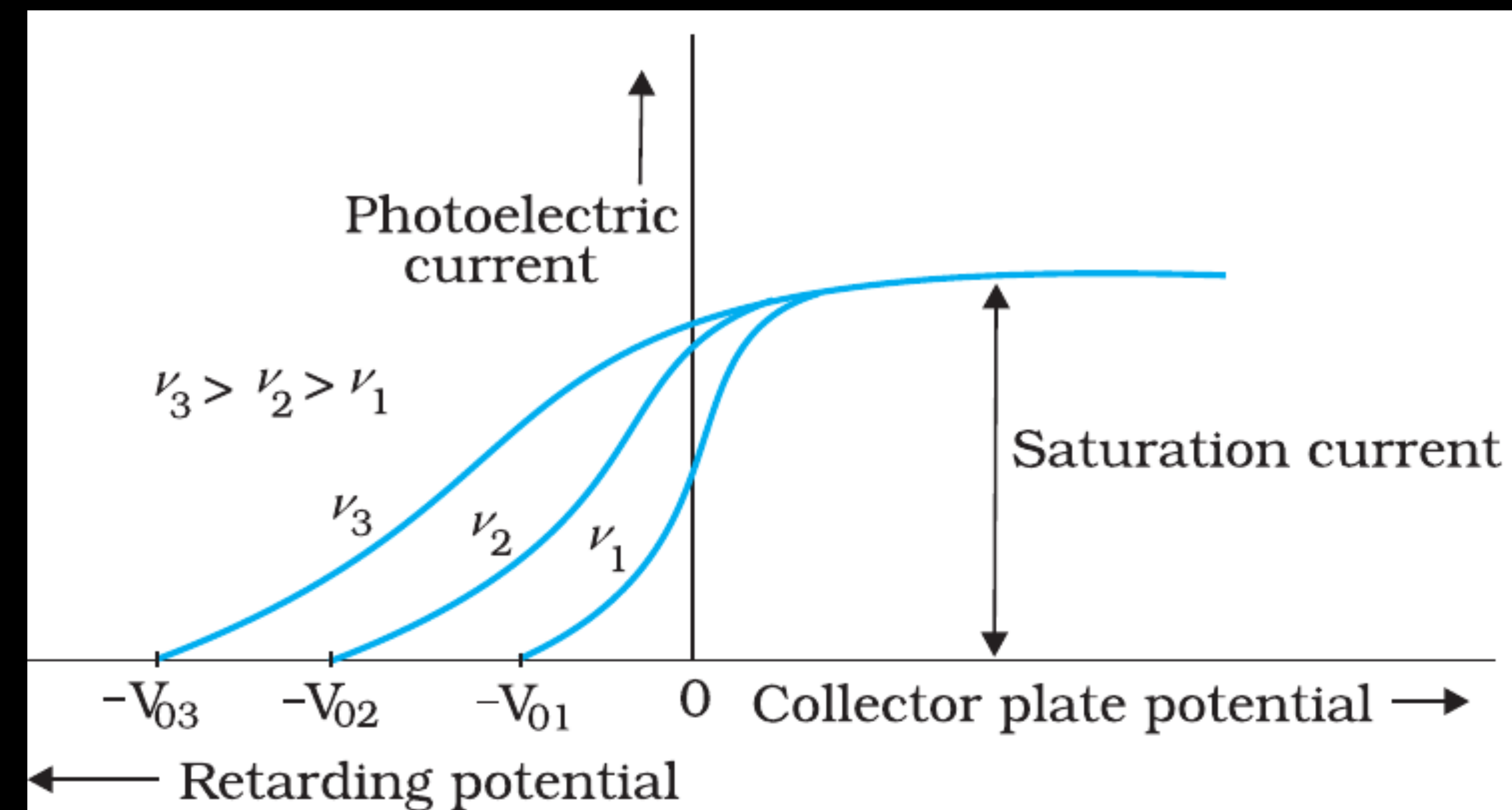


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
- * We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

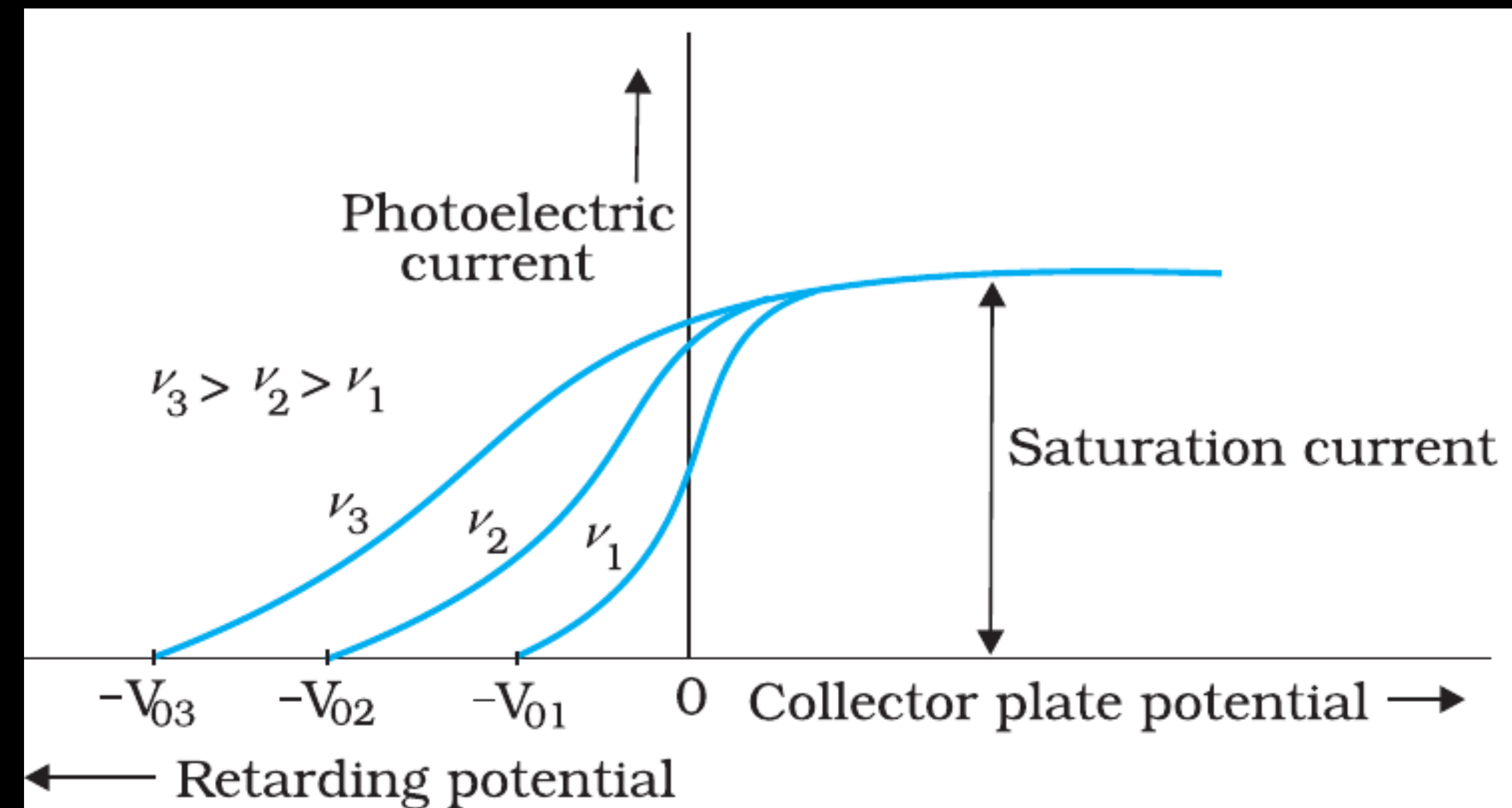


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
- * We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.
- * The energy of the emitted electrons depends on the frequency of the incident radiations.

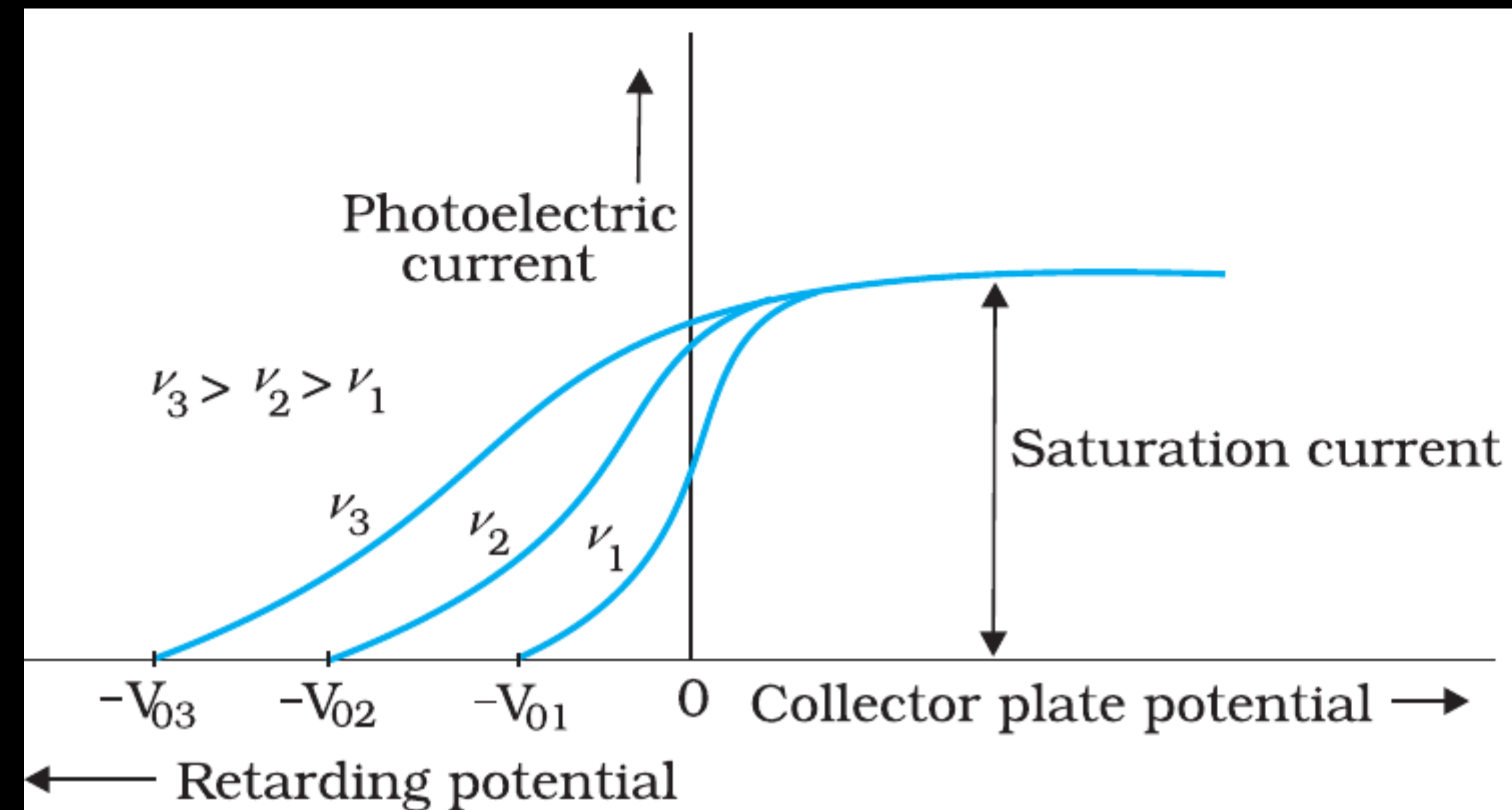


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
- * We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.
- * The energy of the emitted electrons depends on the frequency of the incident radiations.
- * The stopping potential is more negative for higher frequencies of incident radiation.

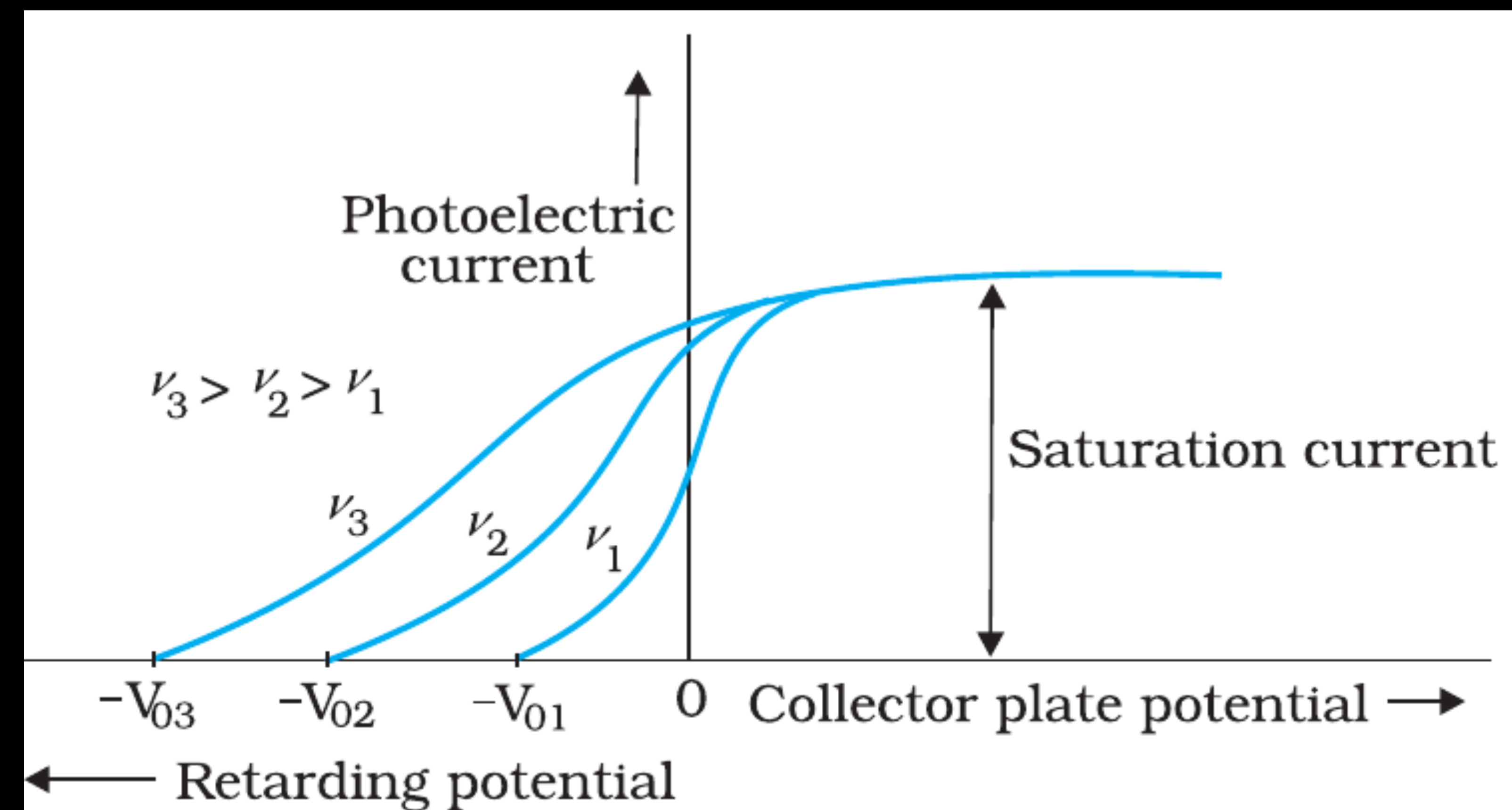


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
- * We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.
- * The energy of the emitted electrons depends on the frequency of the incident radiations.
- * The stopping potential is more negative for higher frequencies of incident radiation.
- * This implies that greater the frequency of incident light, greater is the maximum kinetic energy of the photoelectrons. Consequently, we need greater retarding potential to stop them.

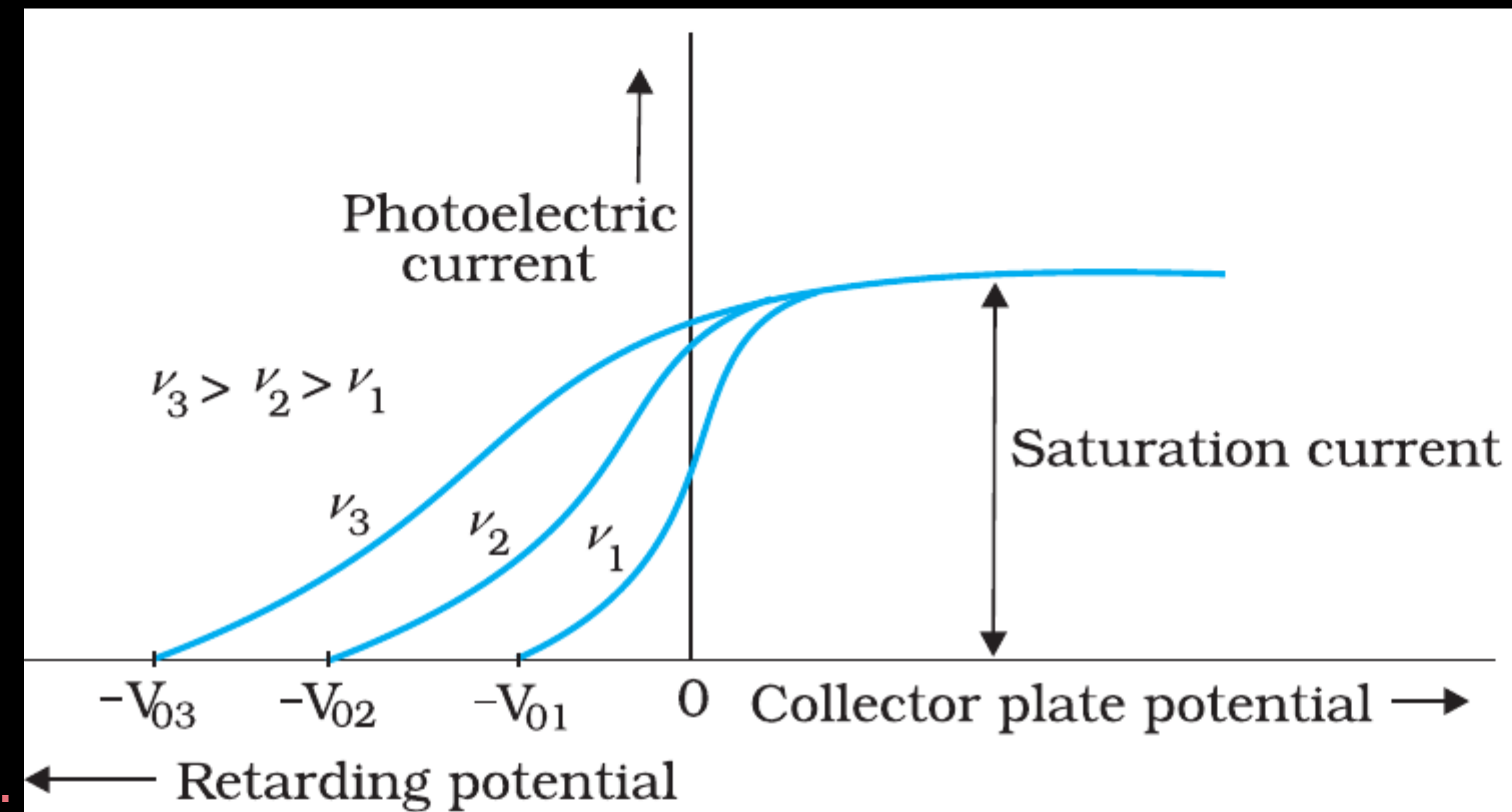


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
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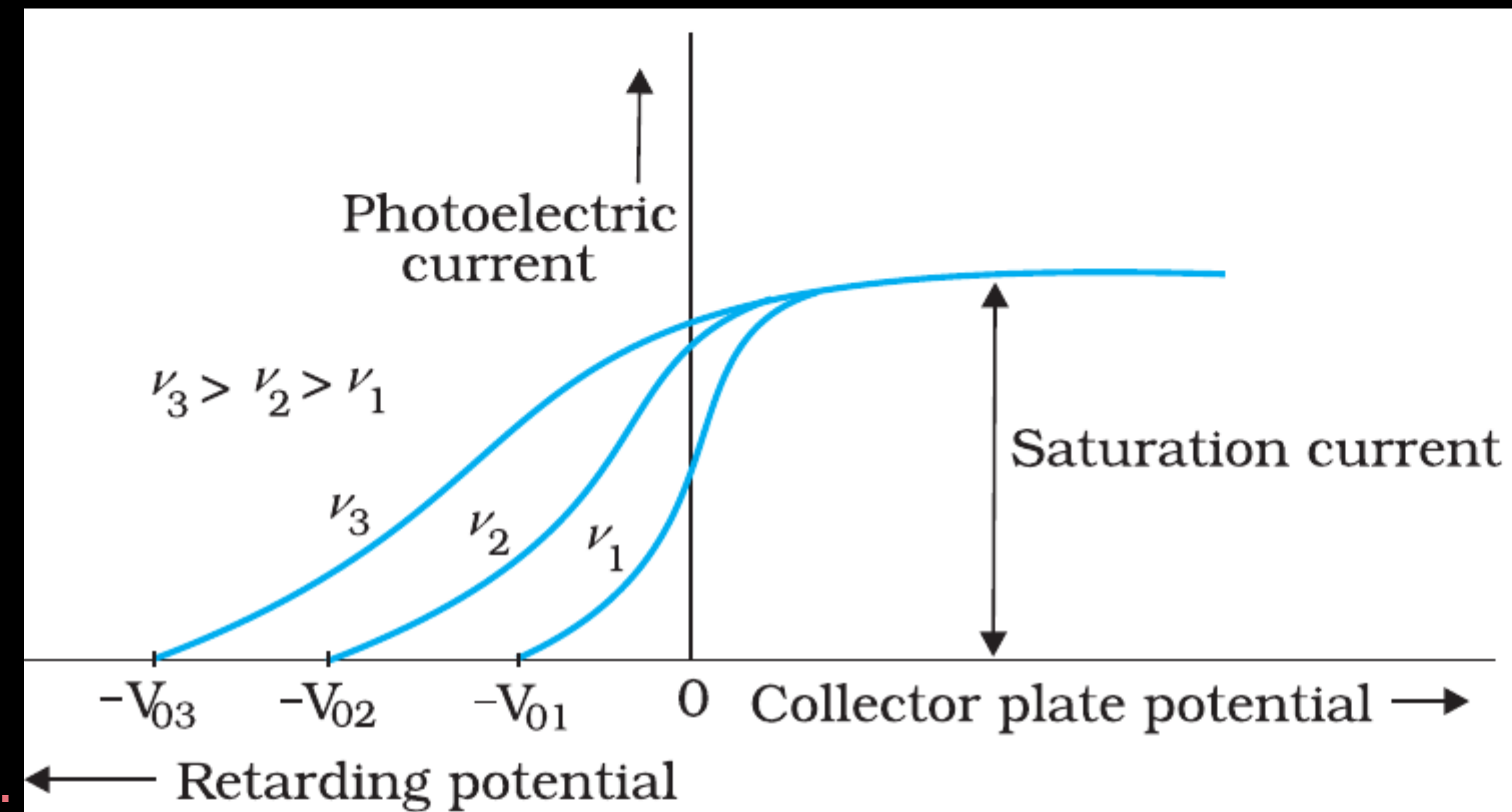


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

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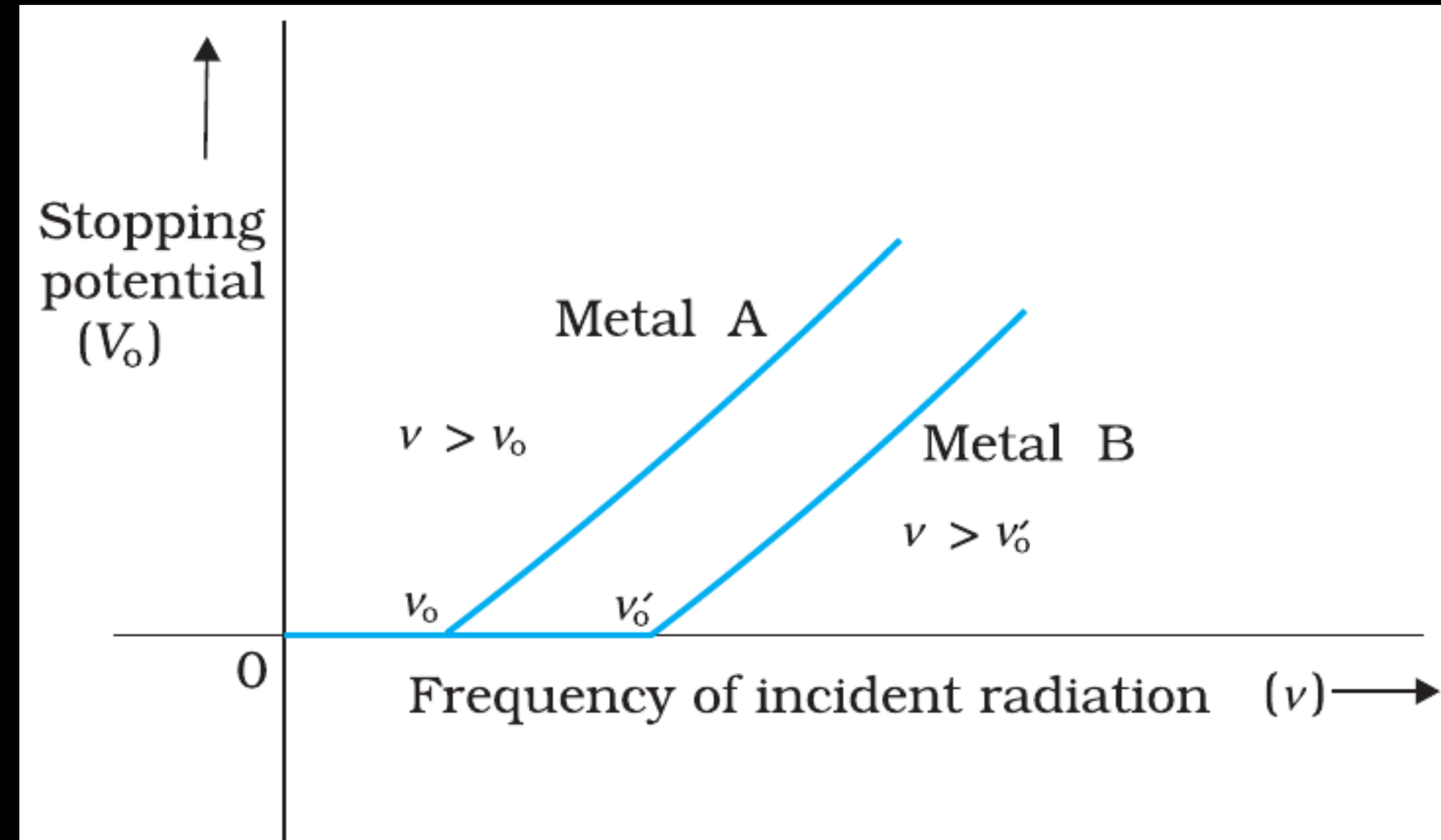


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

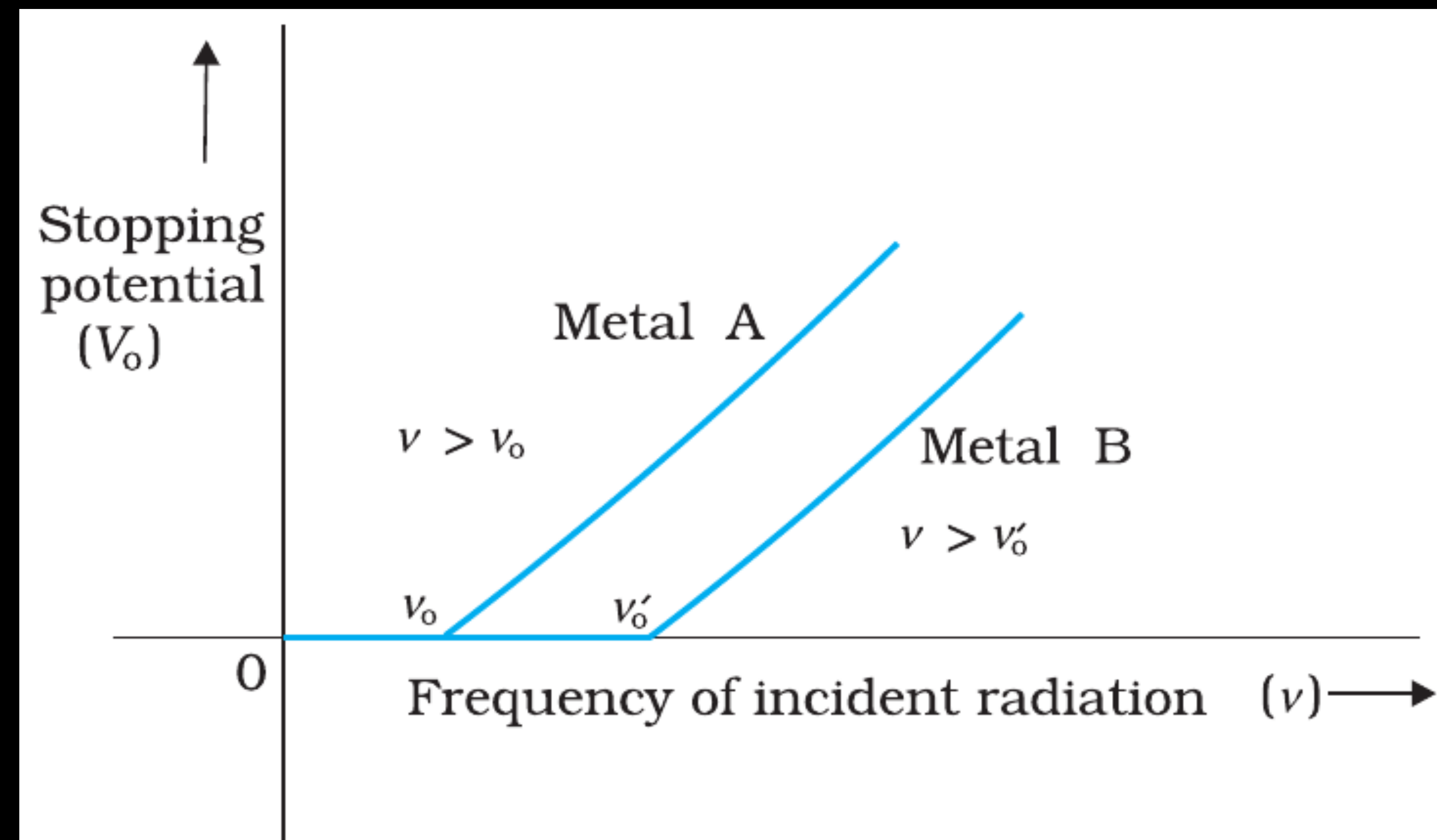


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

- (i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.
- (ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

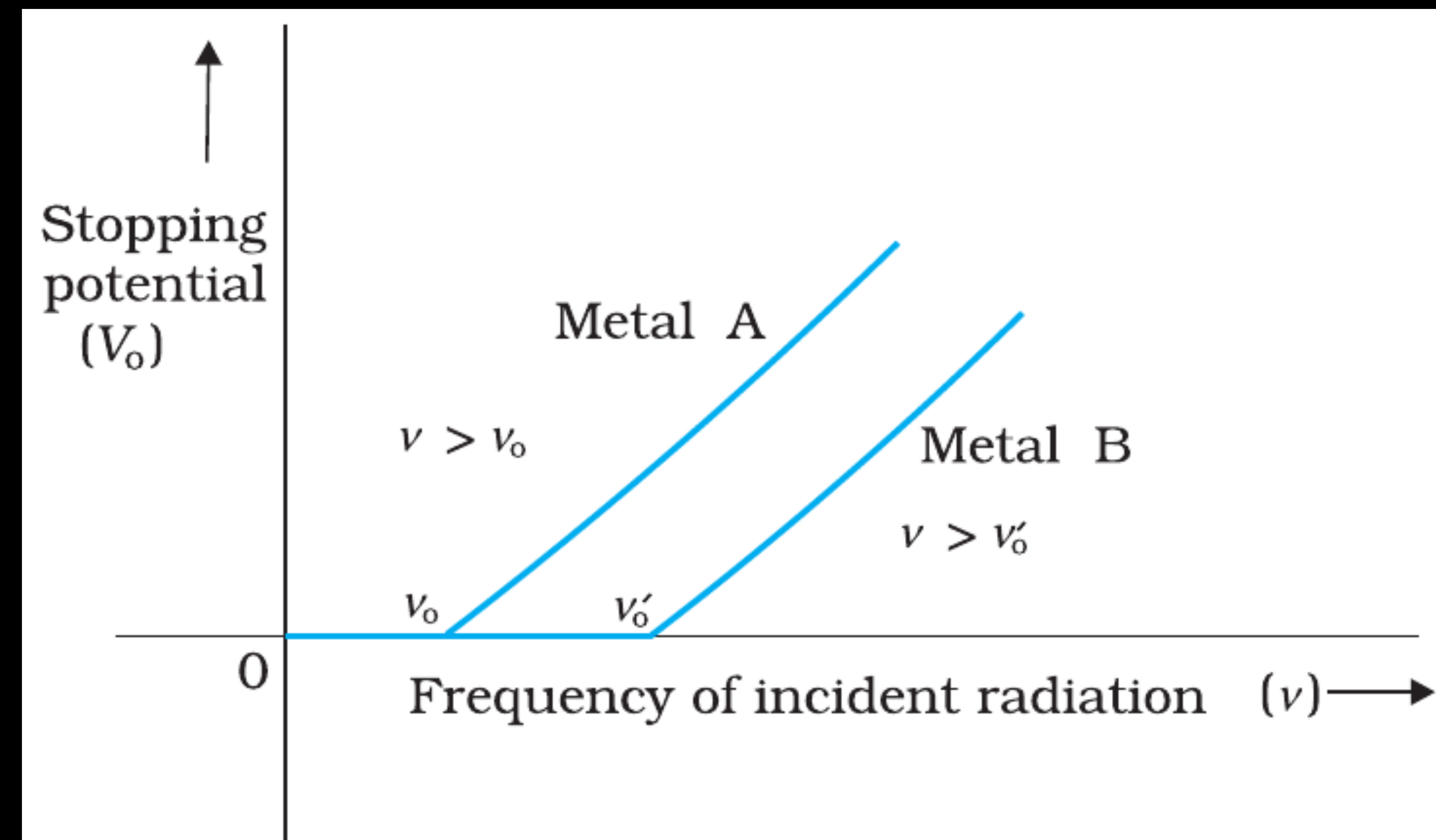


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* These observations have two implications:

(i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

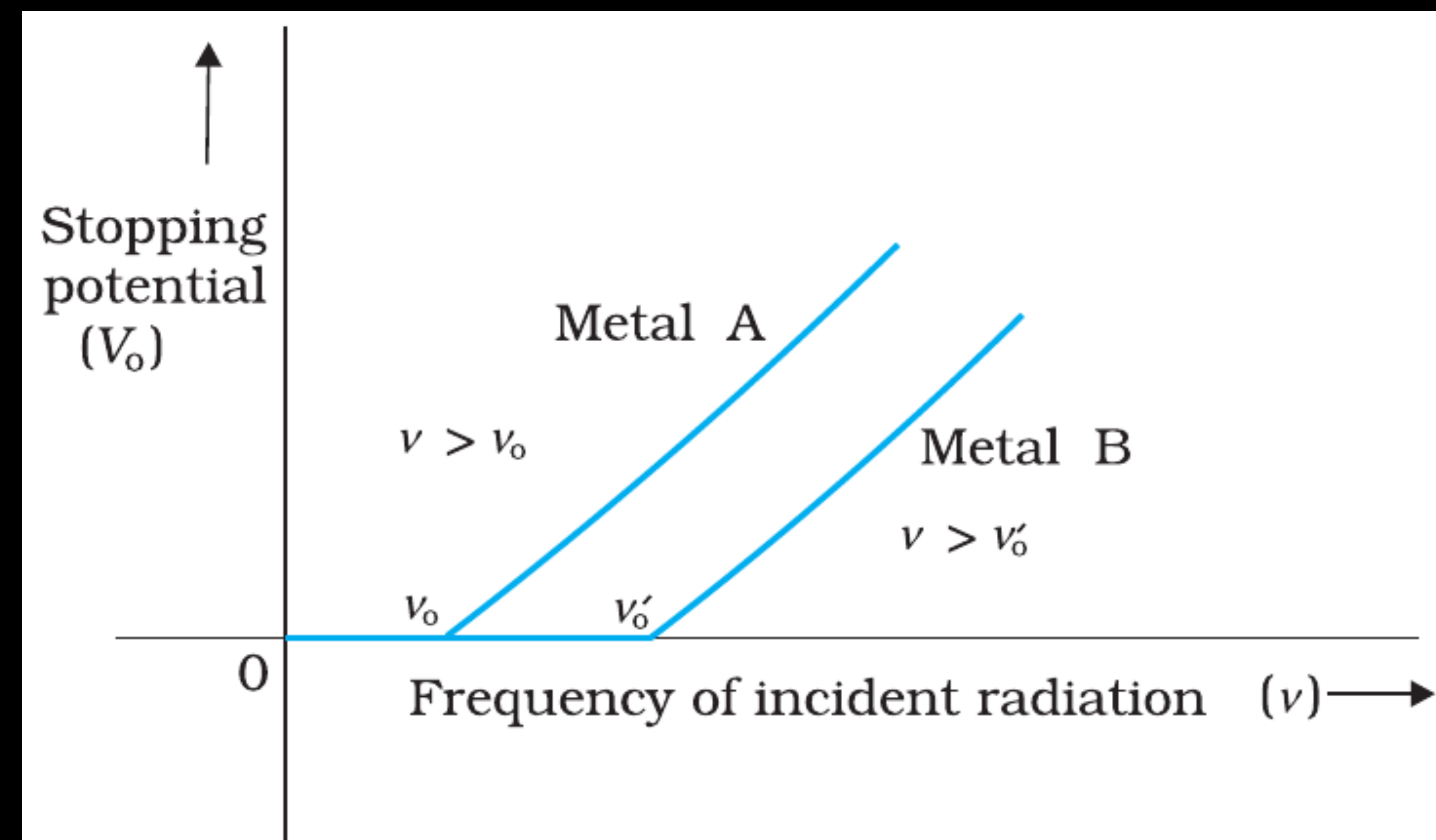


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EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

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* These observations have two implications:

(i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

(ii) For a frequency ν of incident radiation, lower than the cut-off frequency ν_0 , no photoelectric emission is possible even if the intensity is large.

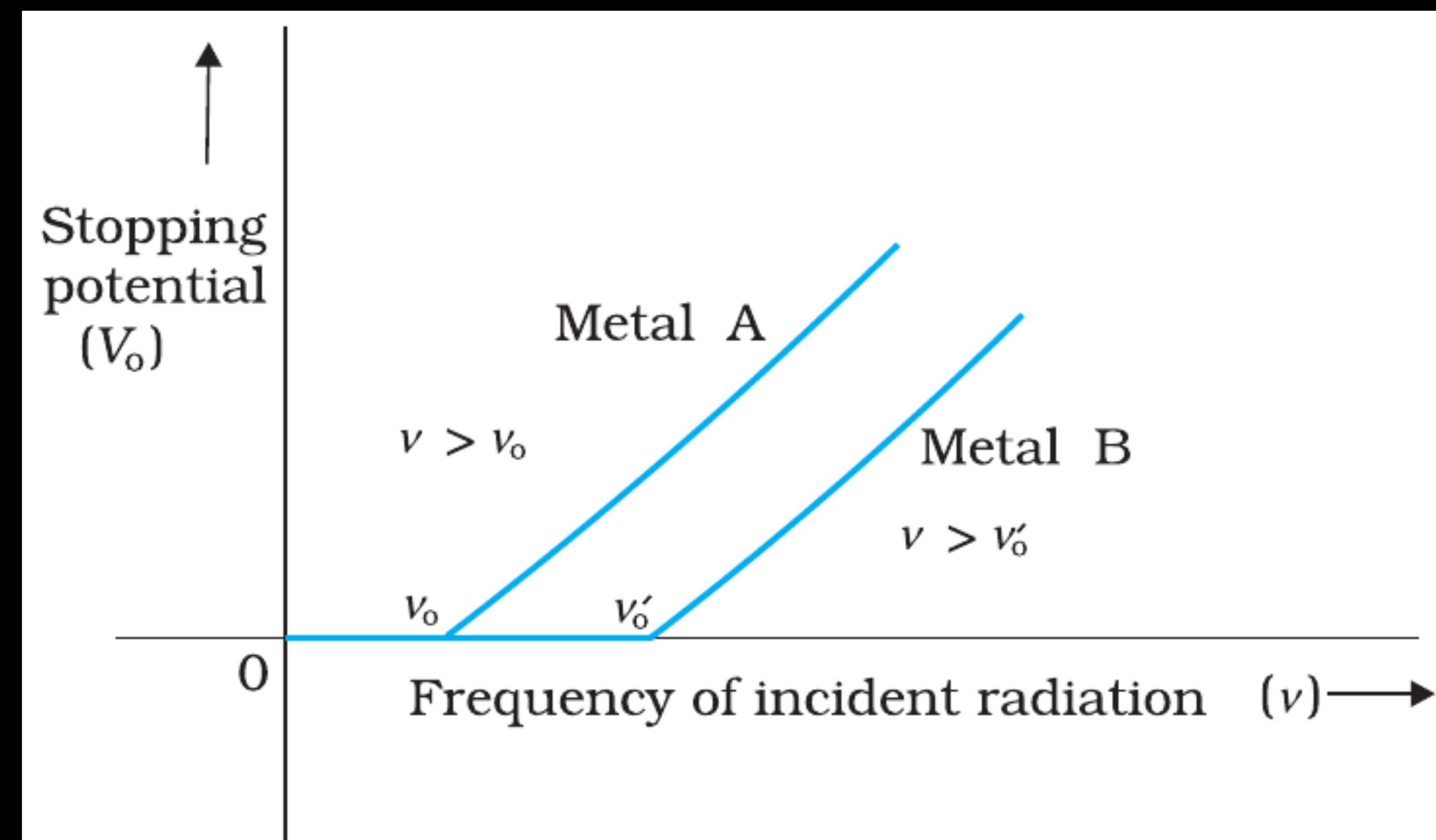


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

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(ii) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero.

* These observations have two implications:

(i) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation, but is independent of its intensity.

(ii) For a frequency ν of incident radiation, lower than the cut-off frequency ν_0 , no photoelectric emission is possible even if the intensity is large.

* This minimum, cut-off frequency ν_0 , is called the threshold frequency. It is different for different metals.

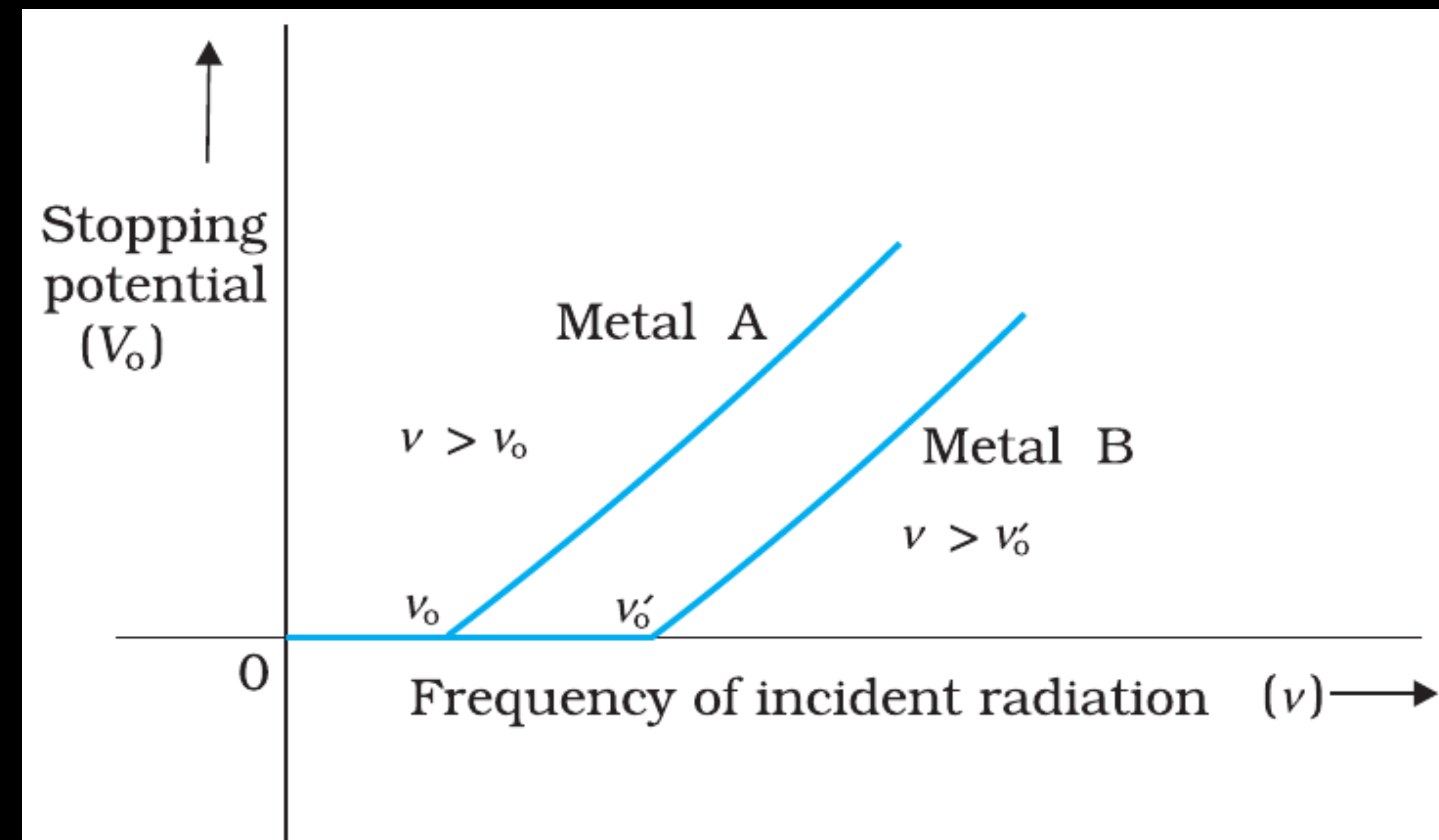


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

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- (i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

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- (i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.
- (ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

- (i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.
- (ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.
- (iii) For a given photosensitive material, there exists a certain minimum cut-off frequency of the incident radiation, called the threshold frequency, below which no emission of photoelectrons takes place, no matter how intense the incident light is. Above the threshold frequency, the stopping potential or equivalently the maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident radiation, but is independent of its intensity.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(4) Summary

- (i) For a given photosensitive material and frequency of incident radiation (above the threshold frequency), the photoelectric current is directly proportional to the intensity of incident light.
- (ii) For a given photosensitive material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation whereas the stopping potential is independent of its intensity.
- (iii) For a given photosensitive material, there exists a certain minimum cut-off frequency of the incident radiation, called the threshold frequency, below which no emission of photoelectrons takes place, no matter how intense the incident light is. Above the threshold frequency, the stopping potential or equivalently the maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident radiation, but is independent of its intensity.
- (iv) The photoelectric emission is an instantaneous process without any apparent time lag ($\sim 10^{-9}$ s or less), even when the incident radiation is made exceedingly dim.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

PARTICLE NATURE OF LIGHT: THE PHOTON

WAVE NATURE OF MATTER: DE-BROGLIE HYPOTHESIS
